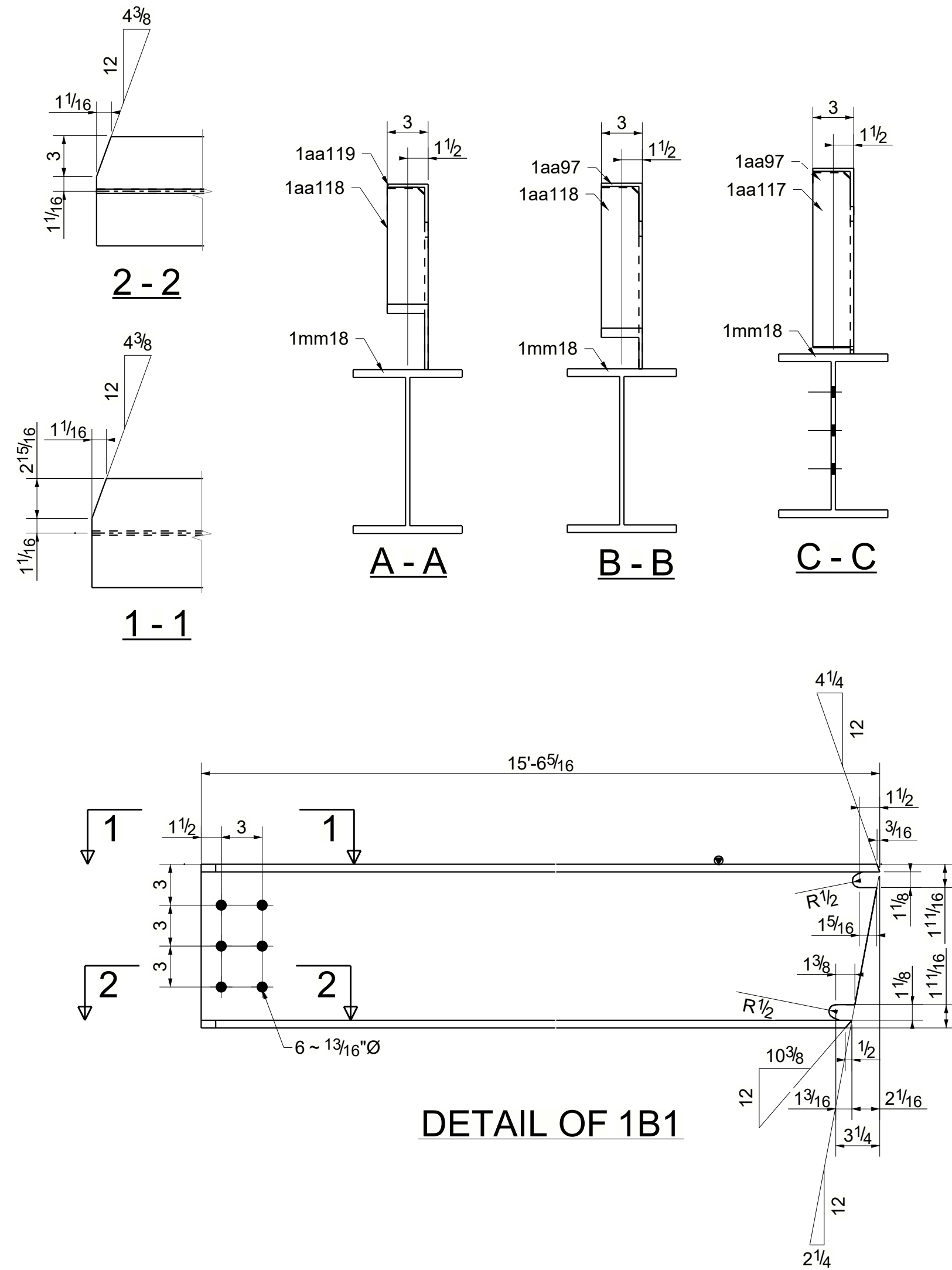
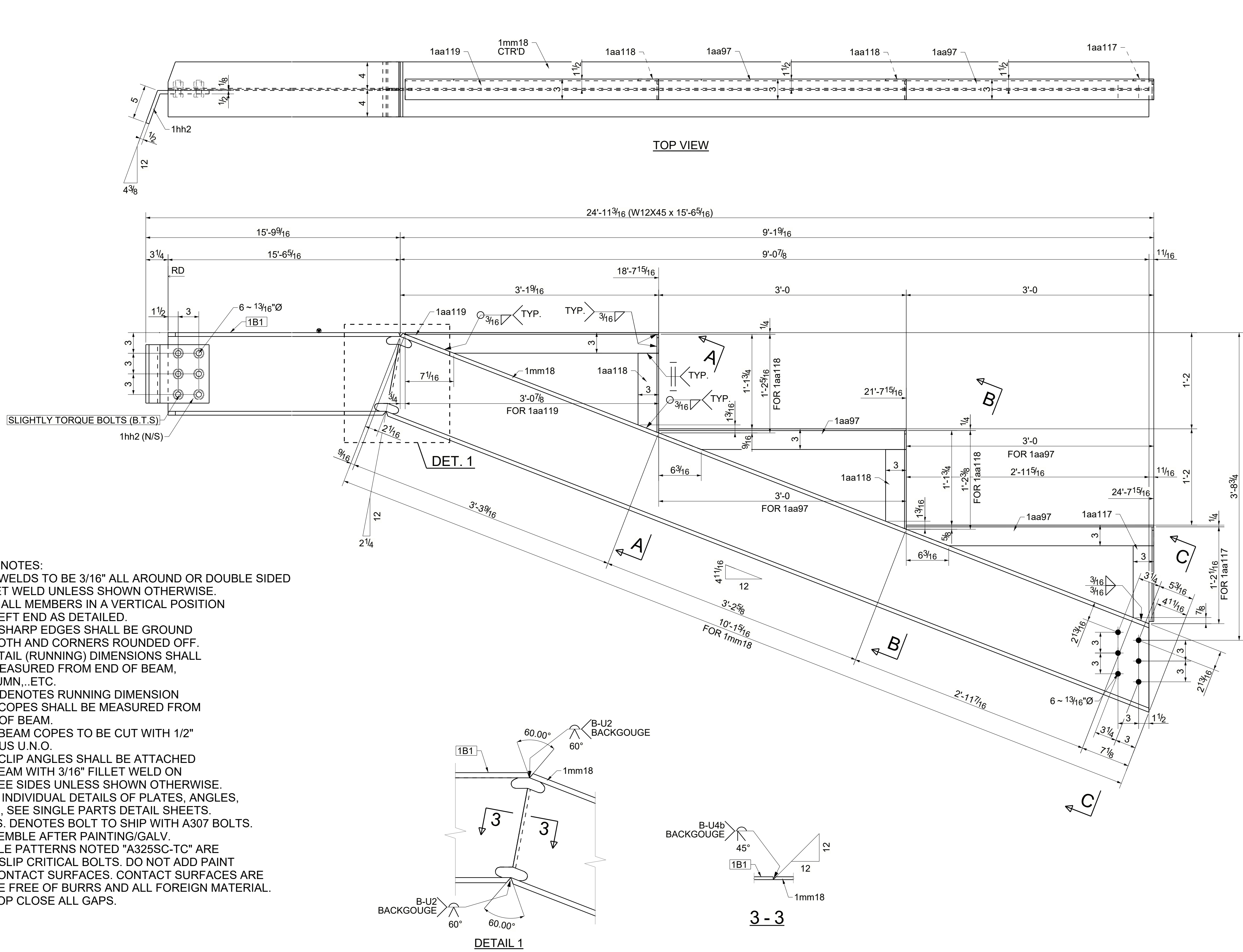
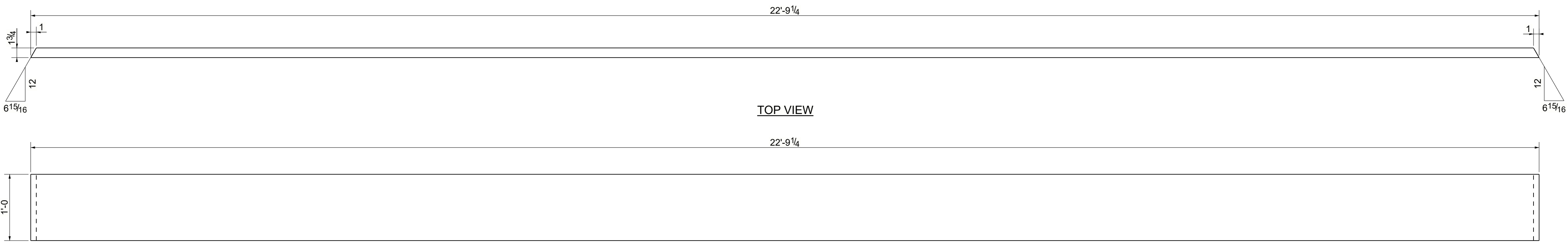




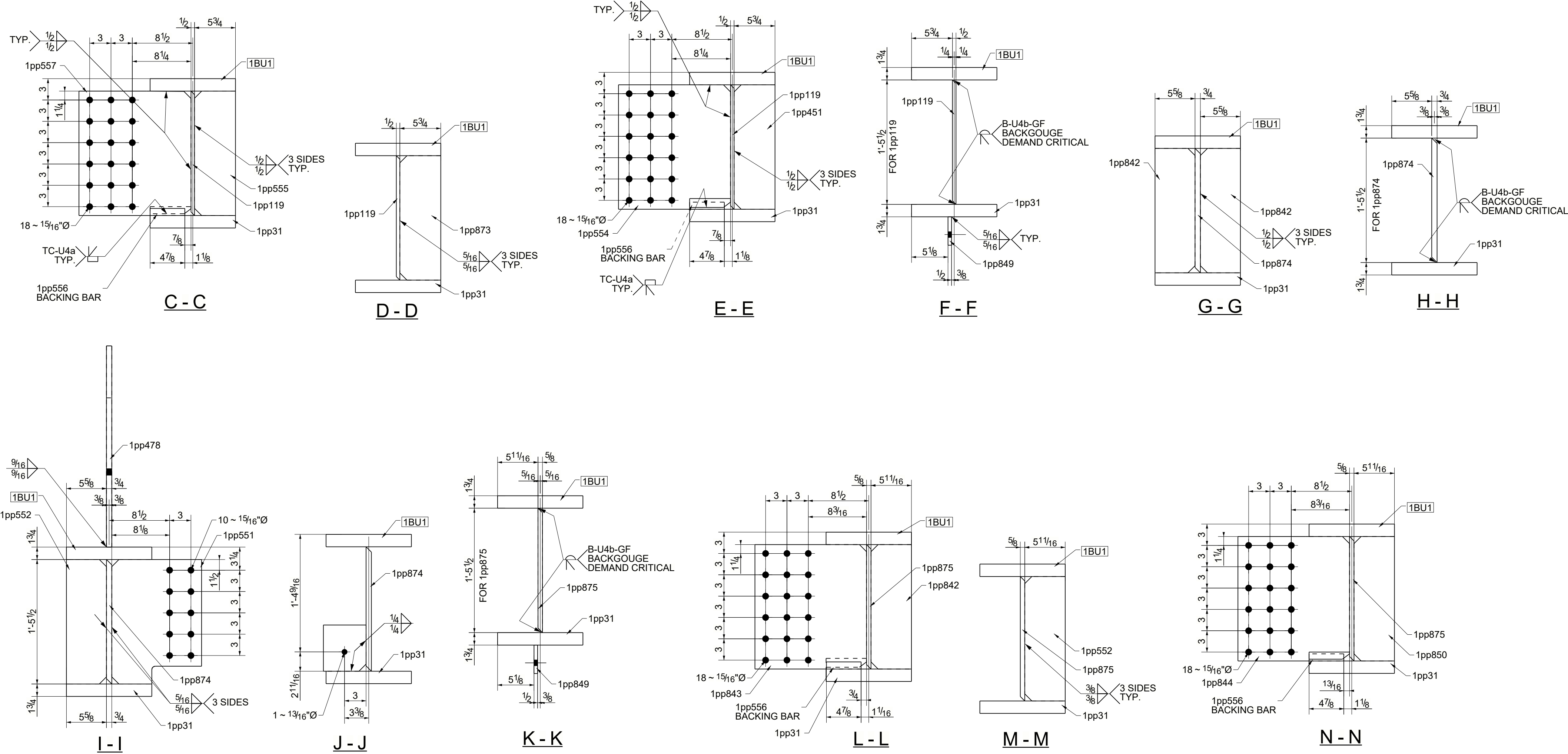
BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1B1	1	BEAM			SQ-1/BEV-1
1B1	1	W12X45	15'-6 5/16"	692	A992
1aa97	2	L3X3X1/4	3'-0"	29	A36
1aa117	1	L3X3X1/4	1'-2 1/16"	6	A36
1aa118	2	L3X3X1/4	1'-2 5/16"	12	A36
1aa119	1	L3X3X1/4	3'-0 7/8"	15	A36
1hh2	1	BPL1/2"X8 1/2"	0'-11 13/16"	15	A572-50
1mm18	1	W12X45	10'-1 5/16"	451	A992
TOTAL WEIGHT THIS DRAWING				1219	

ONE - BEAM - 1B1



DETAIL OF 1BU1



WORK THIS SHEET WITH 1BU1

DEMAND CRITICAL WELD NOTE:
WELDS NOTED "DEMAND CRITICAL" ARE
TO BE PER SECTION A3.4B & 12.3 OF
AISC-341-16.

BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1BU1	1	PLATE			BEV-2
1BU1	1	PL1 3/4"X12"	22'-9 1/4"	1621	[SFRS] A572-50
1pp31	1	PL1 3/4"X12"	22'-9 1/4"	1621	A572-50
1pp119	1	PL1/2"X17 1/2"	4'-8 5/8"	134	A572-50
1pp451	1	PL1 1/4"X5 3/4"	1'-5 1/2"	35	A572-50
1pp478	1	PL3/4"X28 5/16"	4'-7 1/16"	286	A572-50
1pp551	1	PL5/8"X12 5/8"	1'-5 1/2"	36	A572-50
1pp552	3	PL5/8"X5 5/8"	1'-5 1/2"	52	A572-50
1pp554	1	PL1 1/4"X15 3/4"	1'-5 1/2"	96	A572-50
1pp555	1	PL1"X5 3/4"	1'-5 1/2"	28	A572-50
1pp556	4	PL1/4"X1"	0'-4 7/8"	1	A36
1pp557	1	PL1"X15 3/4"	1'-5 1/2"	77	A572-50
1pp842	5	PL1 1/4"X5 11/16"	1'-5 1/2"	173	A572-50
1pp843	1	PL1 1/4"X15 11/16"	1'-5 1/2"	95	A572-50
1pp844	1	PL1"X15 11/16"	1'-5 1/2"	76	A572-50
1pp849	2	PL1/2"X4"	0'-6"	7	A572-50
1pp850	1	PL1"X5 11/16"	1'-5 1/2"	28	A572-50
1pp873	3	PL1/2"X5 3/4"	1'-5 1/2"	42	A572-50
1pp874	1	PL3/4"X17 1/2"	14'-3"	609	A572-50
1pp875	1	PL5/8"X17 1/2"	3'-9 5/8"	133	A572-50
1pp887	1	PL3/8"X6"	0'-6 1/2"	4	A36
TOTAL WEIGHT THIS DRAWING				5157	

SHOP NOTES:

- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
- TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
- ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
- ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN, ETC.
- "RD" DENOTES RUNNING DIMENSION
- ALL COPIES SHALL BE MEASURED FROM END OF BEAM
- ALL BEAM COPIES TO BE CUT WITH 1/2" RADIUS U.N.O.
- FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
- B.T.S. DENOTES BOLT TO SHIP"SNUG TIGHT" WITH NOTED BOLTS. ASSEMBLE AFTER PAINTING/GALV.
- HOLE PATTERNS NOTED "A325SC-TC" OR "A490SC-TC" ARE FOR SLIP CRITICAL BOLTS.
- PROVIDE SHERWIN WILLIAMS ZINC CLAD 4100 ORGANIC ZINC-RICH EPOXY PRIMER AT CONTACT SURFACES.
- SURFACE PREP: SSPC-SP8
- THE MINIMUM AREA OF THIS PAINT IS THE HOLE DIAMETER +2" AT HATCH AREAS SHOWN THUS:

SEISMIC SHOP NOTE :

SUMMARY of SFRS Requirements
(REF: S0.1 Section G)

PROTECTED ZONE NOTE: (from Section F4.5C AISC 341)

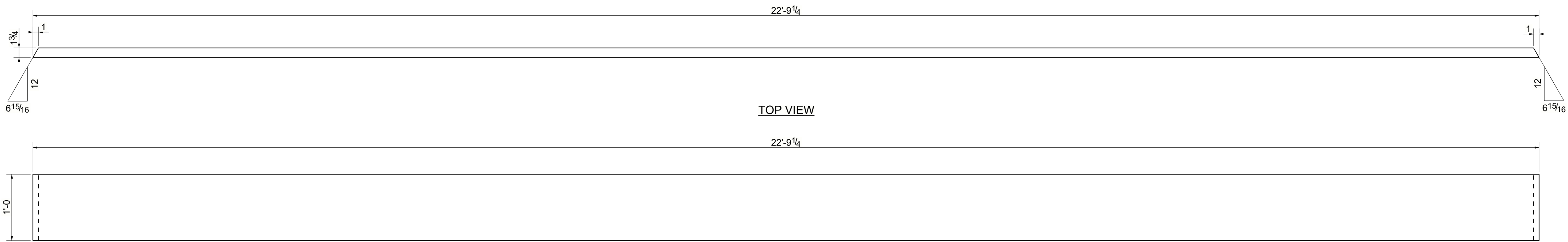
HATCHED AREAS SHOWN INDICATE SFRS PROTECTED ZONES THAT INCLUDE THE STEEL CORE OF BRACES AND ELEMENTS THAT CONNECT STEEL CORE TO BEAMS AND COLUMNS, AND SHALL SATISFY THE REQUIREMENTS OF SECTION D1.3 (AISC 341)

THE PROTECTED ZONE OF BRACING MEMBERS INCLUDES THE CENTER ONE-QUARTER OF THE BRACE LENGTH, AND A ZONE ADJACENT TO EACH CONNECTION EQUAL TO THE LARGEST DIMENSION OF THE BRACE CROSS SECTION MINIMUM. THE PROTECTED ZONE ALSO INCLUDES GUSSET PLATES AND AREAS WHERE BOLTS, WELDS, ANGLES, ETC. CONNECT THE BRACE TO THE GUSSET PLATE.

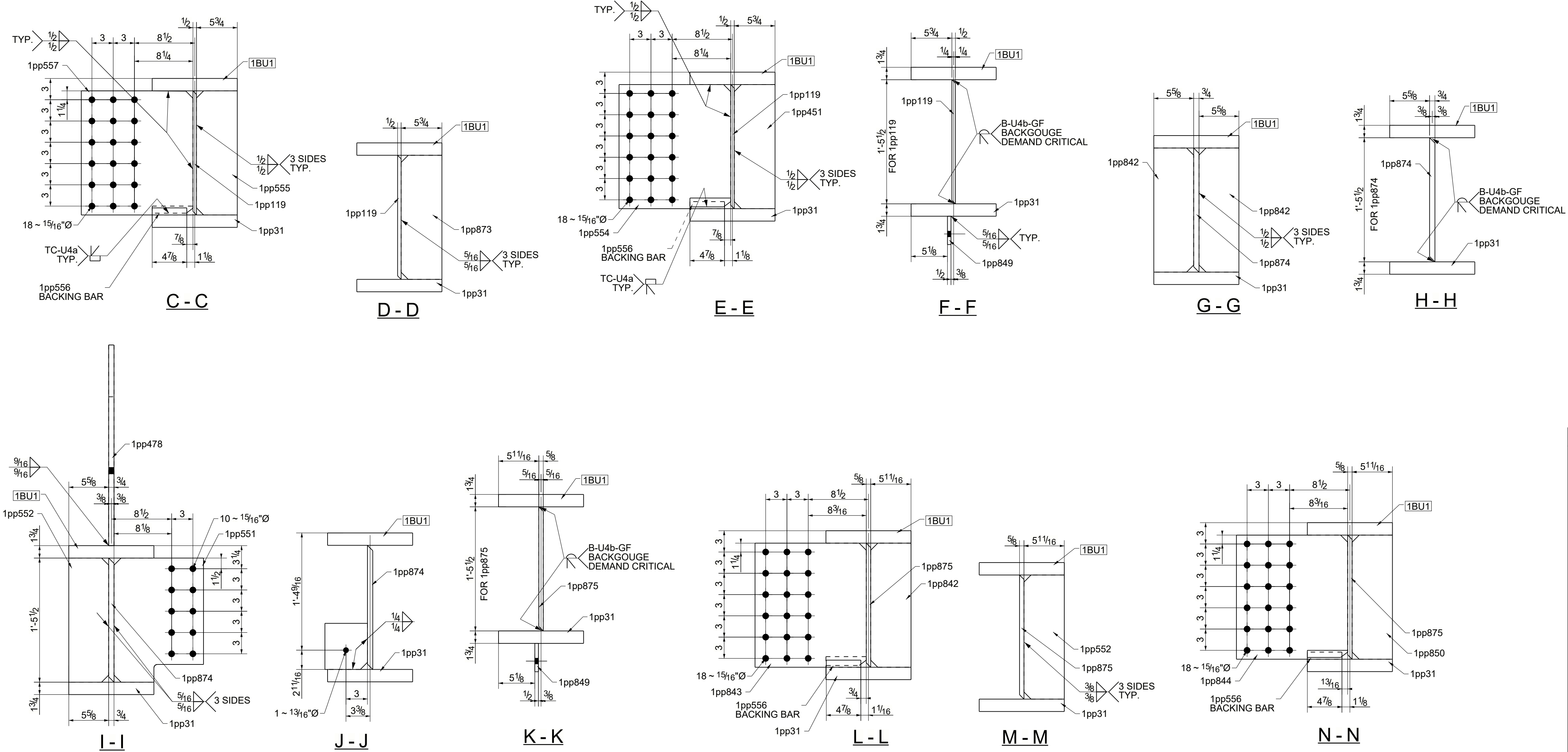
WITHIN THE PROTECTED ZONE, DISCONTINUITIES CREATED BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING AND THERMAL CUTTING ARE PROHIBITED.

WELDED, BOLTED, SCREWED OR SHOT-IN ATTACHMENTS OR OTHER ATTACHMENTS ARE PROHIBITED WITHIN THE PROTECTED ZONE.

ONE - PLATE - 1BU1



DETAIL OF 1BU1



WORK THIS SHEET WITH 1BU

BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1BU1	1	PLATE			BEV-2
1BU1	1	PL 1 3/4"X12"	22'-9 1/4"	1621	[SFRS] A572-50
1pp31	1	PL 1 3/4"X12"	22'-9 1/4"	1621	A572-50
1pp119	1	PL 1/2"X17 1/2"	4'-8 5/8"	134	A572-50
1pp451	1	PL 1 1/4"X5 3/4"	1'-5 1/2"	35	A572-50
1pp478	1	PL 3/4"X28 5/16"	4'-7 1/16"	286	A572-50
1pp551	1	PL 5/8"X12 5/8"	1'-5 1/2"	36	A572-50
1pp552	3	PL 5/8"X5 5/8"	1'-5 1/2"	52	A572-50
1pp554	1	PL 1 1/4"X15 3/4"	1'-5 1/2"	96	A572-50
1pp555	1	PL 1"X5 3/4"	1'-5 1/2"	28	A572-50
1pp556	4	PL 1/4"X1"	0'-4 7/8"	1	A36
1pp557	1	PL 1"X15 3/4"	1'-5 1/2"	77	A572-50
1pp842	5	PL 1 1/4"X5 11/16"	1'-5 1/2"	173	A572-50
1pp843	1	PL 1 1/4"X15 11/16"	1'-5 1/2"	95	A572-50
1pp844	1	PL 1"X15 11/16"	1'-5 1/2"	76	A572-50
1pp849	2	PL 1/2"X4"	0'-6"	7	A572-50
1pp850	1	PL 1"X5 11/16"	1'-5 1/2"	28	A572-50
1pp873	3	PL 1/2"X5 3/4"	1'-5 1/2"	42	A572-50
1pp874	1	PL 3/4"X17 1/2"	14'-3"	609	A572-50
1pp875	1	PL 5/8"X17 1/2"	3'-9 5/8"	133	A572-50
1pp887	1	PL 3/8"X6"	0'-6 1/2"	4	A36
TOTAL WEIGHT THIS DRAWING				5157	

SHOP NOTES:

- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
- TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
- ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
- ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN, ETC.
- "RD" DENOTES RUNNING DIMENSION
- ALL COPIES SHALL BE MEASURED FROM END OF BEAM
- ALL BEAM COPIES TO BE CUT WITH 1/2" RADIUS U.N.O.
- FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
- B.T.S. DENOTES BOLT TO SHIP"SNUG TIGHT" WITH NOTED BOLTS. ASSEMBLE AFTER PAINTING/GALV.
- HOLE PATTERNS NOTED "A325SC-TC" OR "A490SC-TC" ARE FOR SLIP CRITICAL BOLTS.
- PROVIDE SHERWIN WILLIAMS ZINC CLAD 4100 ORGANIC ZINC-RICH EPOXY PRIMER AT CONTACT SURFACES.
- SURFACE PREP: SSPC-SP8
- THE MINIMUM AREA OF THIS PAINT IS THE HOLE DIAMETER +2" AT HATCH AREAS SHOWN THUS:

SEISMIC SHOP NOTE :

SUMMARY of SFRS Requirements
(REF: S0.1 Section G)

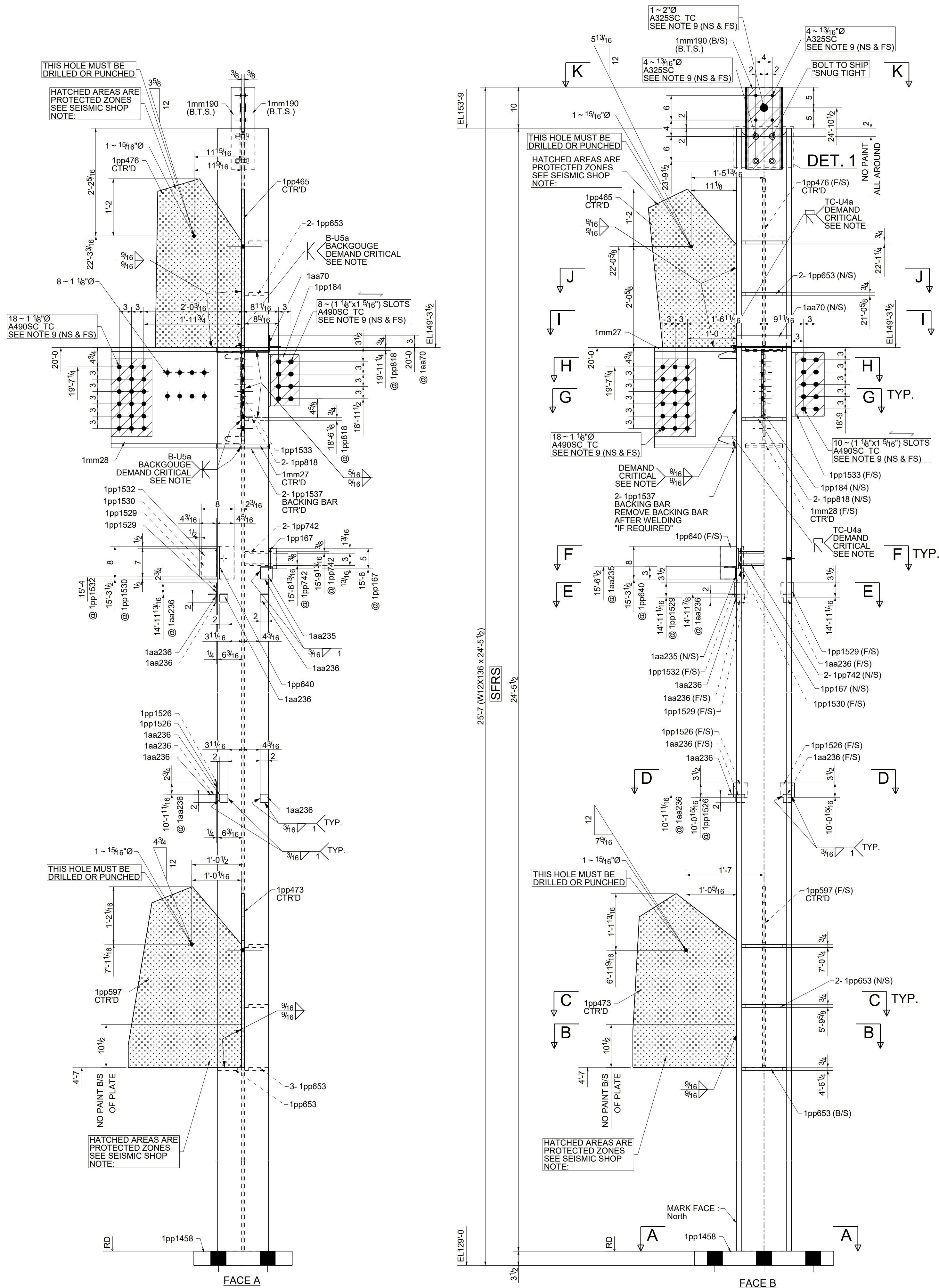
PROTECTED ZONE NOTE: (from Section F4.5C AISC 341)

HATCHED AREAS SHOWN INDICATE SFRS PROTECTED ZONES THAT INCLUDE THE STEEL CORE OF BRACES AND ELEMENTS THAT CONNECT STEEL CORE TO BEAMS AND COLUMNS, AND SHALL SATISFY THE REQUIREMENTS OF SECTION D1.3 (AISC 341)

THE PROTECTED ZONE OF BRACING MEMBERS INCLUDES THE CENTER ONE-QUARTER OF THE BRACE LENGTH, AND A ZONE ADJACENT TO EACH CONNECTION EQUAL TO THE LARGEST DIMENSION OF THE BRACE CROSS SECTION MINIMUM. THE PROTECTED ZONE ALSO INCLUDES GUSSET PLATES AND AREAS WHERE BOLTS, WELDS, ANGLES, ETC. CONNECT THE BRACE TO THE GUSSET PLATE.

WITHIN THE PROTECTED ZONE, DISCONTINUITIES CREATED BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING AND THERMAL CUTTING ARE PROHIBITED.

WELDED, BOLTED, SCREWED OR SHOT-IN ATTACHMENTS OR OTHER ATTACHMENTS ARE PROHIBITED WITHIN THE PROTECTED ZONE.



ONE - COLUMN - 1C59

BILL OF MATERIAL					
MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1C59	1	COLUMN			SQ-2
1C59	1	W12X136	24'-5 1/2"	3321	[SFRS] A992
1aa70	1	L3X3X1/4	1'-1 3/8"	5	A36
1aa235	1	L3X3X1/4	0'-3"	1	A36
1aa236	8	L2X2X3/16	0'-2"	3	A36
1mm27	1	W24X131	1'-8"	219	A992
1mm28	1	W24X103	2'-7 3/4"	273	A992
1mm190	2	C6X15	1'-8"	50	A36
1pp167	1	PL1/2"x2 1/16"	0'-5"	1	A36
1pp184	1	PL1/2"x13 5/16"	1'-4 3/8"	27	A572-50
1pp465	1	PL3/4"x21 9/16"	3'-2 11/16"	142	A572-50
1pp473	1	PL3/4"x25 5/16"	3'-6 7/16"	199	A572-50
1pp476	1	PL3/4"x21"	3'-5 3/16"	161	A572-50
1pp597	1	PL3/4"x27 5/8"	3'-8 1/16"	211	A572-50
1pp640	1	PL3/8"x4"	0'-8"	3	A36
1pp653	6	PL3/4"x5 3/4"	0'-10 7/8"	79	A572-50
1pp742	2	PL3/8"x6 3/16"	0'-7 7/8"	7	A36
1pp818	2	PL3/4"x5 3/4"	0'-10 7/8"	26	A572-50
1pp1458	1	PL3 1/2"x24"	2'-10"	810	A572-50
1pp1526	2	PL1/4"x3 1/2"	0'-3 1/2"	2	A36
1pp1529	2	PL1/4"x3 1/8"	0'-3 1/2"	2	A36
1pp1530	1	PL3/8"x8"	0'-8"	7	A36
1pp1532	1	PL1/4"x4 3/16"	0'-7"	2	A36
1pp1533	1	PL1/2"x8"	1'-3 1/2"	18	A572-50
1pp1537	2	PL1/4"x1"	1'-0 7/8"	2	A36
	4	3/4"Ø A325SC BOLT	0'-2 3/4"	3	Workshop
TOTAL WEIGHT THIS DRAWING				5573	

SEISMIC SHOP NOTE :

SUMMARY of SFRS Requirements
(REF: S0.1 Section G)

PROTECTED ZONE NOTE: (from Section F4.5C AISC 341)

HATCHED AREAS SHOWN INDICATE SFRS PROTECTED ZONES THAT INCLUDE THE STEEL CORE OF BRACES AND ELEMENTS THAT CONNECT STEEL CORE TO BEAMS AND COLUMNS, AND SHALL SATISFY THE REQUIREMENTS OF SECTION D1.3 (AISC 341)

THE PROTECTED ZONE OF BRACING MEMBERS INCLUDES THE CENTER ONE-QUARTER OF THE BRACE LENGTH, AND A ZONE ADJACENT TO EACH CONNECTION EQUAL TO THE LARGEST DIMENSION OF THE BRACE CROSS SECTION MINIMUM. THE PROTECTED ZONE ALSO INCLUDES GUSSET PLATES AND AREAS WHERE BOLTS, WELDS, ANGLES, ETC. CONNECT THE BRACE TO THE GUSSET PLATE.

WITHIN THE PROTECTED ZONE, DISCONTINUITIES CREATED BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING AND THERMAL CUTTING ARE PROHIBITED.

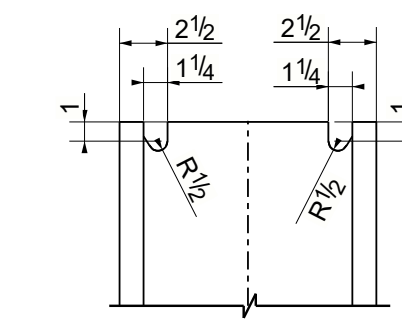
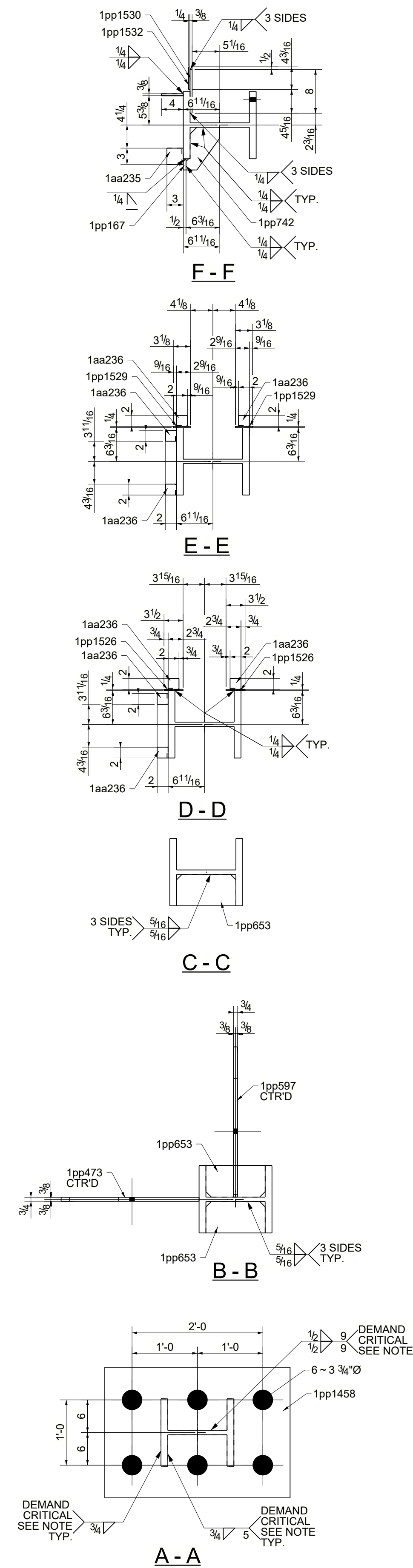
WELDED, BOLTED, SCREWED OR SHOT-IN ATTACHMENTS OR OTHER ATTACHMENTS ARE PROHIBITED WITHIN THE PROTECTED ZONE.

REFER TO SPECIFICATIONS FOR TESTING REQUIREMENTS FOR BASE PLATES WITH THICKNESS GREATER THAN OR EQUAL TO 2".

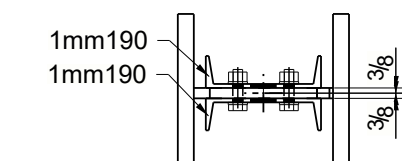
DEMAND CRITICAL WELD NOTE:
WELDS NOTED "DEMAND CRITICAL" ARE TO BE PER SECTION A3.4B & 12.3 OF AISC-341-16.

- SHOP NOTES:
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN, ETC.
 - "RD" DENOTES RUNNING DIMENSION
 - ALL COPES SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPES TO BE CUT WITH 1/2" RADIUS U.N.O.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
 - B.T.S.DENOTES BOLT TO SHIP"SNUG TIGHT" WITH NOTED BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - HOLE PATTERNS NOTED "A325SC-TC" OR "A490SC-TC" ARE FOR SLIP CRITICAL BOLTS. PROVIDE SHERWIN WILLIAMS ZINC CLAD 4100 ORGANIC ZINC-RICH EPOXY PRIMER AT CONTACT SURFACES. SURFACE PREP: SSPC-SP6
 - THE MINIMUM AREA OF THIS PAINT IS THE HOLE DIAMETER +2" AT HATCH AREAS SHOWN THUS:

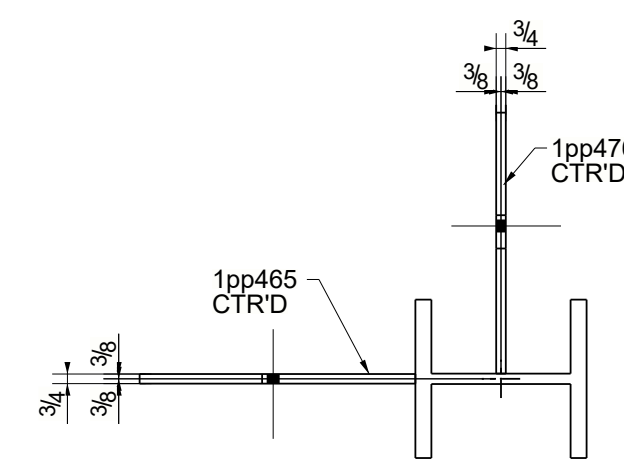
WORK THIS SHEET WITH 1C59-1



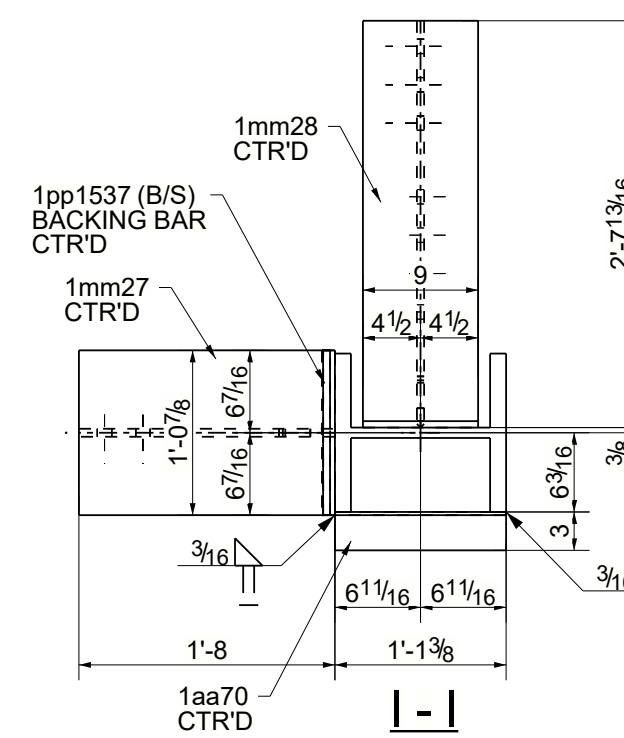
DETAIL 1



K - K

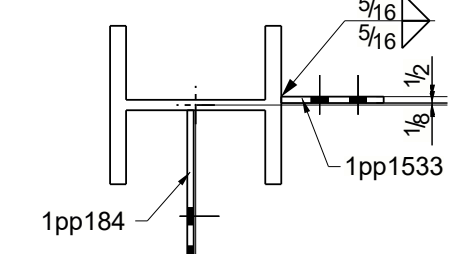


J - J

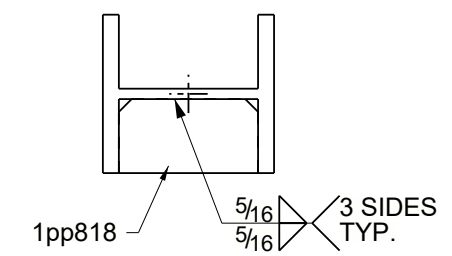


1

WORK THIS SHEET WITH 1C59



H - H



G - G

ONE - COLUMN - 1C59

BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1C62	1	COLUMN			SQ-2
1C62	1	W12X120	24'-6 1/2"	2940	(SFRS) A992
1aa60	1	L3X3X1/4	0'-3"	1	A36
1aa236	12	L2X2X3/16	0'-2"	5	A36
1mm27	1	W24X131	1'-8"	219	A992
1mm71	1	W24X131	2'-4"	307	A992
1mm177	1	W24X131	2'-4"	307	A992
1mm188	1	C9X15	1'-8"	25	A36
1mm189	1	C9X15	1'-8"	25	A36
1pp167	1	PL1/2"x2 1/16"	0'-5"	1	A36
1pp436	1	PL3/4"x30 5/16"	2'-6 15/16"	152	A572-50
1pp443	1	PL3/4"x32 9/16"	3'-8"	225	A572-50
1pp444	1	PL3/4"x31 7/16"	2'-10 5/8"	180	A572-50
1pp459	1	PL3/4"x38 15/16"	3'-9 13/16"	291	A572-50
1pp466	1	PL3/4"x21 9/16"	3'-2 5/8"	142	A572-50
1pp472	1	PL3/4"x25 7/16"	3'-6 5/16"	199	A572-50
1pp629	2	PL1/4"x5 1/2"	0'-6 1/16"	5	A36
1pp635	1	PL3/4"x4"	0'-8"	6	A36
1pp663	1	PL3/8"x4 1/2"	0'-8"	4	A36
1pp743	2	PL3/8"x6 1/16"	0'-7 15/16"	7	A36
1pp1459	1	PL3"x20"	2'-6"	510	A572-50
1pp1479	6	PL3/4"x5 13/16"	0'-10 7/8"	80	A572-50
1pp1526	8	PL1/4"x3 1/2"	0'-3 1/2"	7	A36
1pp1537	2	PL1/4"x1"	1'-0 7/8"	2	A36
	4	3/4"Ø A325SC BOLT	0'-2 3/4"	3	Workshop
TOTAL WEIGHT THIS DRAWING				5642	

SEISMIC SHOP NOTE :

SUMMARY OF SFRS Requirements

(REF: S0.1 Section G)

PROTECTED ZONE NOTE:

(from Section F4.5C AISC 341)

HATCHED AREAS SHOWN INDICATE SFRS PROTECTED ZONES THAT INCLUDE THE STEEL CORE OF BRACES AND ELEMENTS THAT CONNECT STEEL CORE TO BEAMS AND COLUMNS, AND SHALL SATISFY THE REQUIREMENTS OF SECTION D1.3 (AISC 341)

THE PROTECTED ZONE OF BRACING MEMBERS INCLUDES THE CENTER ONE-QUARTER OF THE BRACE LENGTH, AND A ZONE ADJACENT TO EACH CONNECTION EQUAL TO THE LARGEST DIMENSION OF THE BRACE CROSS SECTION MINIMUM. THE PROTECTED ZONE ALSO INCLUDES GUSSET PLATES AND AREAS WHERE BOLTS, WELDS, ANGLES, ETC. CONNECT THE BRACE TO THE GUSSET PLATE.

WITHIN THE PROTECTED ZONE, DISCONTINUITIES CREATED BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING AND THERMAL CUTTING ARE PROHIBITED.

WELDED, BOLTED, SCREWED OR SHOT-IN ATTACHMENTS OR OTHER ATTACHMENTS ARE PROHIBITED WITHIN THE PROTECTED ZONE.

SHOP NOTES:

1. ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.

2. TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.

3. ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.

4. ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN, ETC.

5. ALL COPIES SHALL BE MEASURED FROM END OF BEAM.

6. ALL BEAM COPIES TO BE CUT WITH 1/2" RADIUS U.N.O.

7. FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.

8. B.T.S. DENOTES BOLT TO SHIP "SNUG TIGHT" WITH NOTED BOLTS. ASSEMBLE AFTER PAINTING/GALV.

9. HOLE PATTERNS NOTED "A325SC-TC" OR "A490SC-TC" ARE FOR SLIP CRITICAL BOLTS.

PROVIDE SHERWIN WILLIAMS ZINC CLAD 4100 ORGANIC ZINC-RICH EPOXY PRIMER AT CONTACT SURFACES.

SURFACE PREP: SSPC-SP6

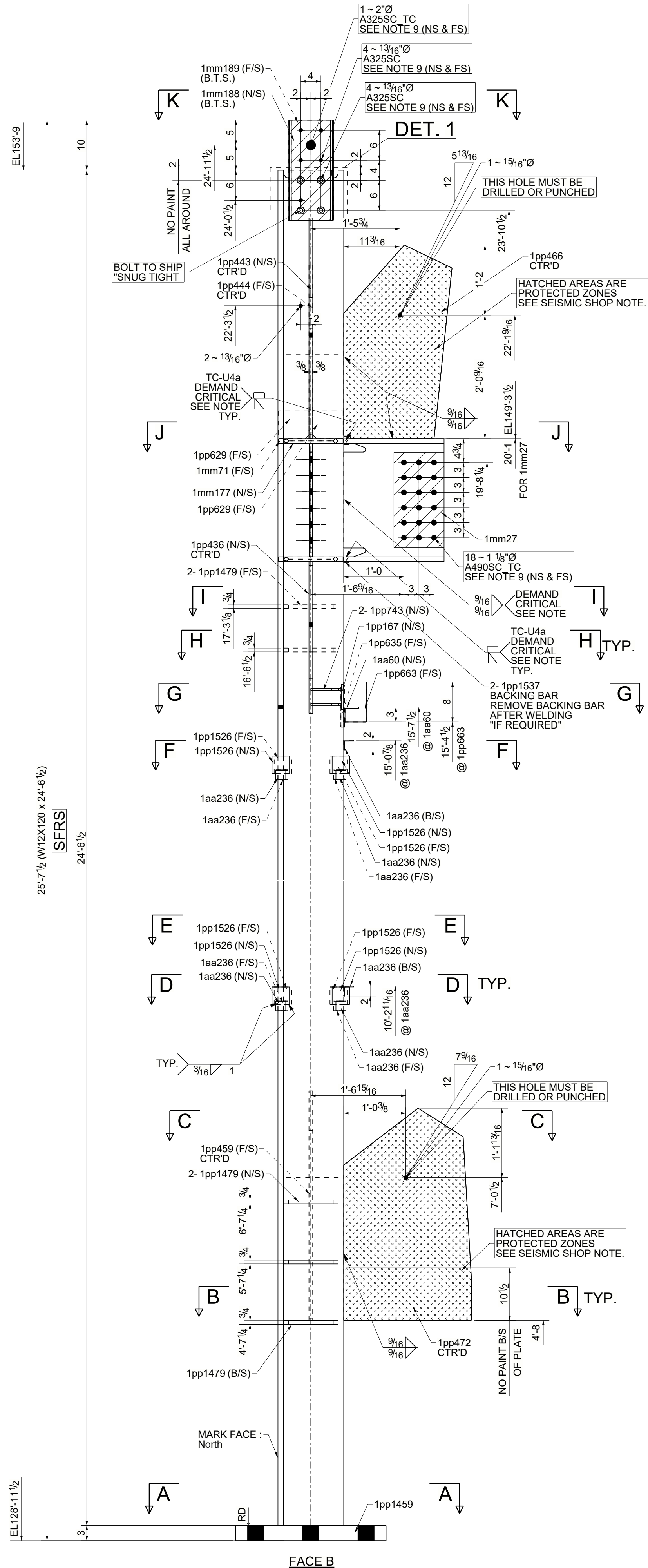
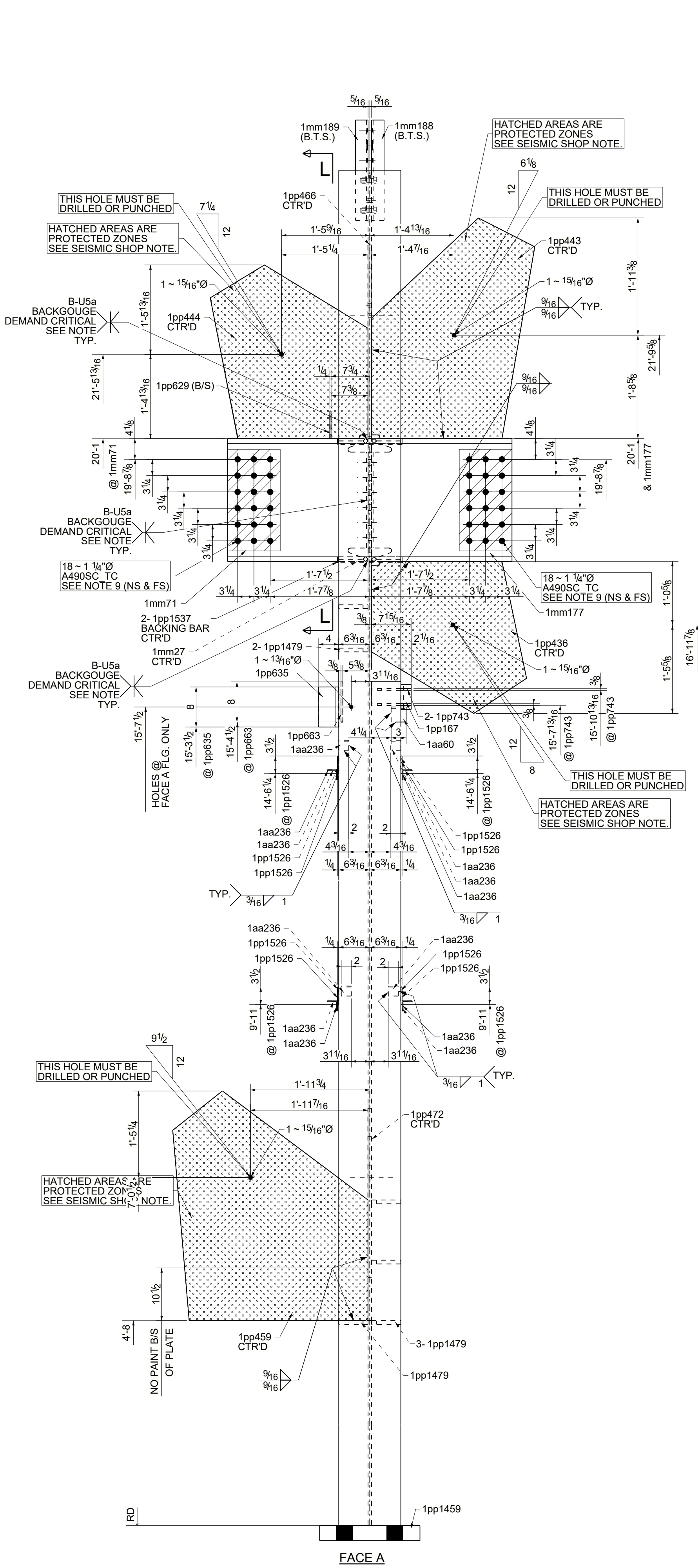
THE MINIMUM AREA OF THIS PAINT IS THE HOLE DIAMETER +2" AT HATCH AREAS SHOWN THUS:

DEMAND CRITICAL WELD NOTE:

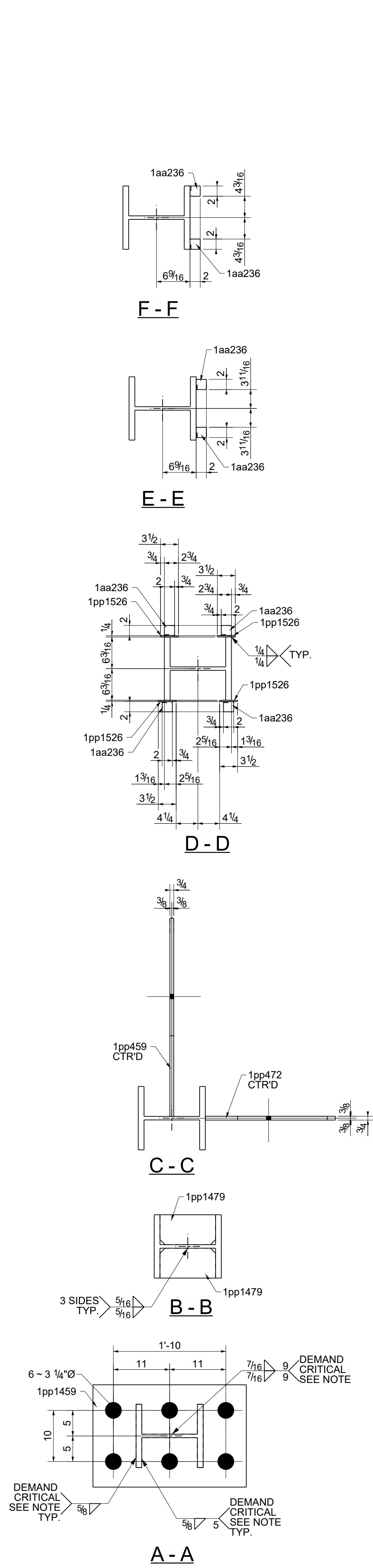
WELDS NOTED "DEMAND CRITICAL" ARE TO BE PER SECTION A3.4B & 12.3 OF AISC-341-16.

REFER TO SPECIFICATIONS FOR TESTING REQUIREMENTS FOR BASE PLATES WITH THICKNESS GREATER THAN OR EQUAL TO 2".

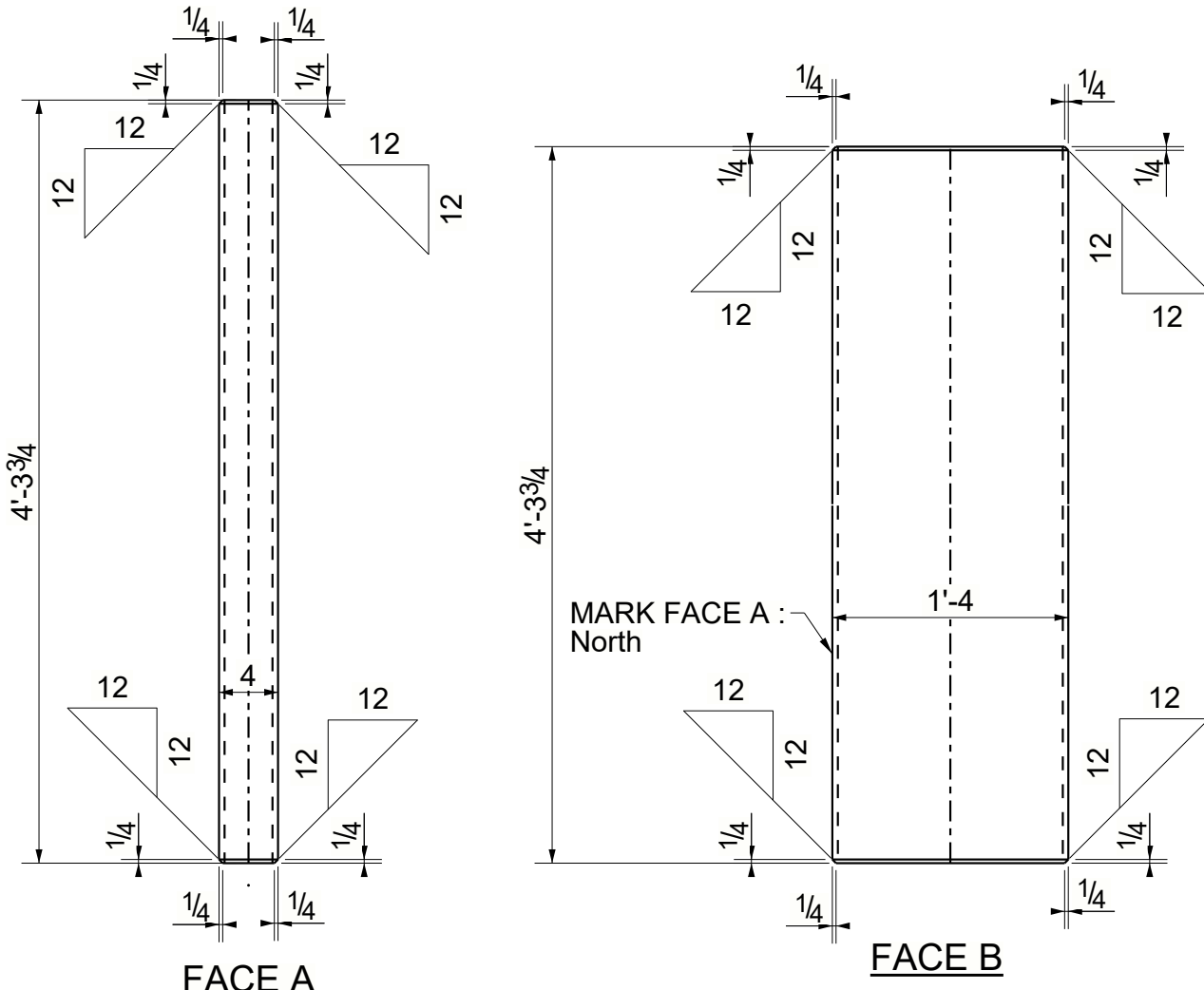
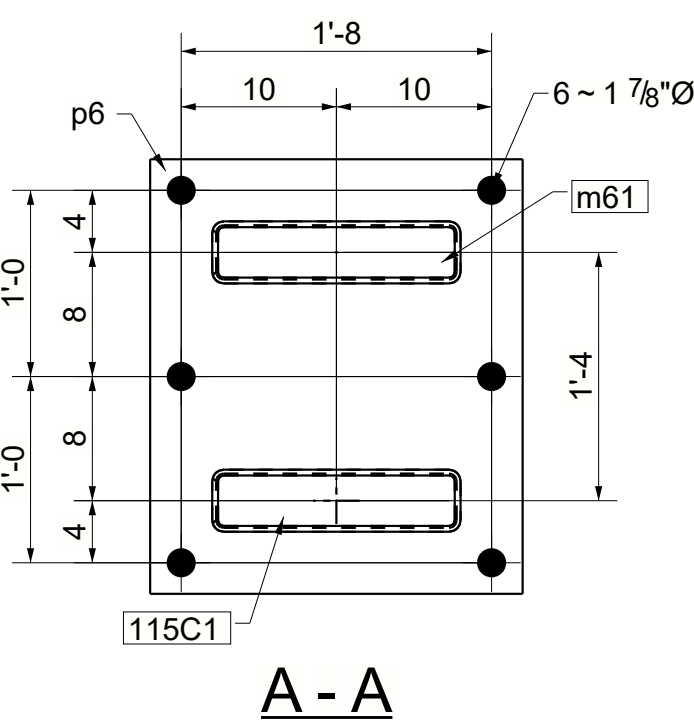
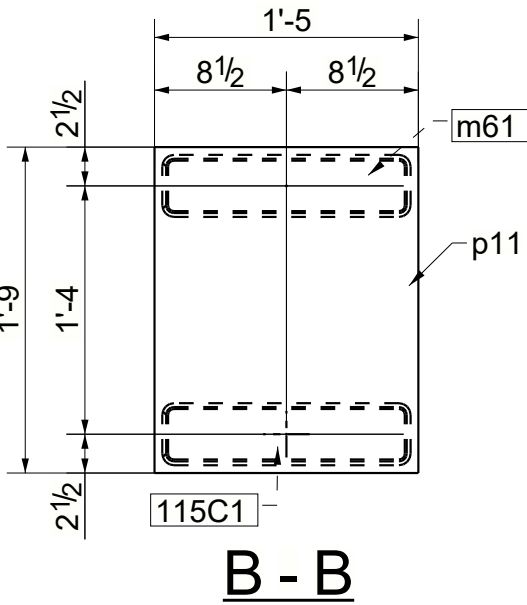
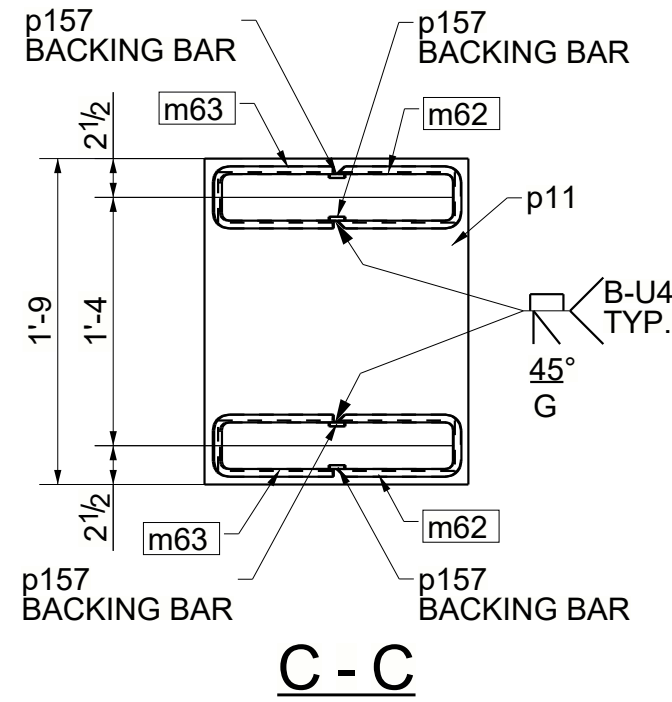
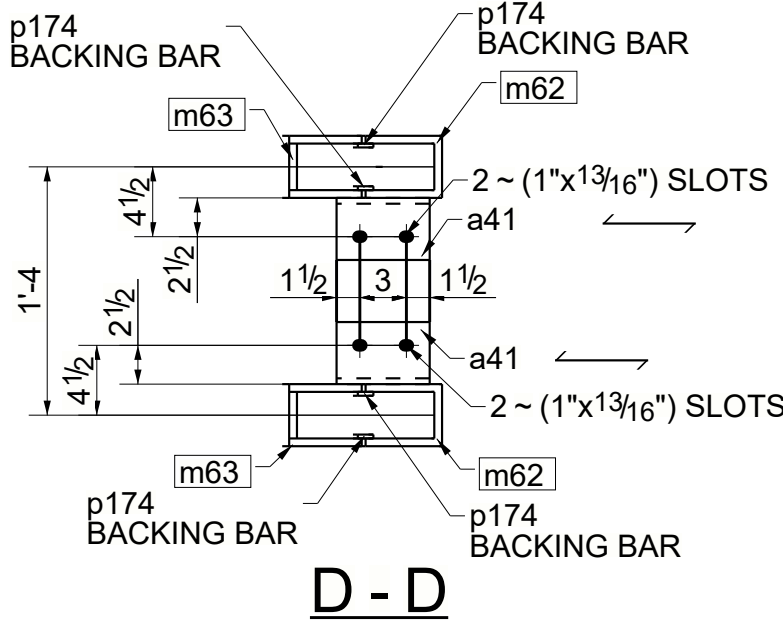
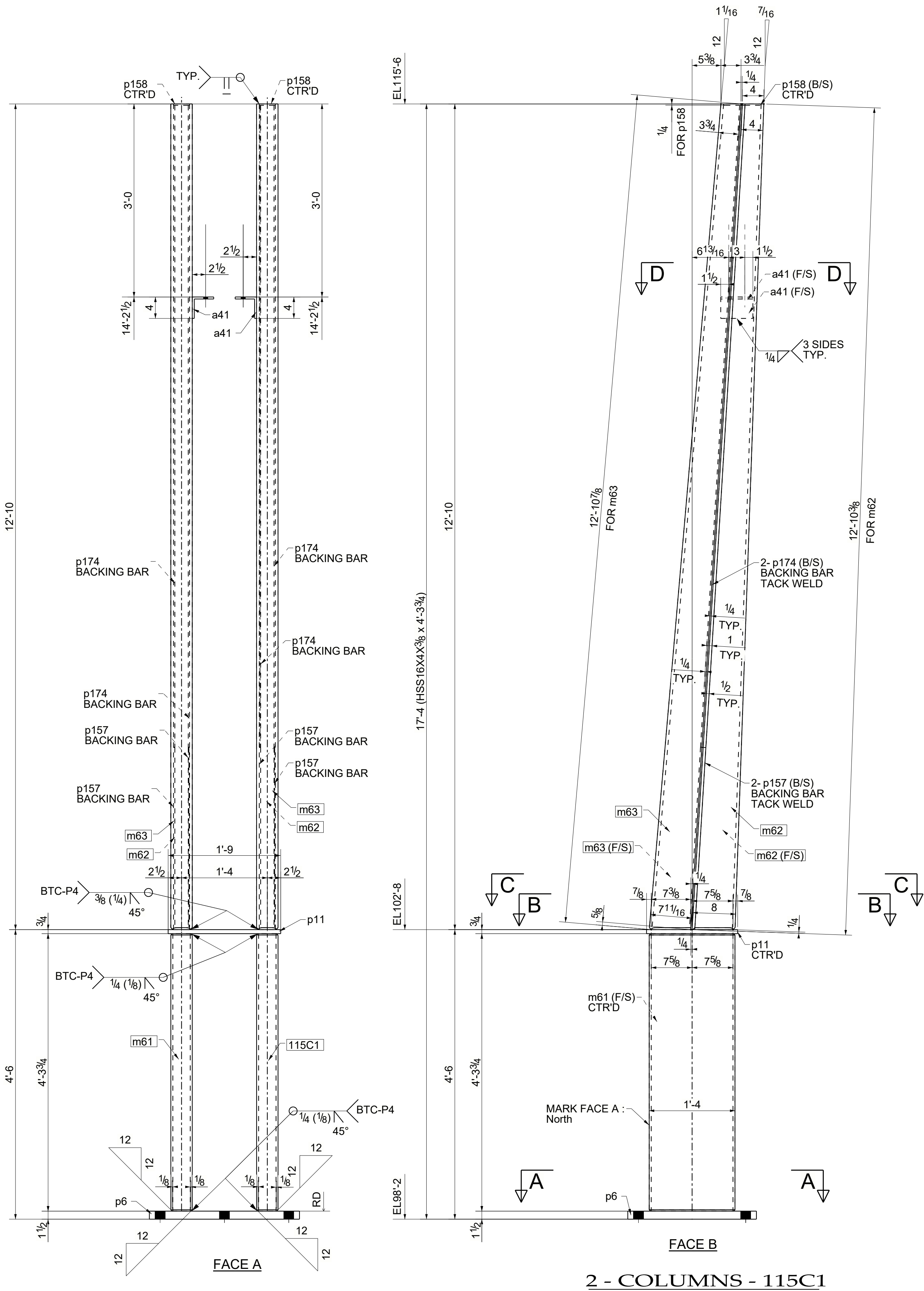
WORK THIS SHEET WITH 1C62-1



ONE - COLUMN - 1C62



ONE - COLUMN - 1C62

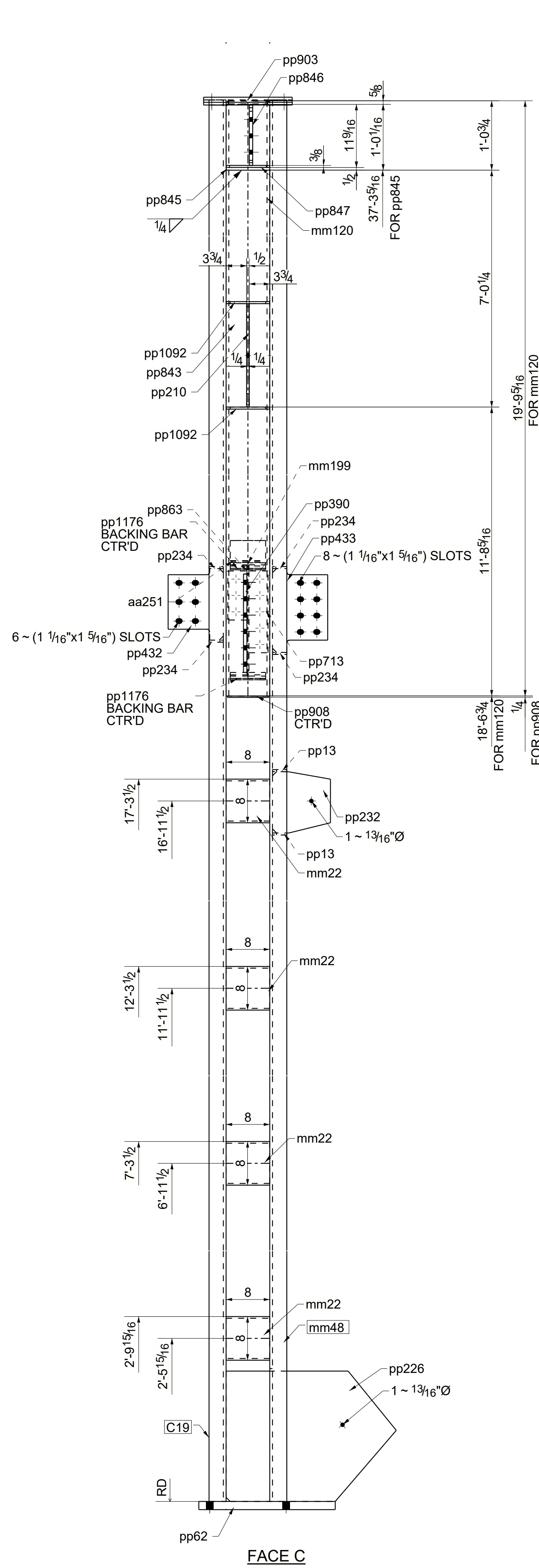
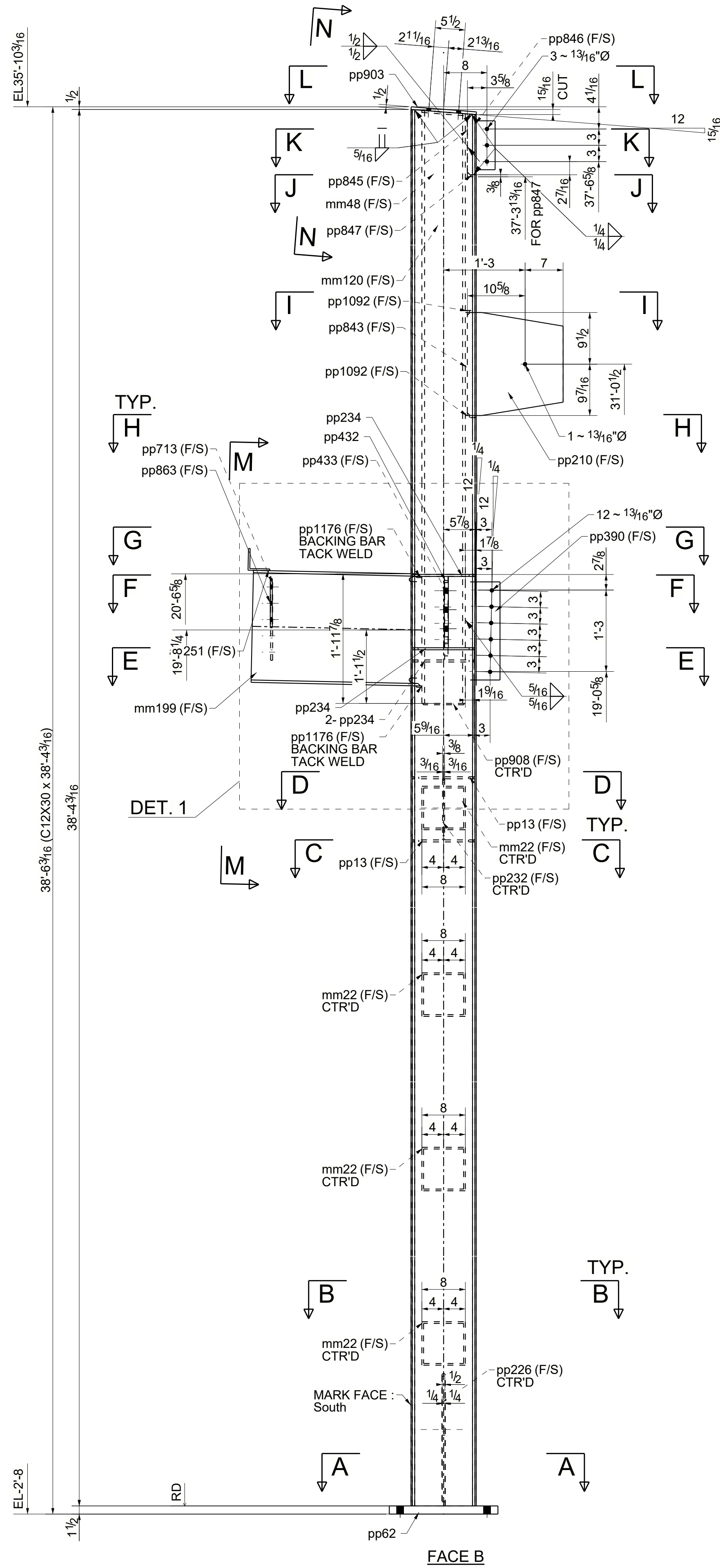
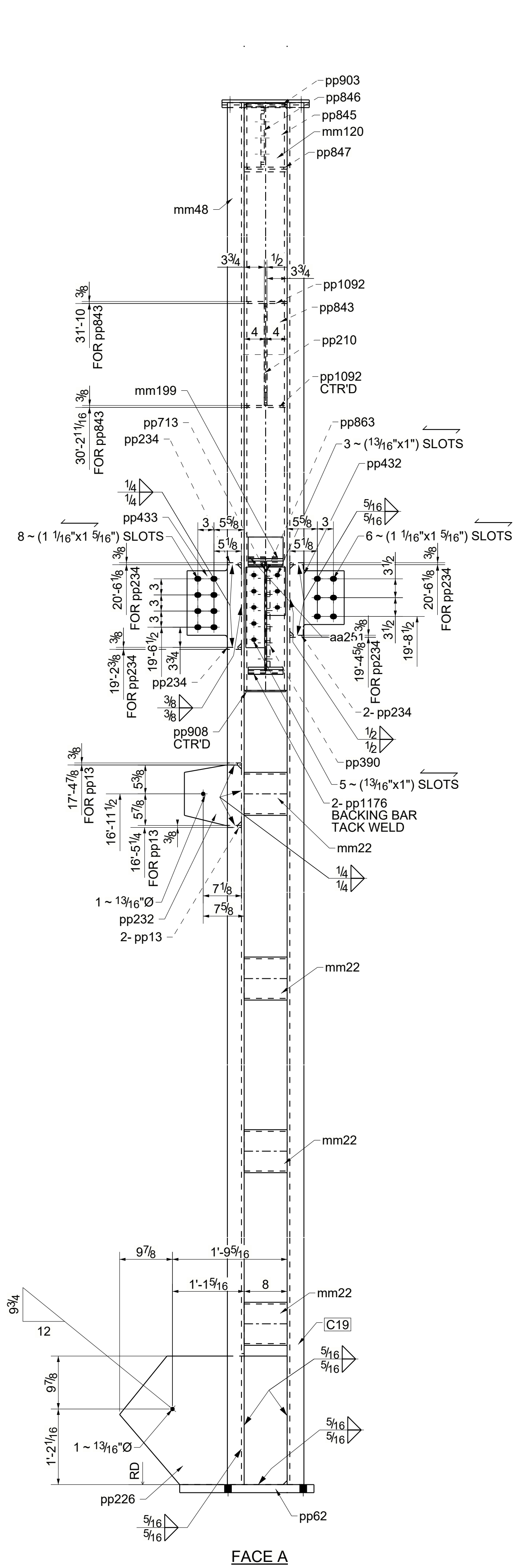


DETAIL OF 115C1

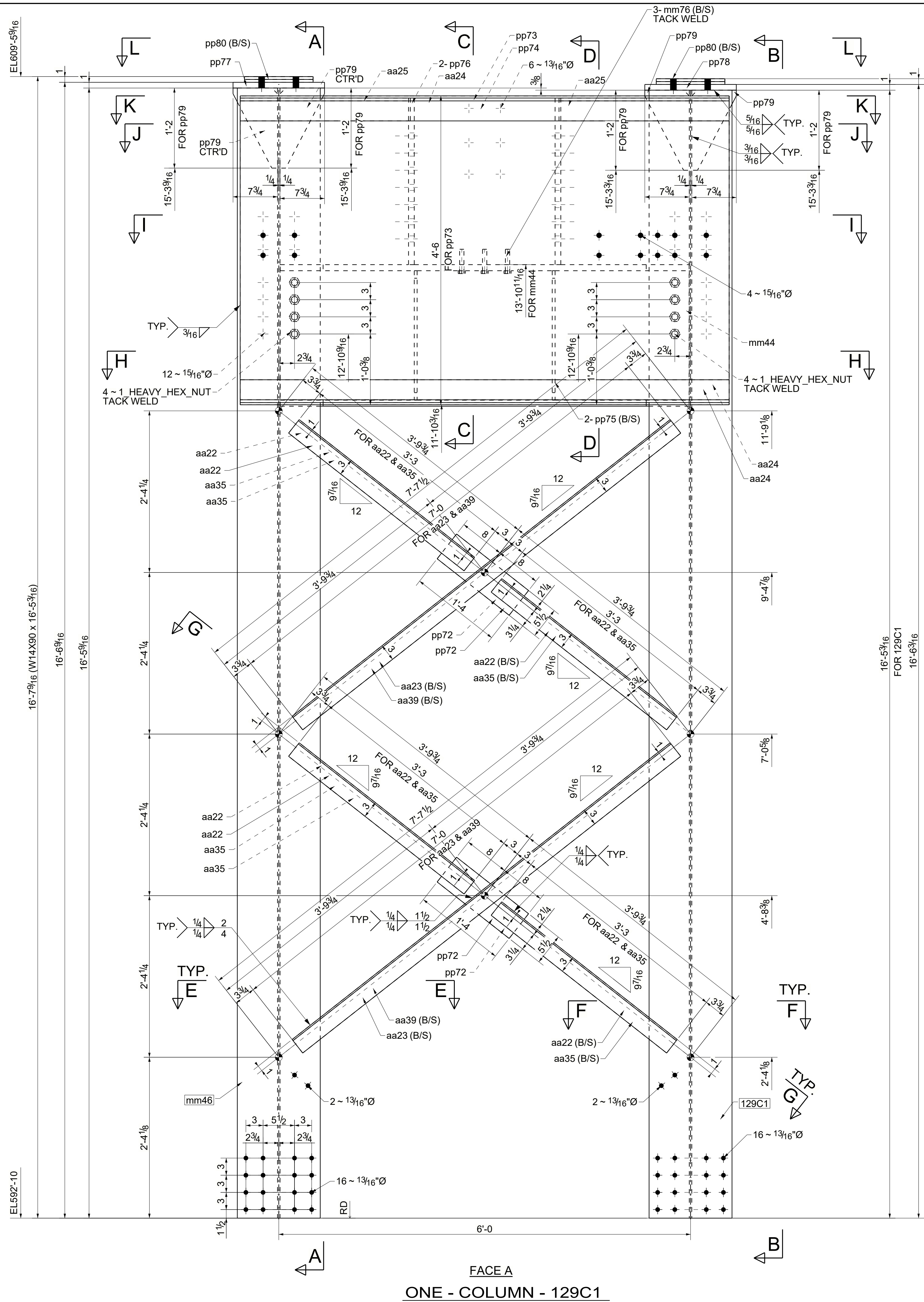
BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
115C1	2	COLUMNS			SQ-2
115C1	2	HSS16X4X3/8	4'-3 3/4"	387	A500-GR.B
a41	4	L4X4X3/8	0'-6"	19	A36
m61	2	HSS16X4X3/8	4'-3 3/4"	387	A500-GR.B
m62	4	HSS12X4X1/2	12'-10 3/8"	2364	A500-GR.B
m63	4	HSS12X4X1/2	12'-10 7/8"	2372	A500-GR.B
p6	2	PL1 1/2"X24"	2'-4"	572	A36
p11	2	PL3/4"X17"	1'-9"	152	A36
p157	8	PL1/4"X1"	2'-10 1/16"	19	A36
p158	4	PL1/4"X3"	0'-7"	6	A36
p174	8	PL1/4"X1"	10'-0"	68	A36
TOTAL WEIGHT THIS DRAWING				6347	

GRID LOCATION	BOTTOM ELEVATION
4/E	-2'-8"



ONE - COLUMN - C19



FACE A
ONE - COLUMN - 129C1

BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
129C1	1	COLUMN			SQ-2
129C1	1	W14X90	16'-5 3/16"	1482	A572-GR.50
aa22	8	L4X3X1/4	3'-3"	150	A36
aa23	4	L4X3X1/4	7'-0"	161	A36
aa24	3	L4X4X1/2	7'-1 1/2"	273	A36
aa25	2	L4X4X1/2	2'-5 3/4"	63	A36
aa35	8	L4X3X1/4	3'-3"	150	A36
aa39	4	L4X3X1/4	7'-0"	161	A36
mm44	1	W24X146	5'-11 1/2"	872	A992
mm46	1	W14X90	16'-5 9/16"	1485	A572-GR.50
pp72	4	FL5/8"X5 1/2"	1'-4"	62	A36
pp73	1	PL1"X54"	7'-1 1/2"	1309	A36
pp74	1	PL1"X54"	7'-1 1/2"	1309	A36
pp75	4	PL1/2"X6 1/8"	1'-10 5/8"	78	A36
pp76	2	PL1"X14"	2'-5 1/2"	234	A36
pp77	1	PL1"X16"	1'-4"	73	A36
pp78	1	PL1"X16"	1'-4"	73	A36
pp79	4	PL1/2"X7 13/16"	1'-2"	37	A36
pp80	4	FL1"X3 1/2"	1'-0"	44	A36
	6	3/4_HEAVY_HEX_NUT		4	
	8	1_HEAVY_HEX_NUT		3	
TOTAL WEIGHT THIS DRAWING				8015	

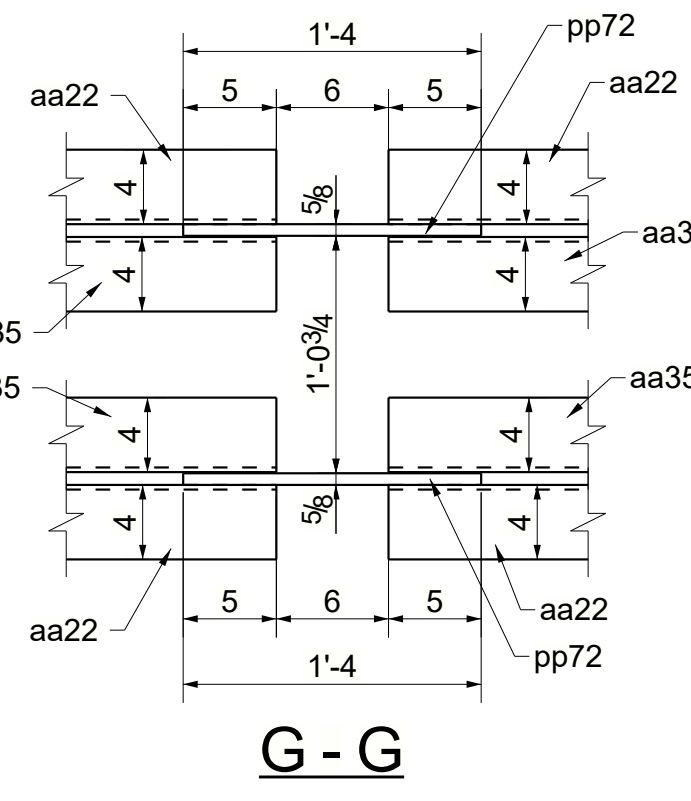
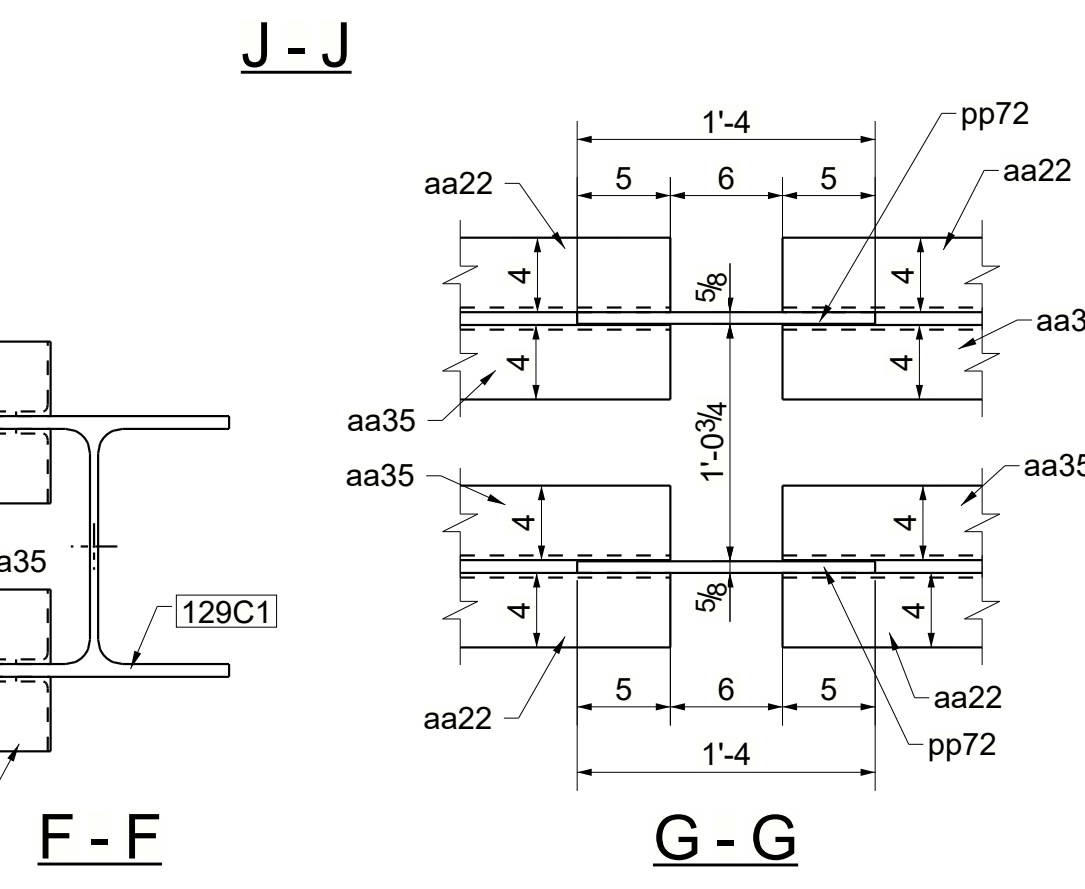
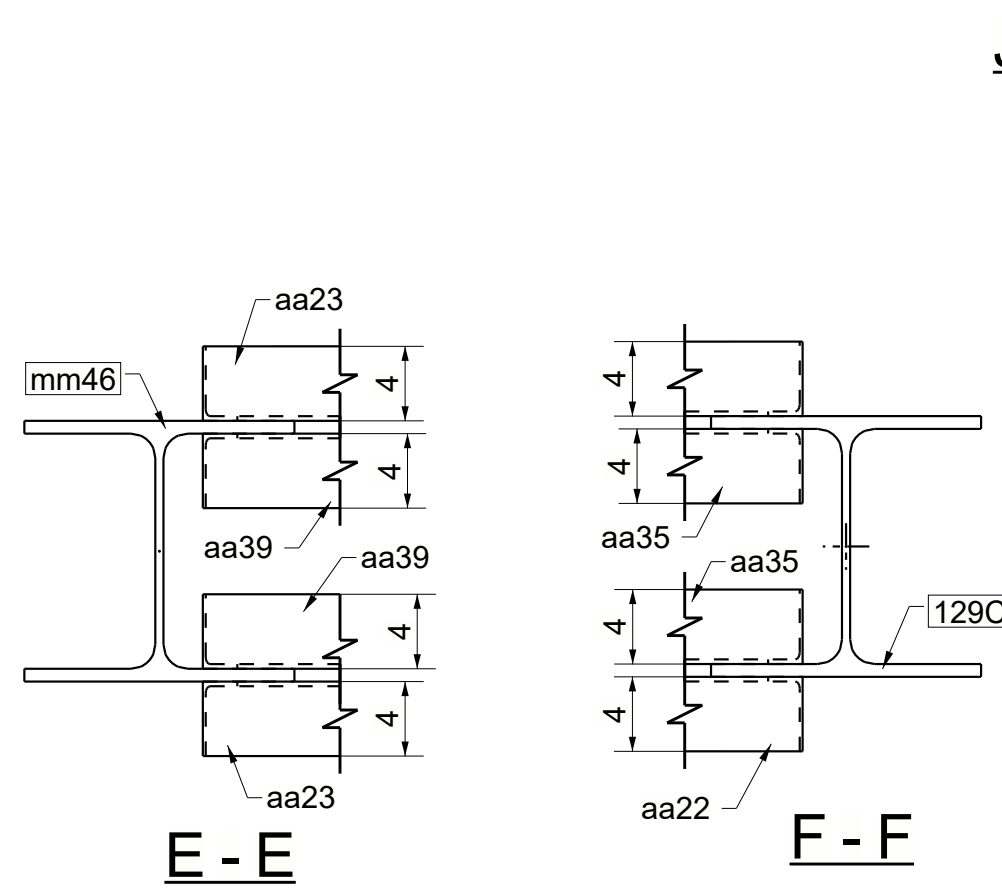
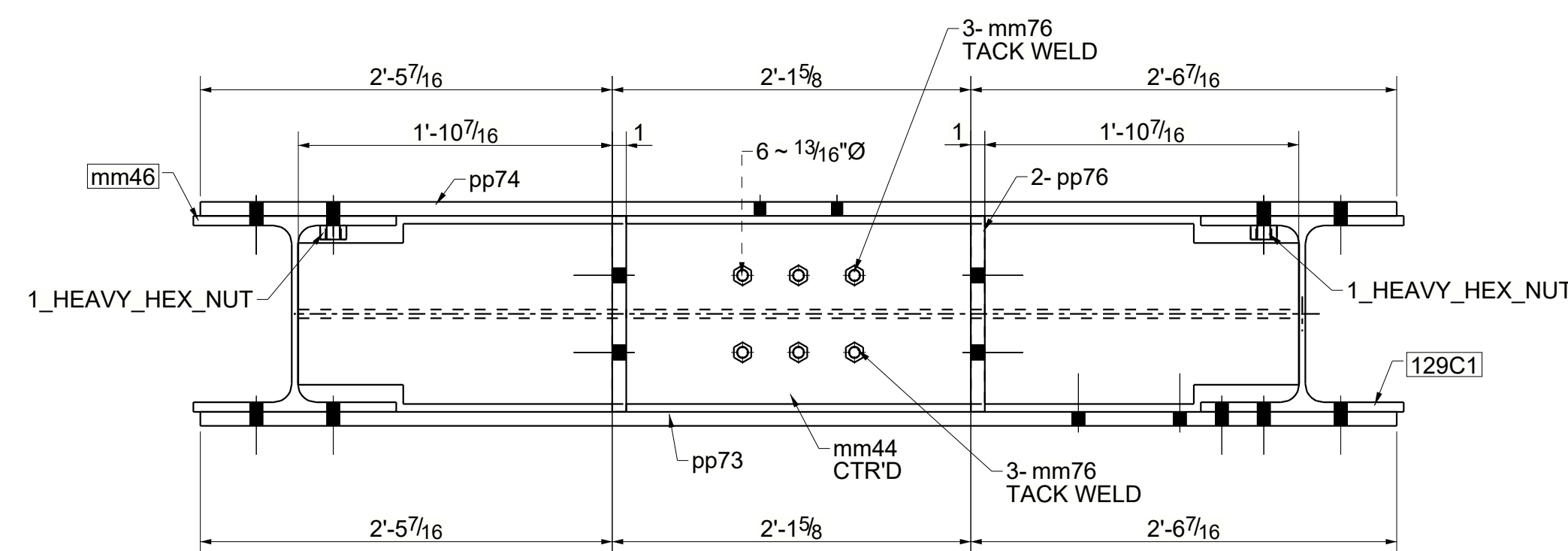
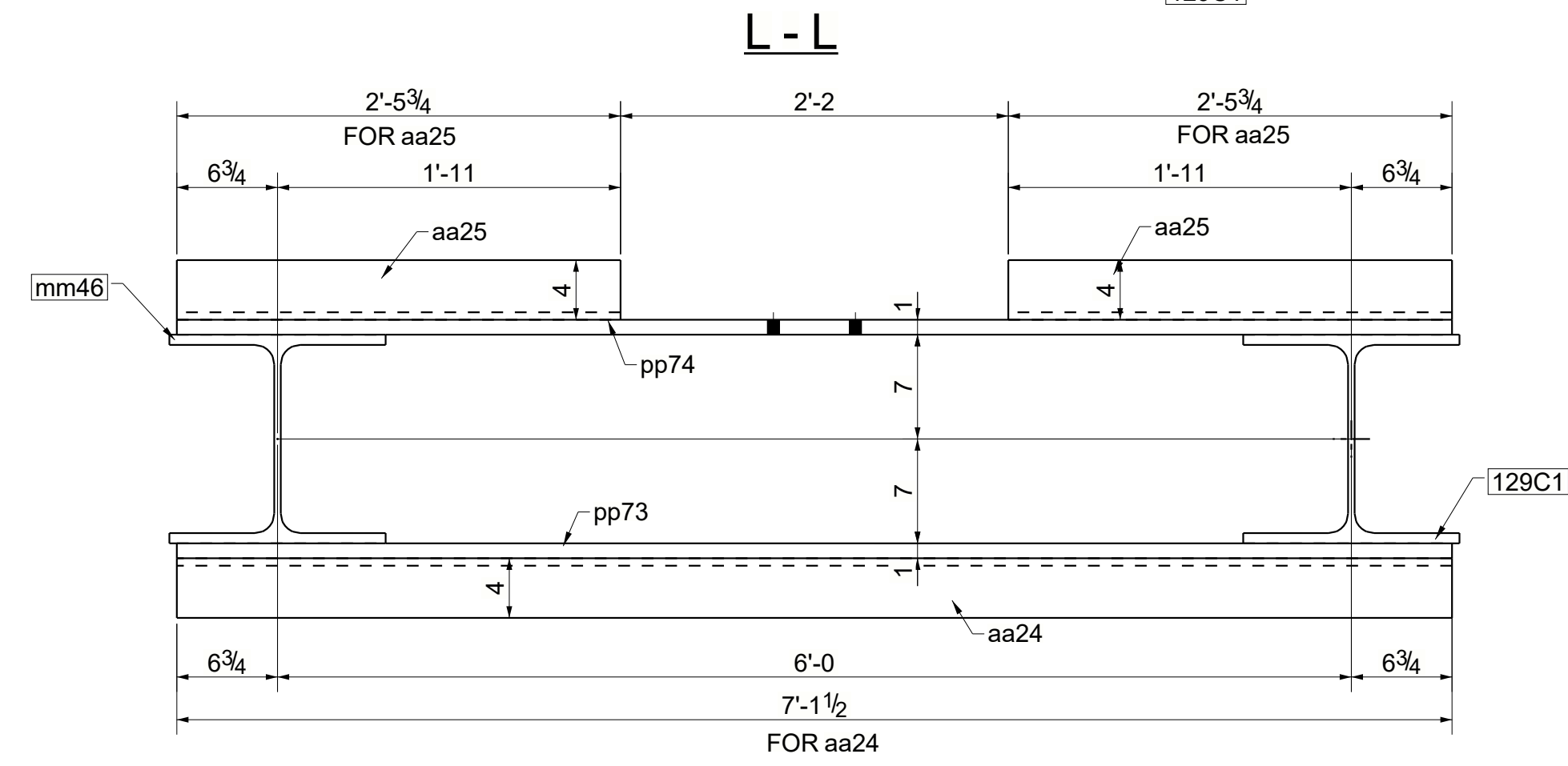
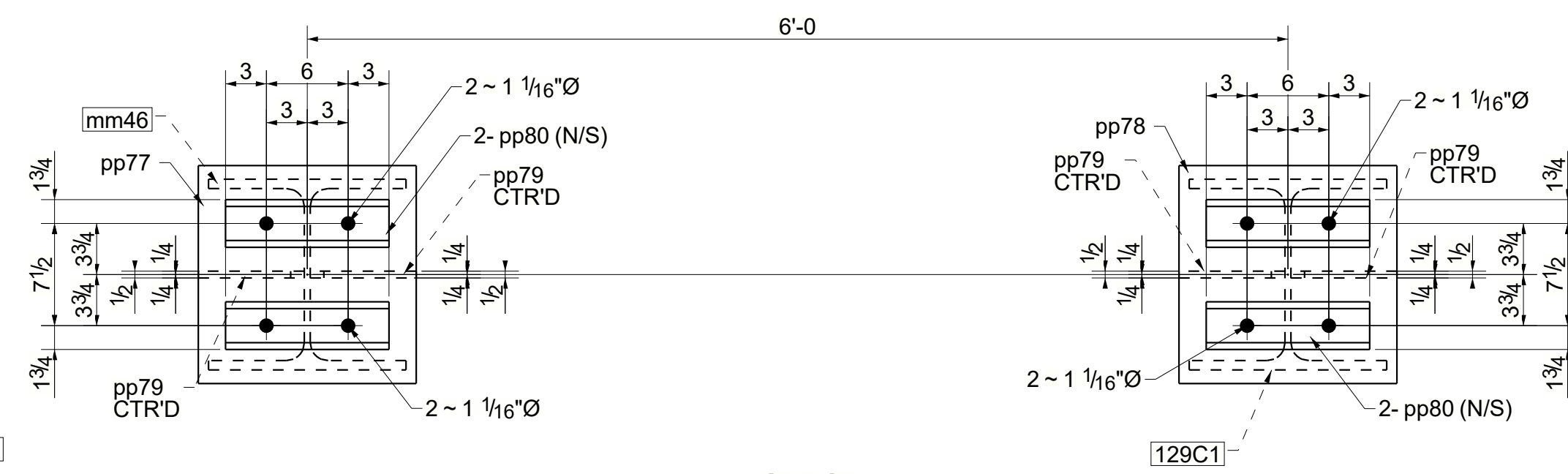
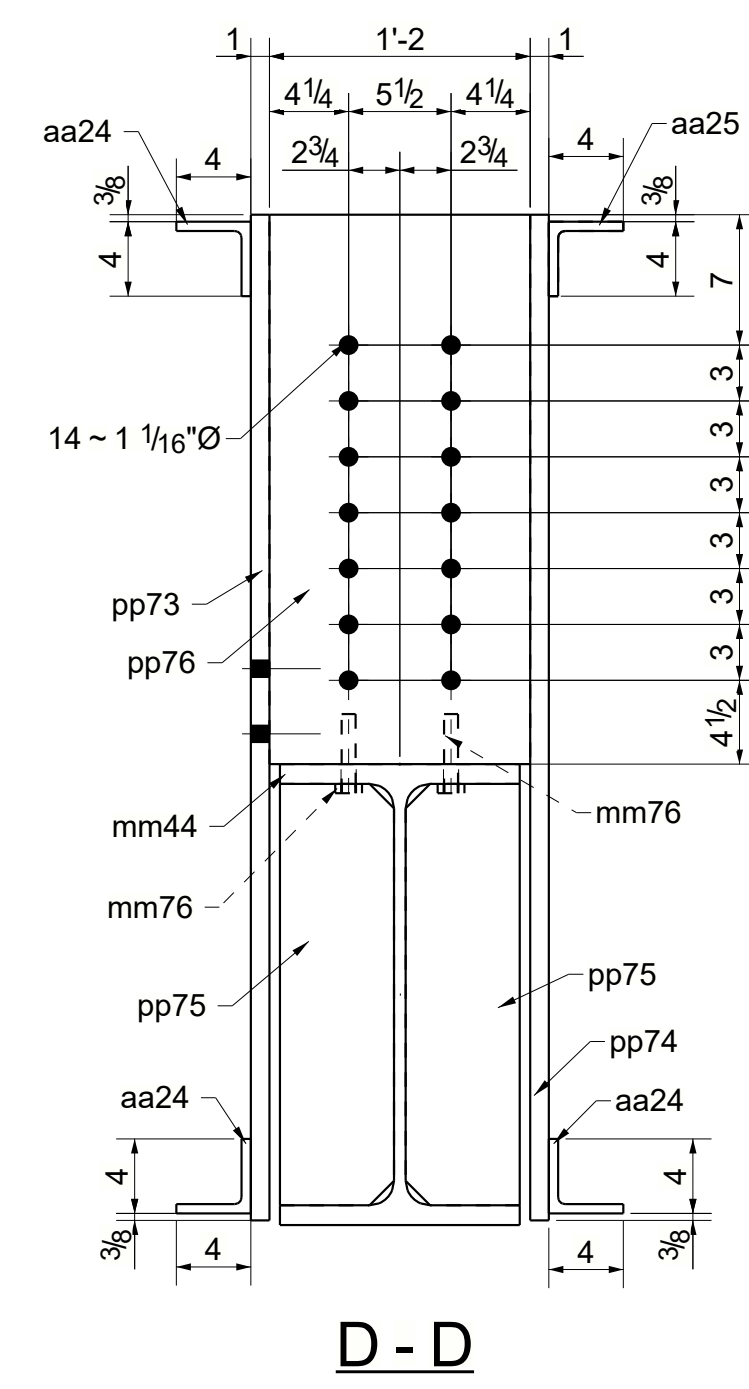
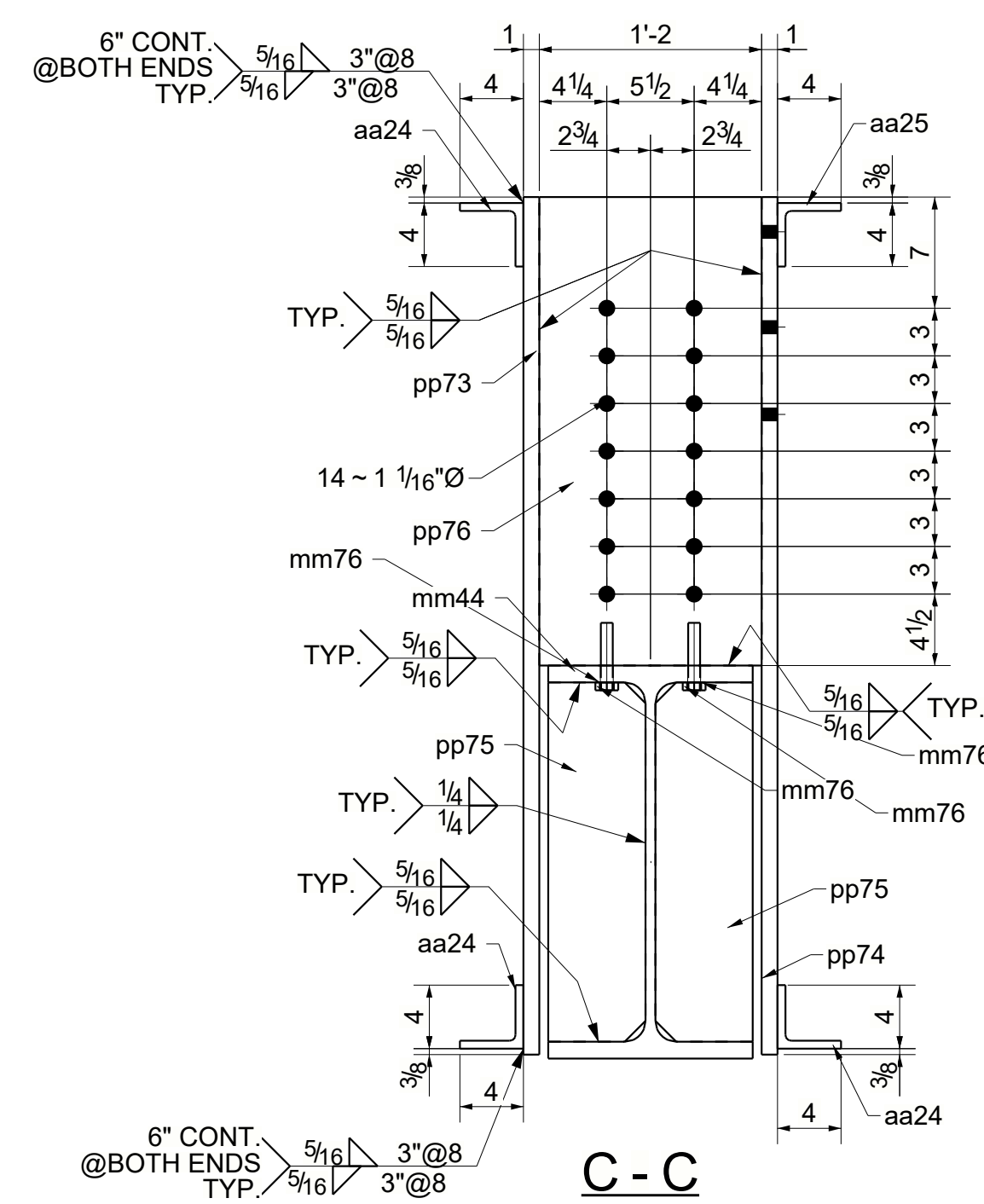
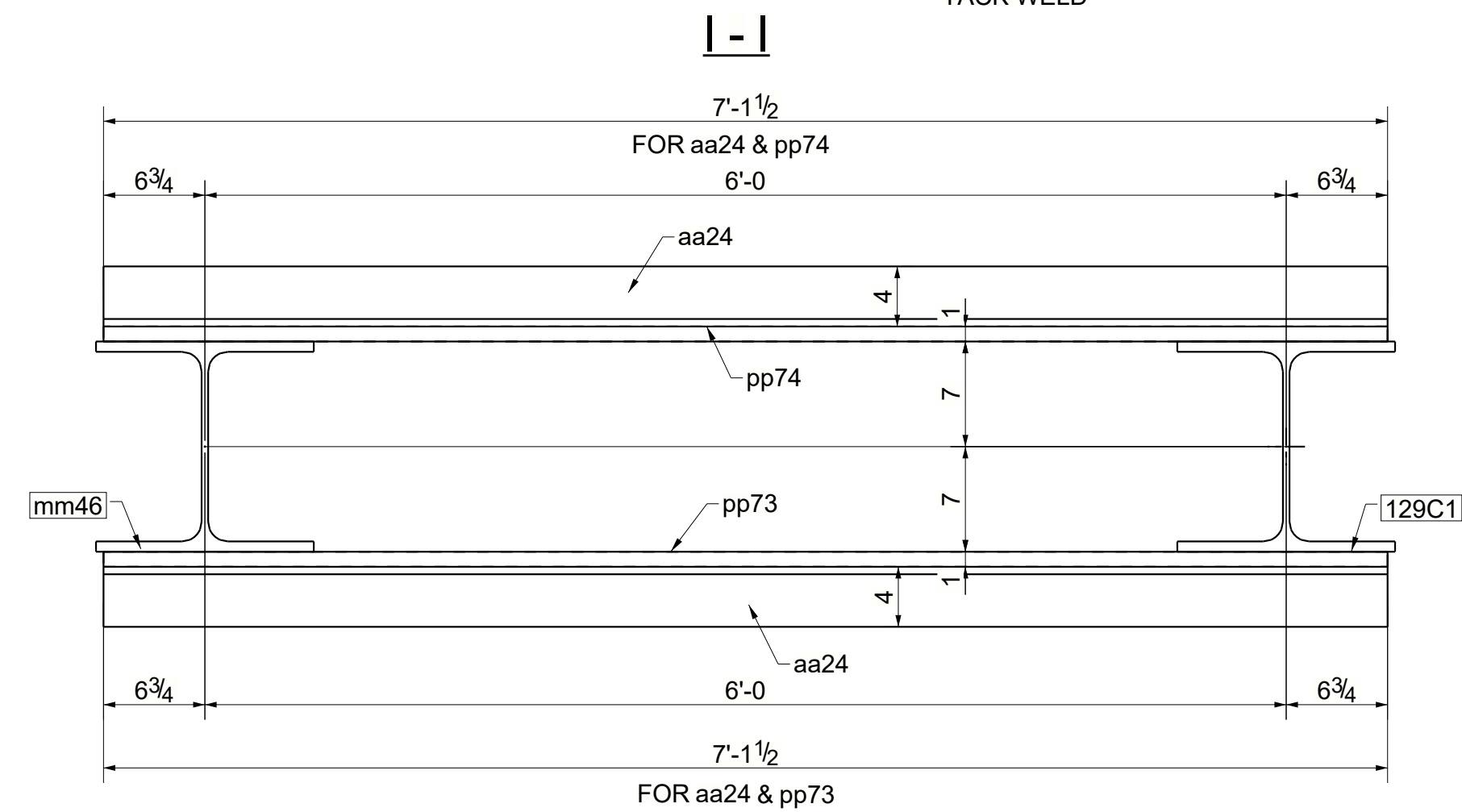
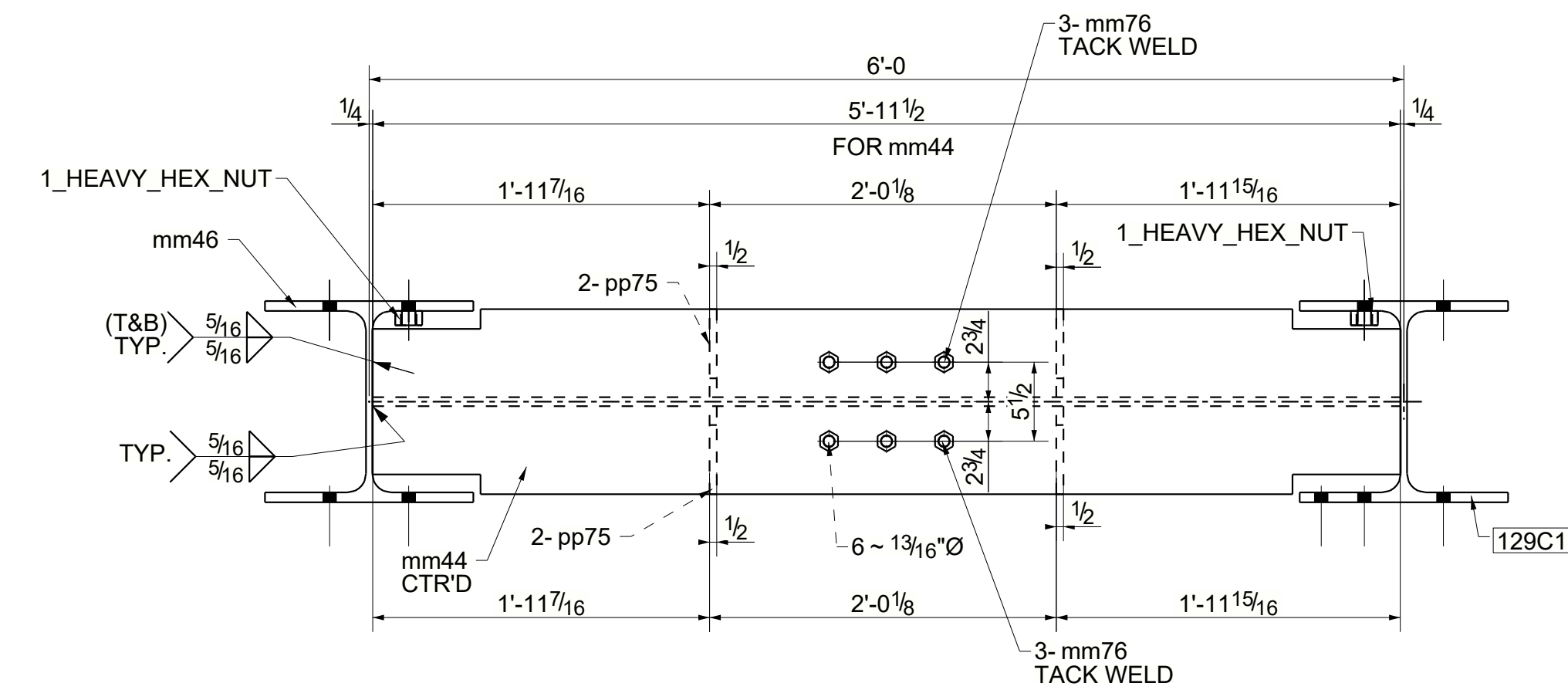
SHOP NOTES :

1. ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
2. TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
3. ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
4. ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,...ETC.
5. ALL COPES SHALL BE MEASURED FROM END OF BEAM.
6. ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
7. FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PART SHEET DETAILS.
8. ALL COPE CORNERS ARE 1/2" RADIUS U.N.O

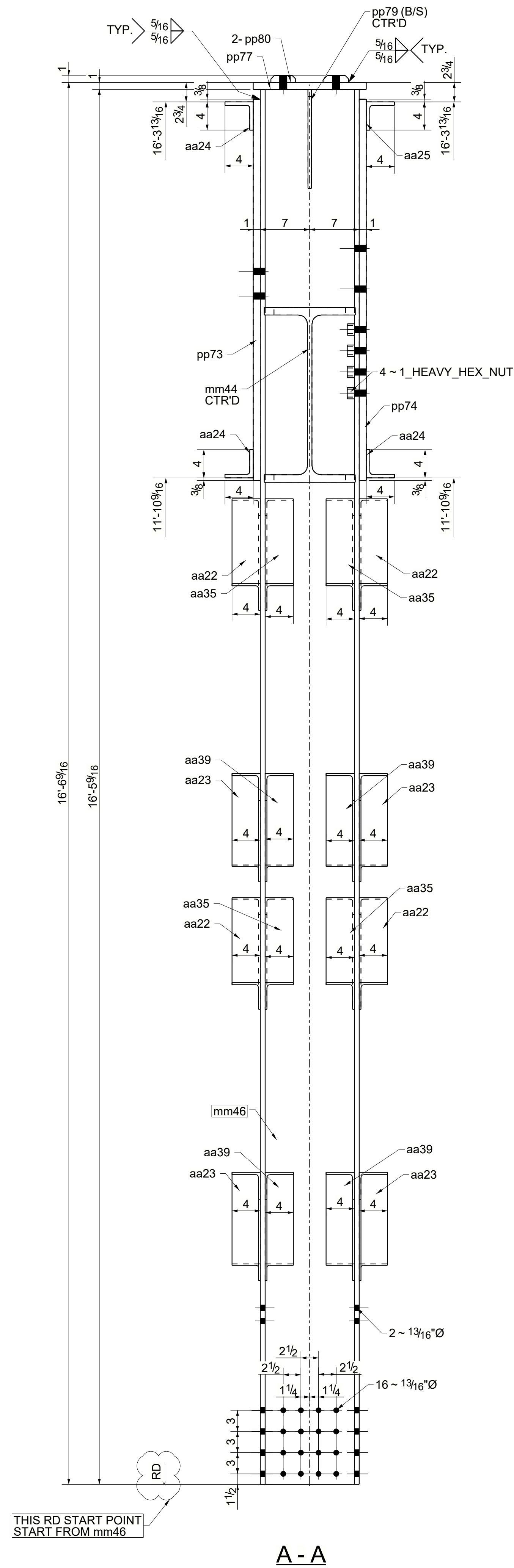
GENERAL SHOP NOTES:

MATERIAL: WF & WT's: A992 (U.N.)
PIPES : A53-GR.B
ALL OTHER MATERIAL: A36
WELD: E70XX
SURFACE PREP: SSPC-SP3
PAINT: ONE OR MORE COATS OF SHERWIN-WILLIAMS KEM BOND HS UNIVERSAL METAL PRIMER #B50WZ4(OFF WHITE) 3.0 D.F.T MIL

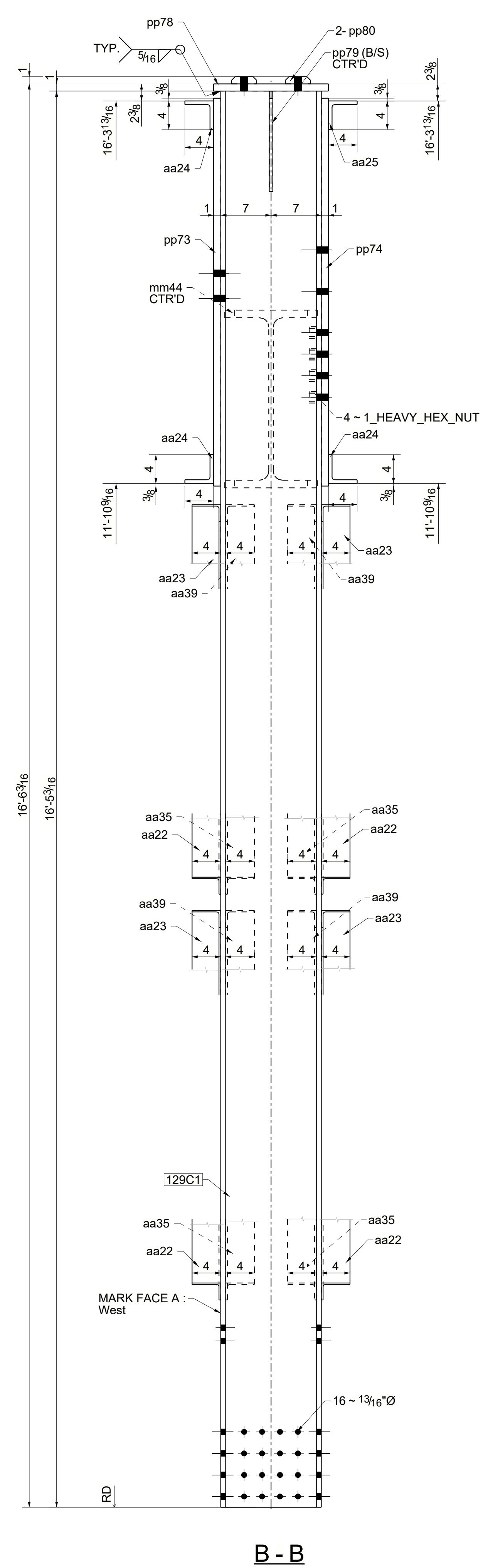
WORK THIS SHEET WITH 129-1, 129-2



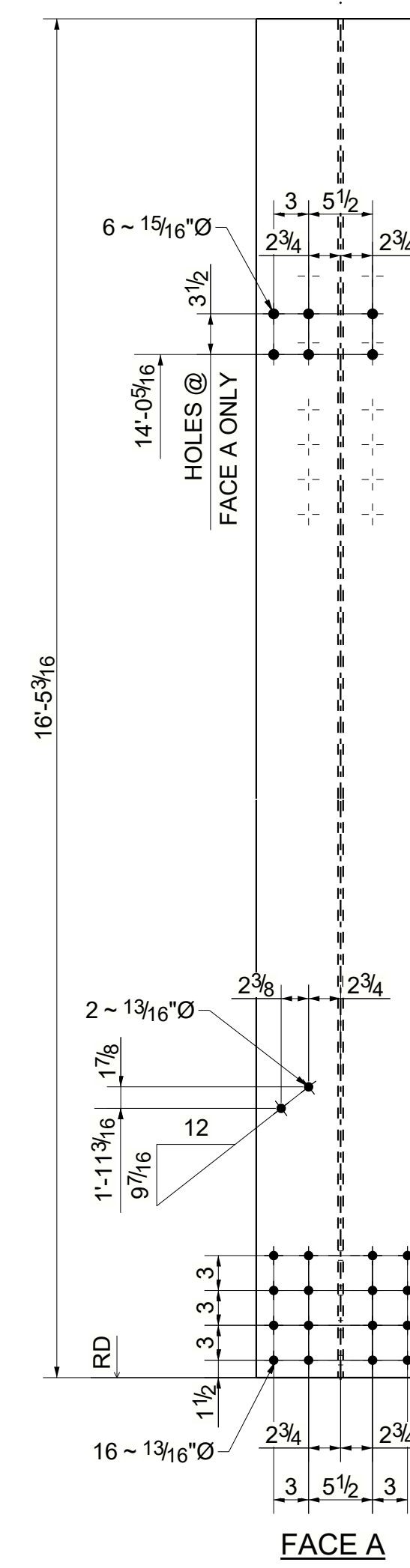
WORK THIS SHEET WITH 129, 129-2



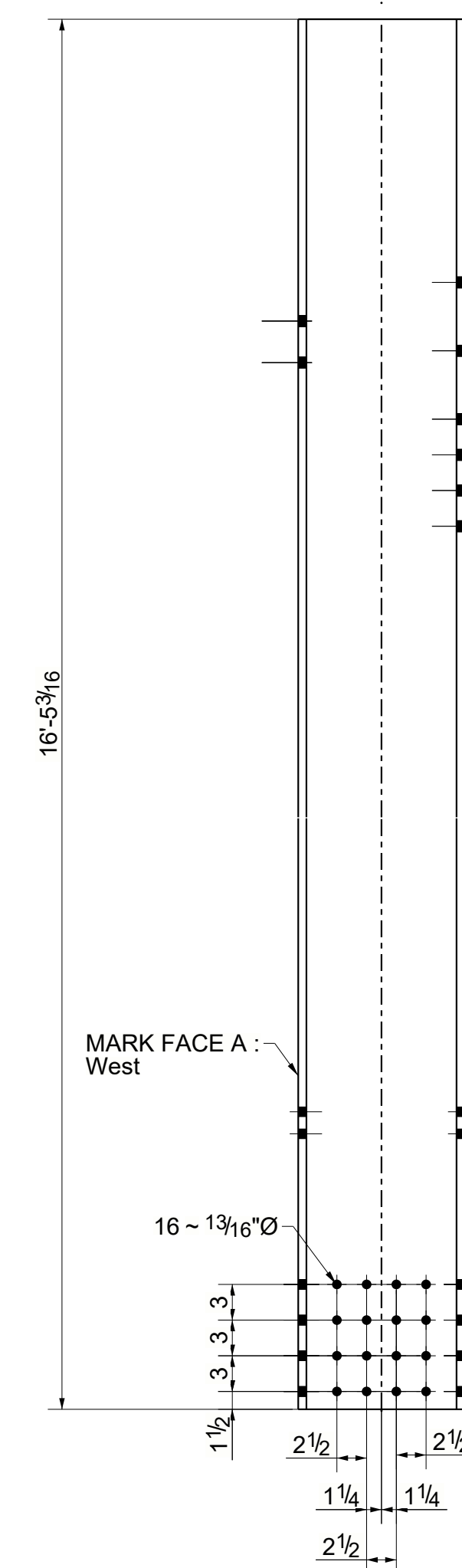
A - A



B - B

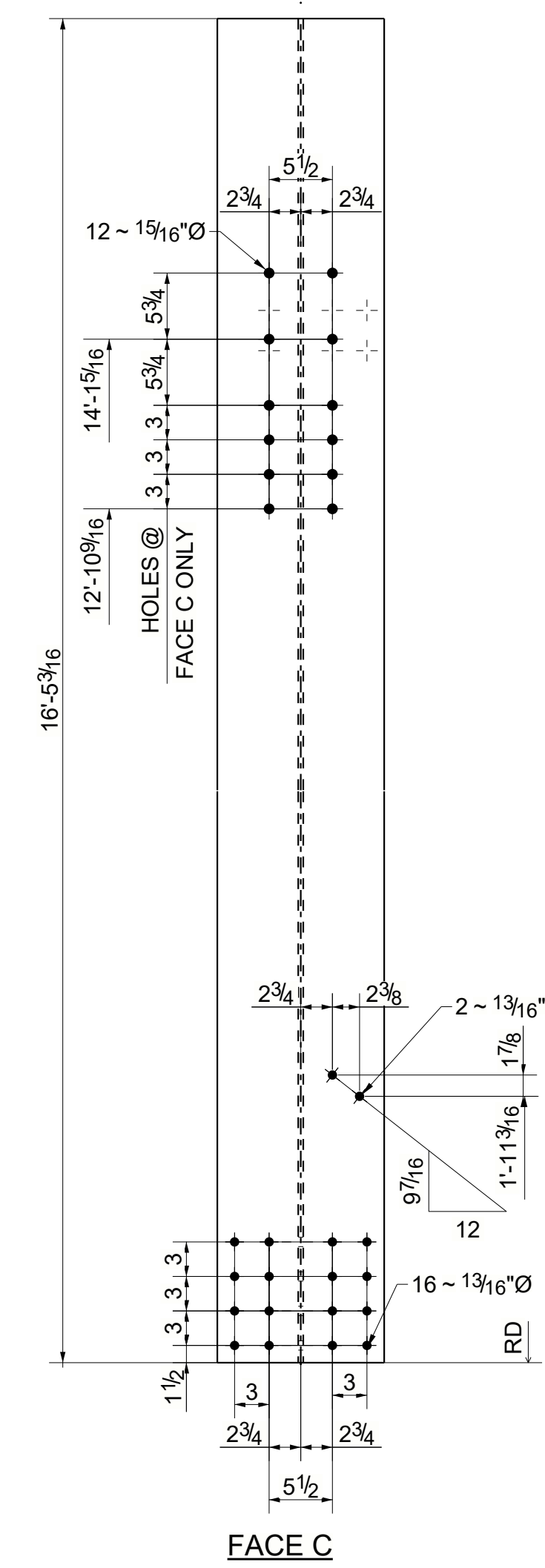


FACE A



FACE B

DETAIL OF 129C1

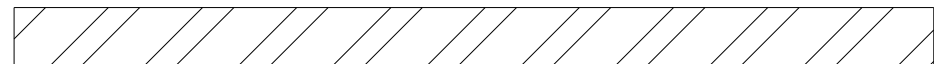


FACE C

WORK THIS SHEET WITH 129, 129-1

BILL OF MATERIAL					
MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
1D77	1	VERTICAL BRACE			SQ-1/BEV-1
1D77	1	W10X60	15'-1 3/4"	912	[SFRS] A992
1pp883	1	PL3/8"X6 3/4"	0'-9"	6	A36
1pp884	1	PL3/8"X4"	0'-9 9/16"	3	A36
TOTAL WEIGHT THIS DRAWING				922	

- SHOP NOTES:
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,...ETC.
 - "RD" DENOTES RUNNING DIMENSION
 - ALL COPEs SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPEs TO BE CUT WITH 1/2" RADIUS U.N.O.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC..., SEE SINGLE PARTS DETAIL SHEETS.
 - B.T.S.DENOTES BOLT TO SHIP"SNUG TIGHT" WITH NOTED BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - HOLE PATTERNS NOTED "A325SC-TC" OR "A490SC-TC" ARE FOR SLIP CRITICAL BOLTS. PROVIDE SHERWIN WILLIAMS ZINC CLAD 4100 ORGANIC ZINC-RICH EPOXY PRIMER AT CONTACT SURFACES. SURFACE PREP: SSPC-SP6 THE MINIMUM AREA OF THIS PAINT IS THE HOLE DIAMETER +2" AT HATCH AREAS SHOWN THUS:



SEISMIC SHOP NOTE :

SUMMARY of SFRS Requirements

(REF: S0.1 Section G)

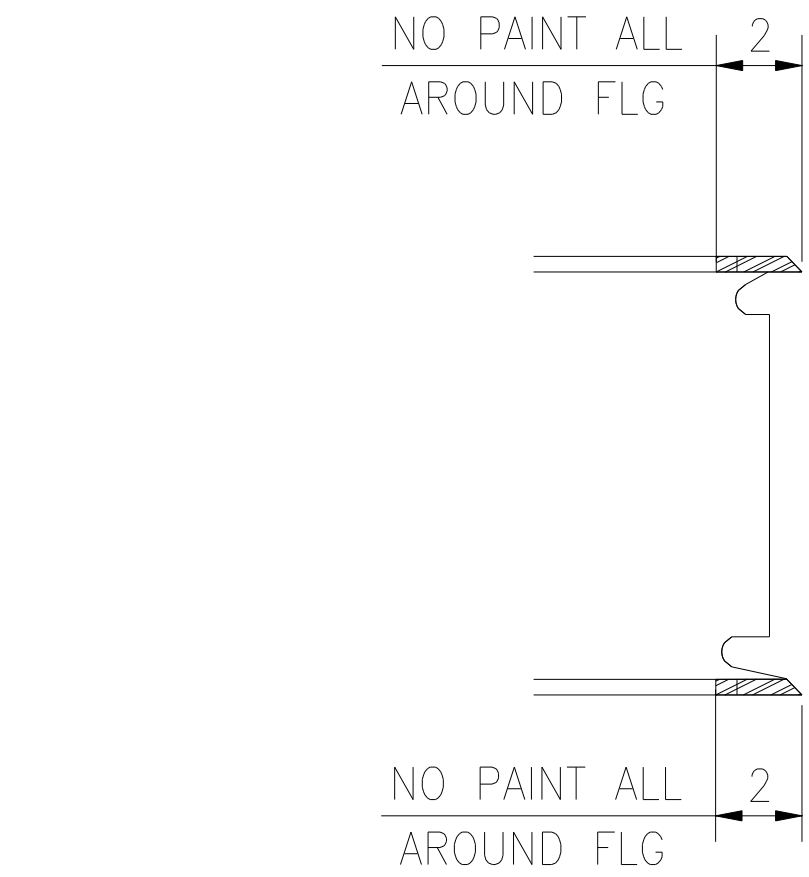
PROTECTED ZONE NOTE: (from Section F4.5C AISC 341)

HATCHED AREAS SHOWN INDICATE SFRS PROTECTED ZONES THAT INCLUDE THE STEEL CORE OF BRACES AND ELEMENTS THAT CONNECT STEEL CORE TO BEAMS AND COLUMNS, AND SHALL SATISFY THE REQUIREMENTS OF SECTION D1.3 (AISC 341)

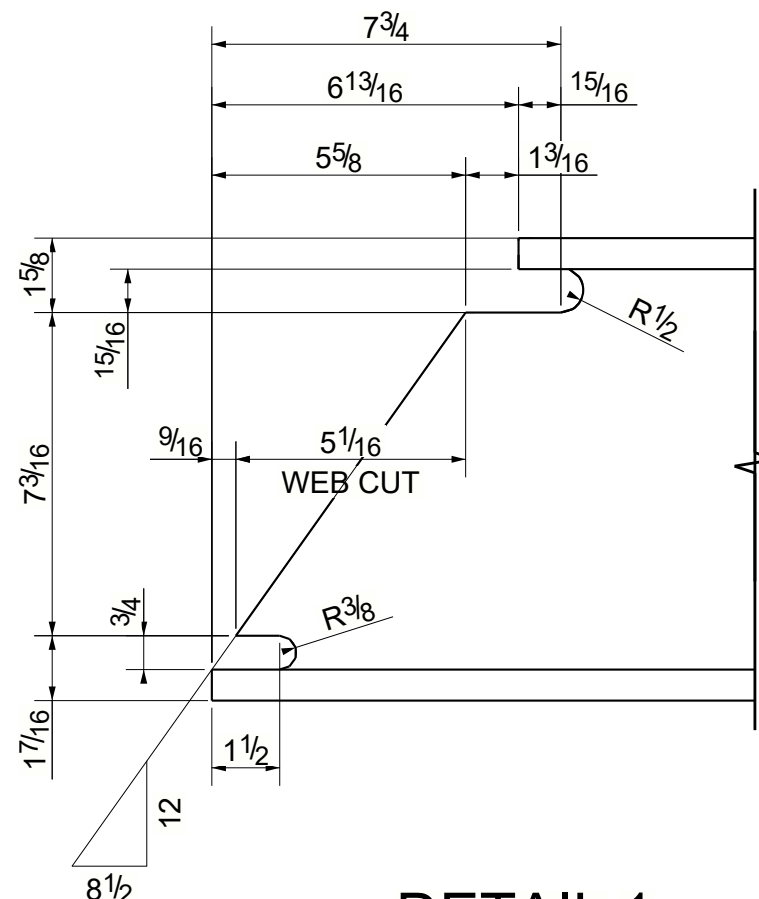
THE PROTECTED ZONE OF BRACING MEMBERS INCLUDES THE CENTER ONE-QUARTER OF THE BRACE LENGTH, AND A ZONE ADJACENT TO EACH CONNECTION EQUAL TO THE LARGEST DIMENSION OF THE BRACE CROSS SECTION MINIMUM. THE PROTECTED ZONE ALSO INCLUDES GUSSET PLATES AND AREAS WHERE BOLTS, WELDS, ANGLES, ETC. CONNECT THE BRACE TO THE GUSSET PLATE.

WITHIN THE PROTECTED ZONE, DISCONTINUITIES CREATED BY FABRICATION OR ERECTION OPERATIONS, SUCH AS TACK WELDS, ERECTION AIDS, AIR-ARC GOUGING AND THERMAL CUTTING ARE PROHIBITED.

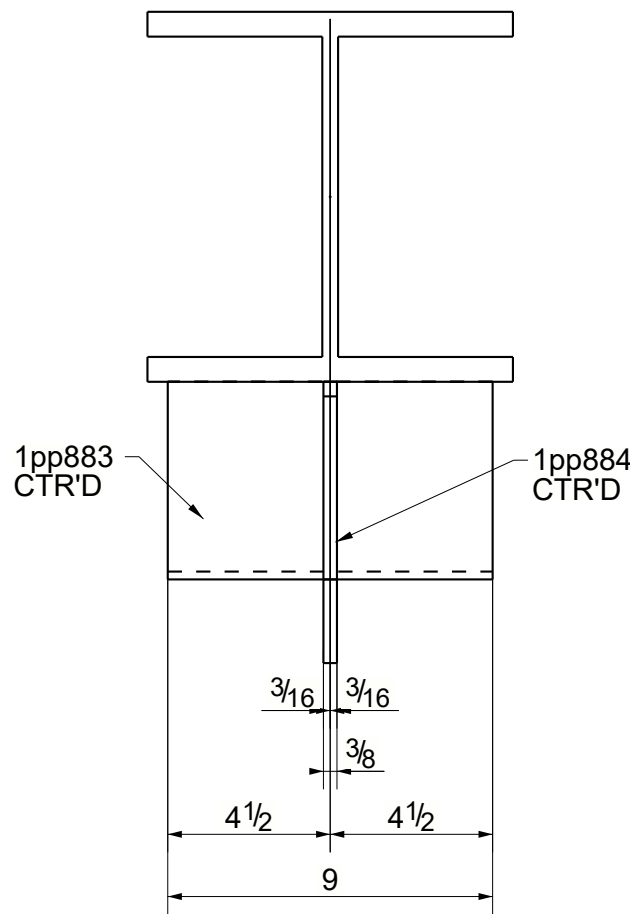
WELDED, BOLTED, SCREWED OR SHOT-IN ATTACHMENTS OR OTHER ATTACHMENTS ARE PROHIBITED WITHIN THE PROTECTED ZONE.



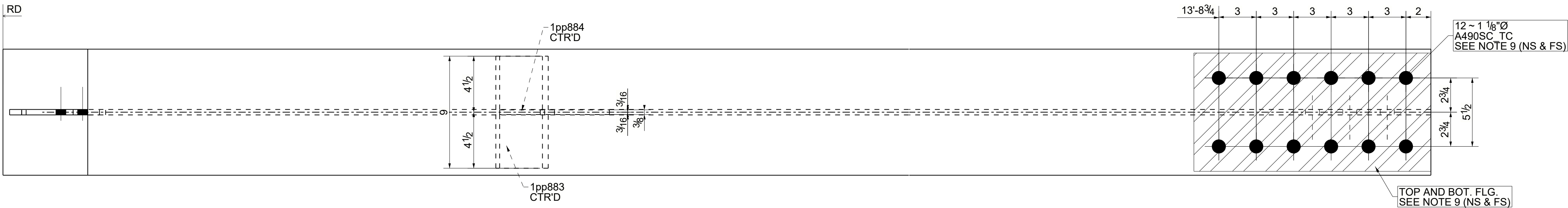
TYP PAINT BLOCK DETAIL
@ MOMENT PREP BRACE END



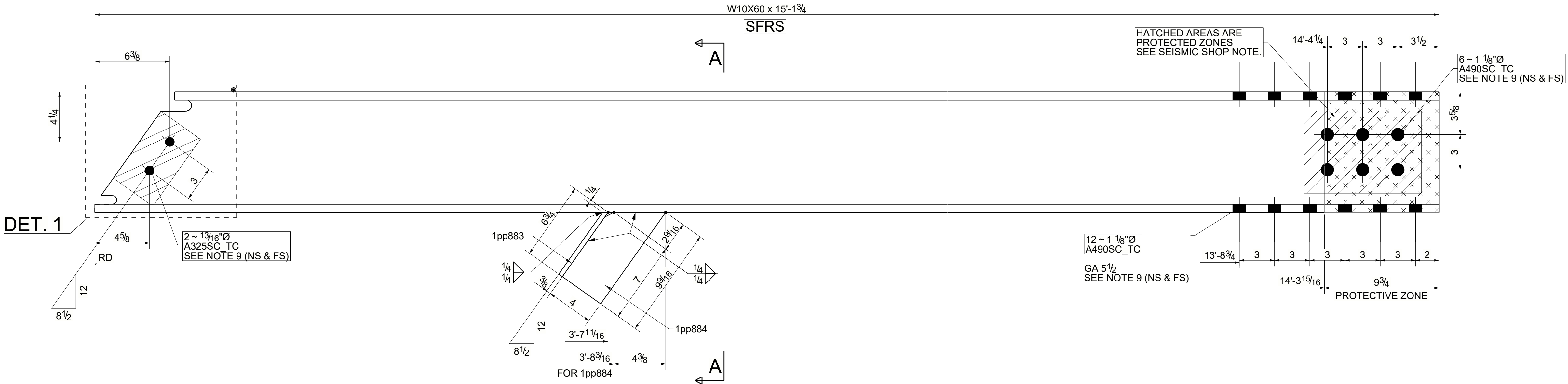
DETAIL 1



A - A

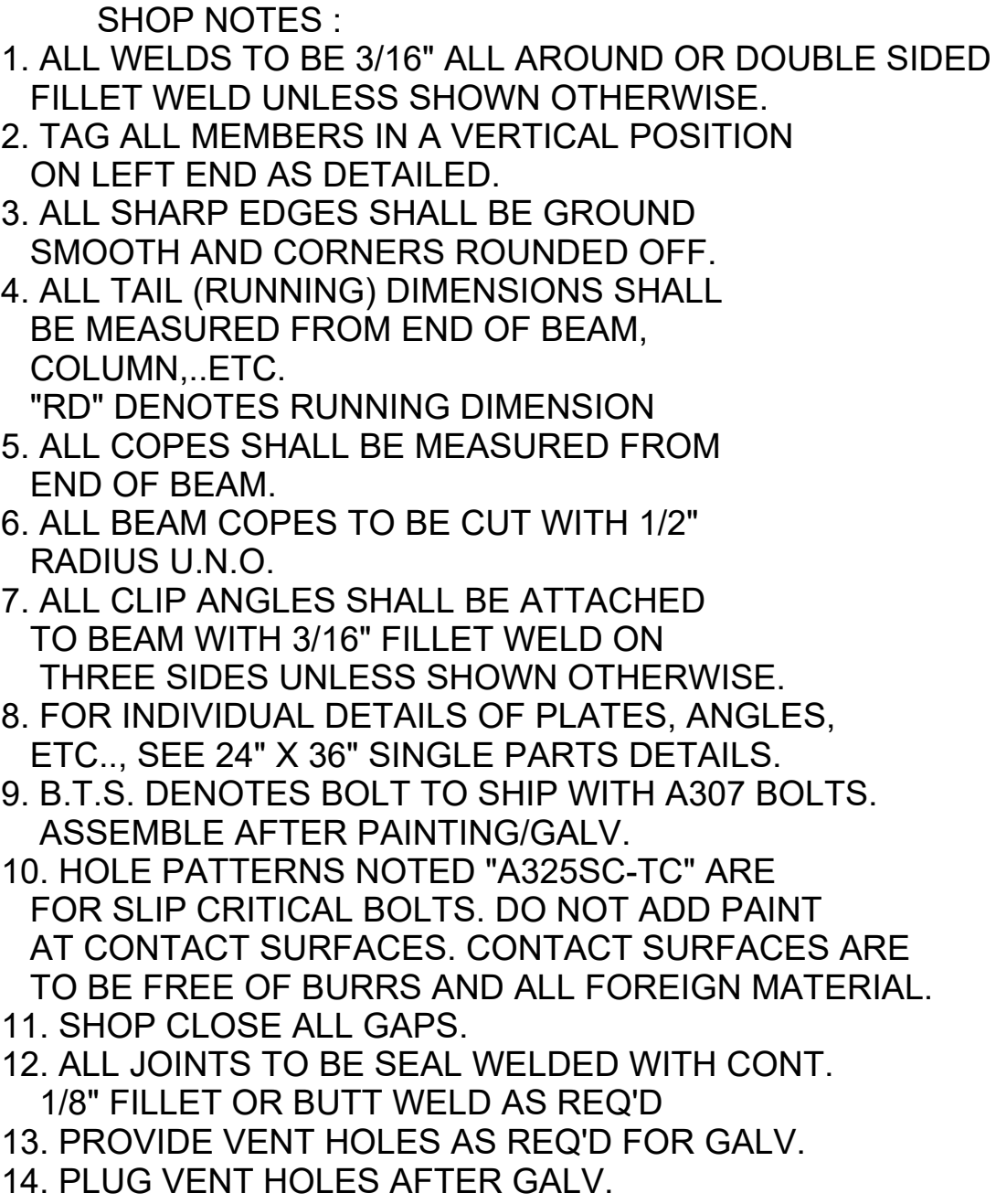


TOP VIEW



ONE - VERTICAL BRACE - 1D77

BILL OF MATERIAL								
NO PCS	DESCRIPTION	MARK	LENGTH	TOTAL WEIGHT	UNIT WEIGHT	WEIGHT	PRELIM MARK	REMARKS
1	W16X36	4A-B20	19'-3 3/4"			697		GALV. A992
1	HSS4X4X1/4	4A-mm1	2'-5 11/16"			28		A500-GR.C
1	HSS4X4X1/4	4A-mm2	2'-5 11/16"			28		A500-GR.C
1	PL3/8""X4 1/2"	4A-pp14	1'-0"			6		A36
4	PL3/8""X3 5/16"	4A-pp20	1'-3"			21		A36
2	PL3/4""X12"	4A-pp33	1'-1 1/4"			59		A36
2	PL1/4""X1 13/16"	4A-pp41	0'-3 5/8"			1		A36
2	PL1/4""X1 1/16"	4A-pp42	0'-3 5/8"			1		A36

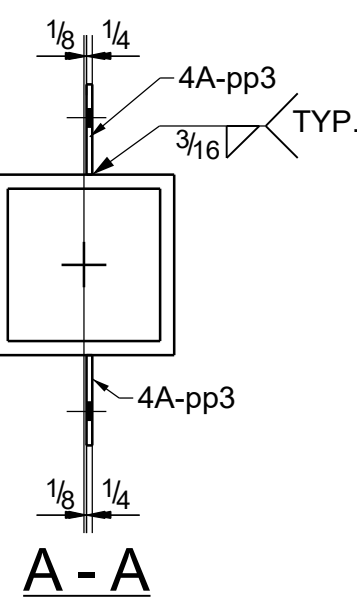
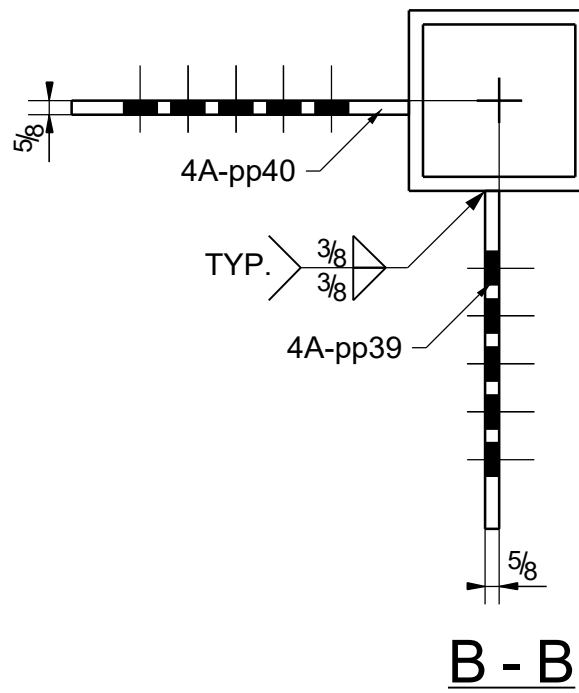
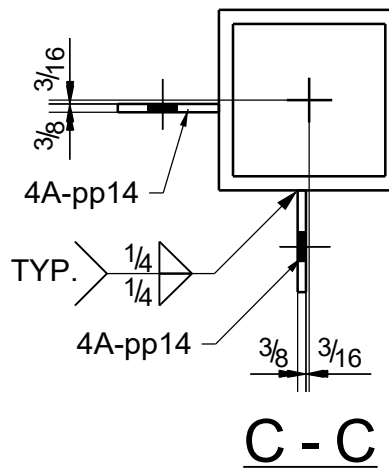
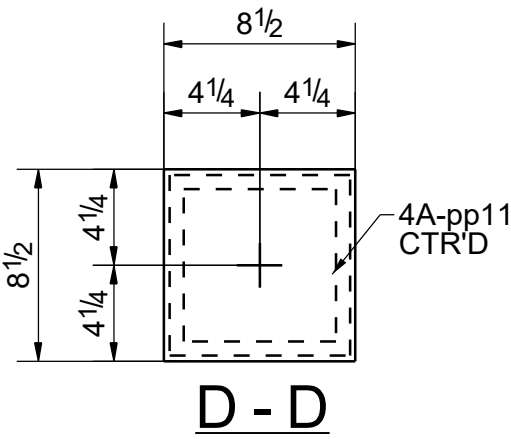
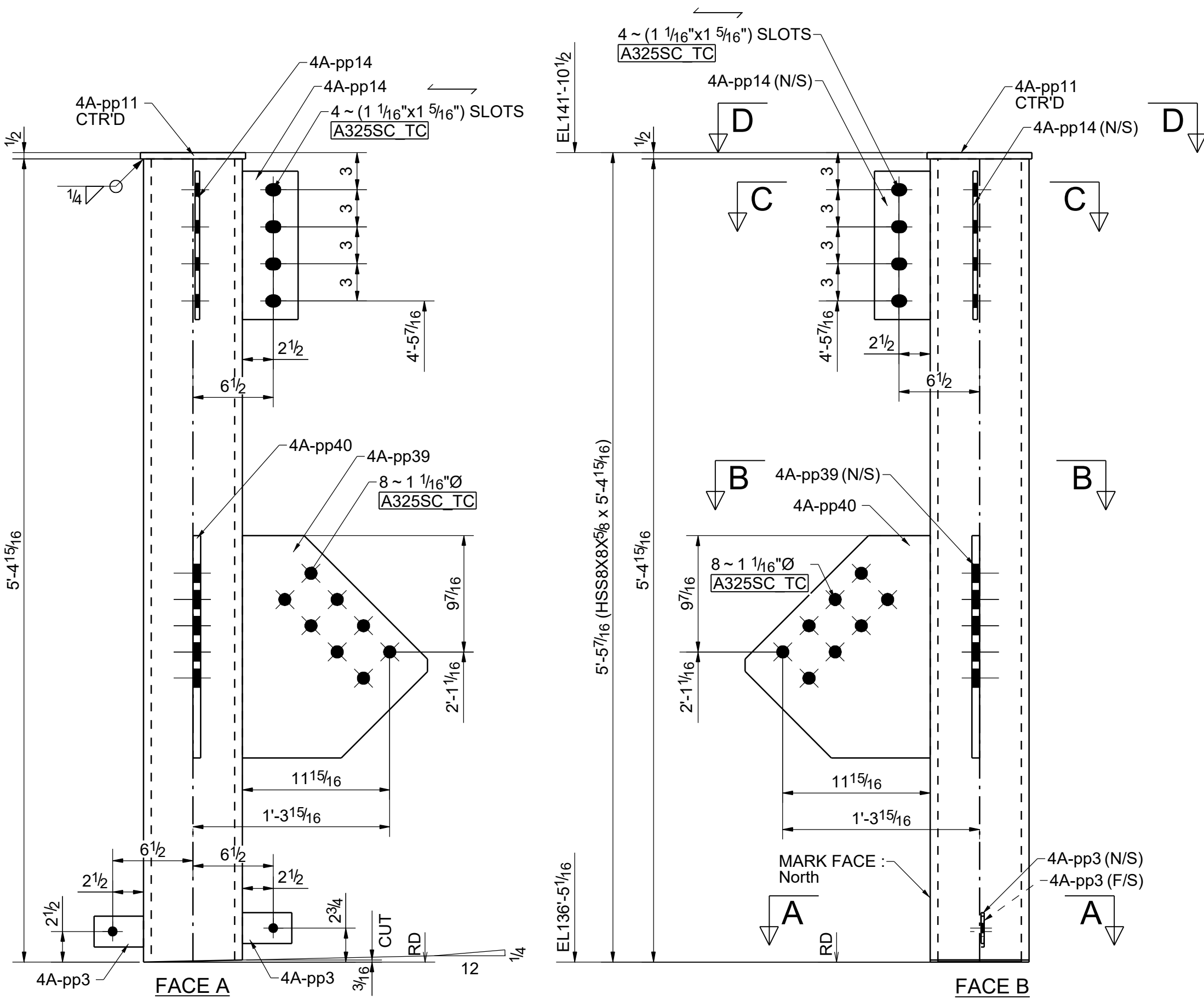


ONE - BEAM - 4A-B20

BILL OF MATERIAL

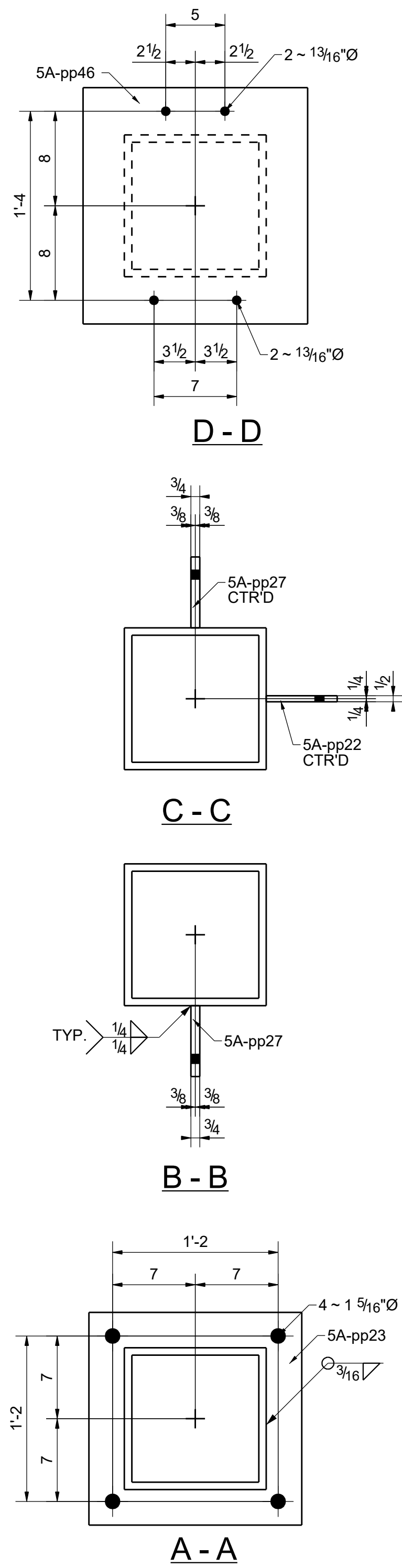
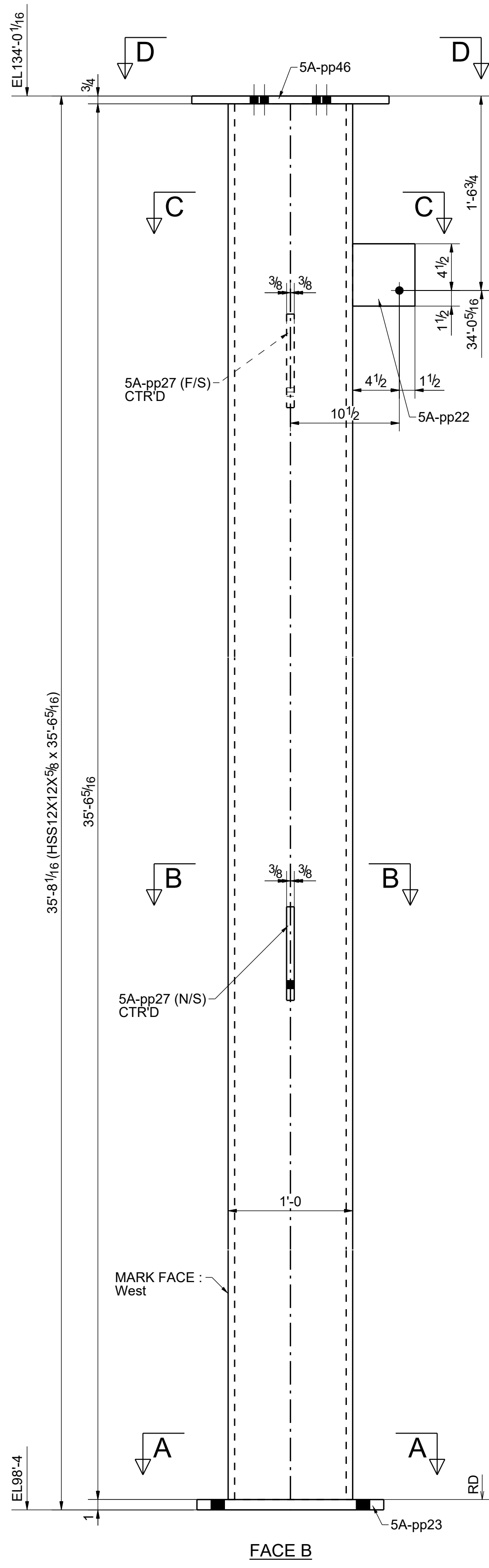
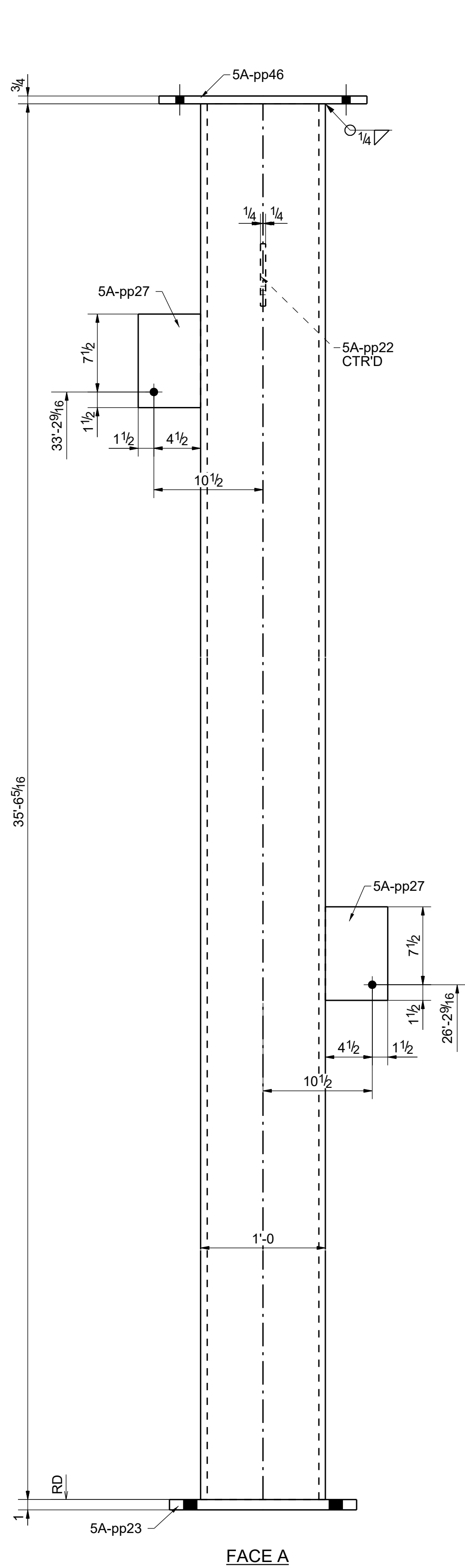
NO PCS	DESCRIPTION	MARK	LENGTH	TOTAL WEIGHT	UNIT WEIGHT	WEIGHT	PRELIM MARK	REMARKS
1	HSS8X8X5/8	4A-C15	5'-4 15/16"			302		GALV. A500-GR.C
2	PL1/4"X2 1/2"	4A-pp3	0'-4"			1		A36
1	PL1/2"X8 1/2"	4A-pp11	0'-8 1/2"			10		A36
2	PL3/8"X4 1/2"	4A-pp14	1'-0"			11		A36
1	PL5/8"X15"	4A-pp39	1'-6"			35		A36
1	PL5/8"X14 15/16"	4A-pp40	1'-6"			35		A36

- SHOP NOTES :
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,...ETC.
 - "RD" DENOTES RUNNING DIMENSION
 - ALL COPES SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPES TO BE CUT WITH 1/2" RADIUS U.N.O.
 - ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC..., SEE 24" X 36" SINGLE PARTS DETAILS.
 - B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - HOLE PATTERNS NOTED "A325SC-TC" ARE FOR SLIP CRITICAL BOLTS. DO NOT ADD PAINT AT CONTACT SURFACES. CONTACT SURFACES ARE TO BE FREE OF BURRS AND ALL FOREIGN MATERIAL.
 - SHOP CLOSE ALL GAPS.
 - ALL JOINTS TO BE SEAL WELDED WITH CONT. 1/8" FILLET OR BUTT WELD AS REQ'D
 - PROVIDE VENT HOLES AS REQ'D FOR GALV.
 - PLUG VENT HOLES AFTER GALV.



ONE - COLUMN - 4A-C15

GRID LOCATION	BOTTOM ELEVATION
G.4/2.2	136'-5 1/16"



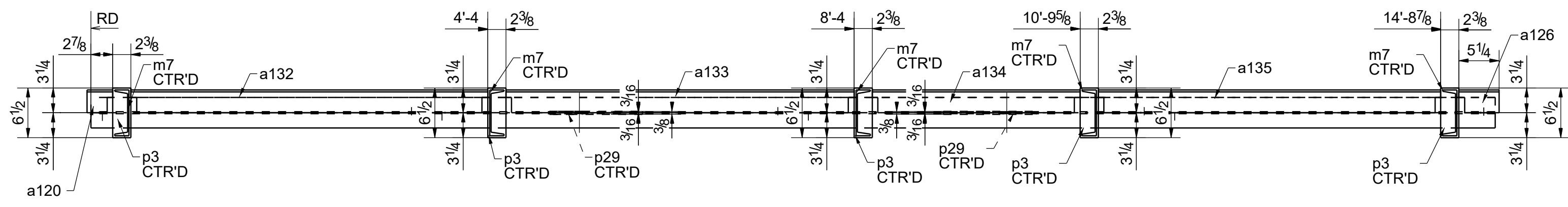
BILL OF MATERIAL

NO PCS	DESCRIPTION	MARK	LENGTH	TOTAL WEIGHT	UNIT WEIGHT	WEIGHT	PRELIM MARK	REMARKS
1	HSS12X12X5/8	5A-C14	35'-6 5/16"			3107		A500-GR.C
1	PL1/2"X6"	5A-pp22	0'-6"			5		A36
1	PL1"X18"	5A-pp23	1'-6"			92		A36
2	PL3/4"X6"	5A-pp27	0'-9"			23		A36
1	PL3/4"X19"	5A-pp46	1'-8"			81		A36

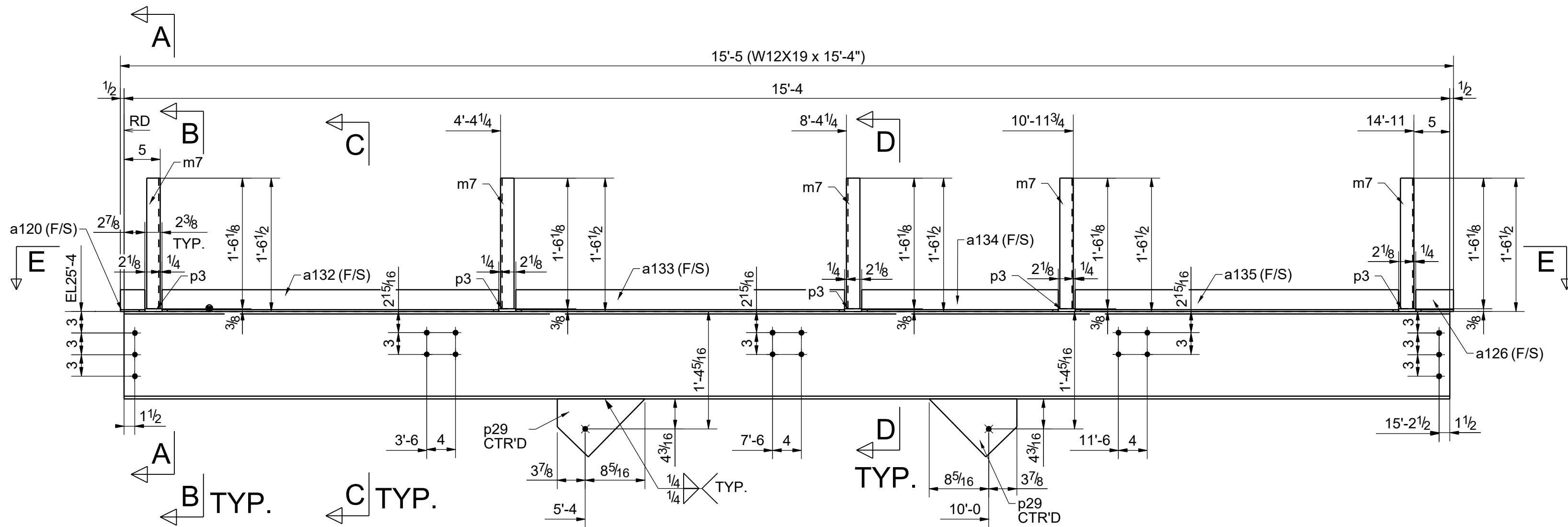
- SHOP NOTES:
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,...ETC.
 - "RD" DENOTES RUNNING DIMENSION
 - ALL COPEs SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPEs TO BE CUT WITH 1/2" RADIUS U.N.O.
 - ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
 - B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - SHOP CLOSE ALL GAPS.
 - HOLE PATTERNS NOTED "A325SC-TC" ARE FOR SLIP CRITICAL BOLTS. DO NOT ADD PAINT AT CONTACT SURFACES. CONTACT SURFACES ARE TO BE FREE OF BURRS AND ALL FOREIGN MATERIAL.

ONE - COLUMN - 5A-C14

GRID LOCATION	BOTTOM ELEVATION
J/7-6.9	98'-4"

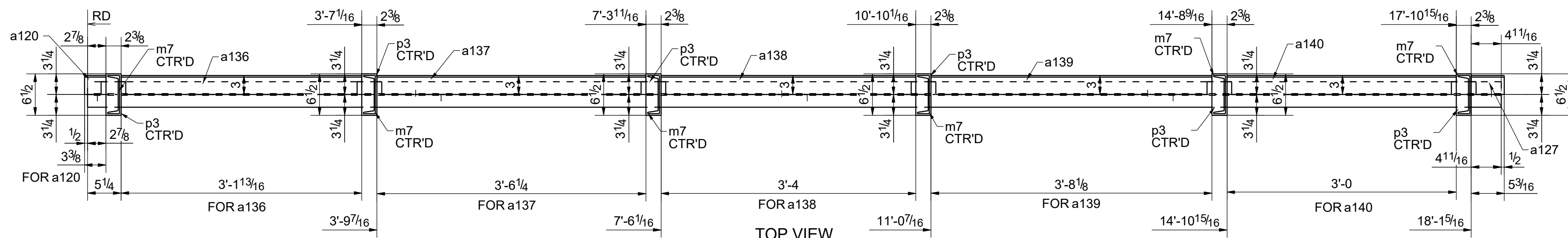


TOP VIEW

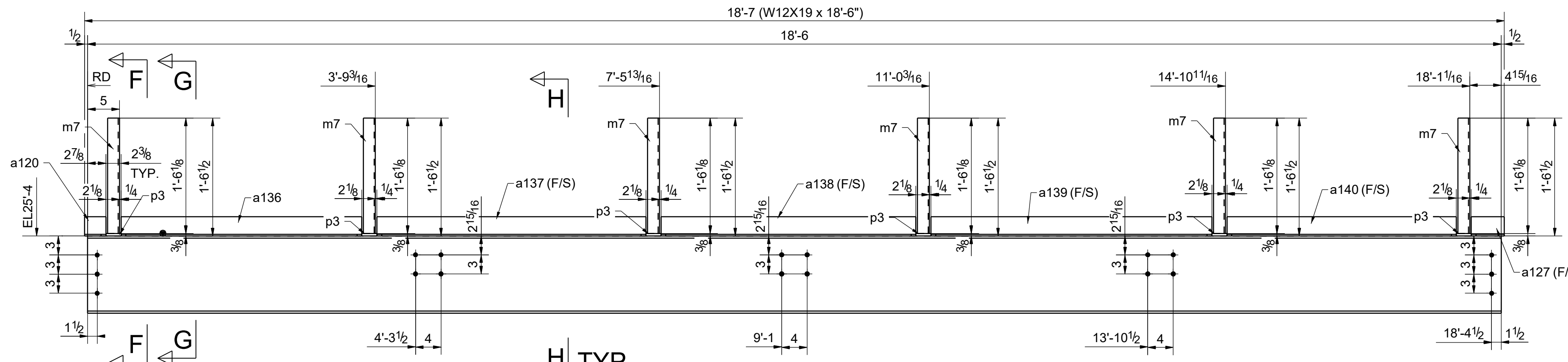


E - E

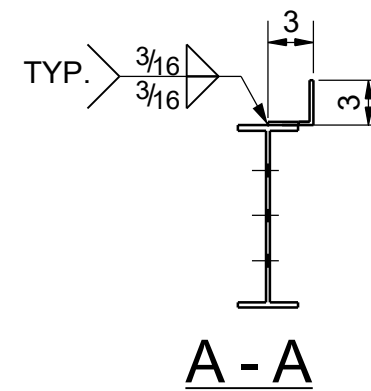
ONE - BEAM - 12B1



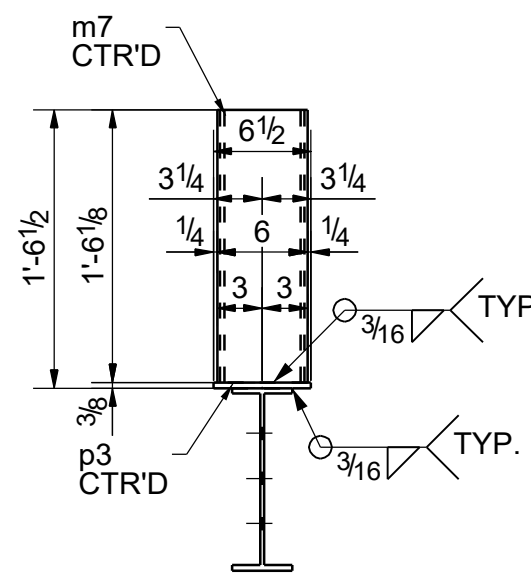
TOP VIEW



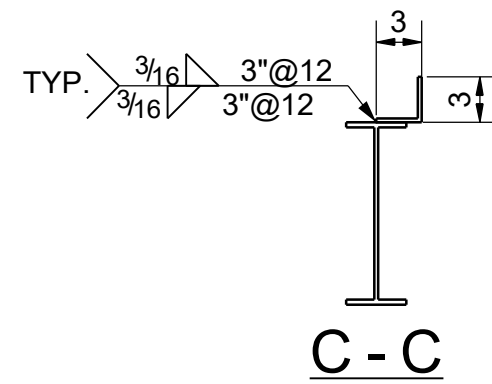
ONE - BEAM - 12B2



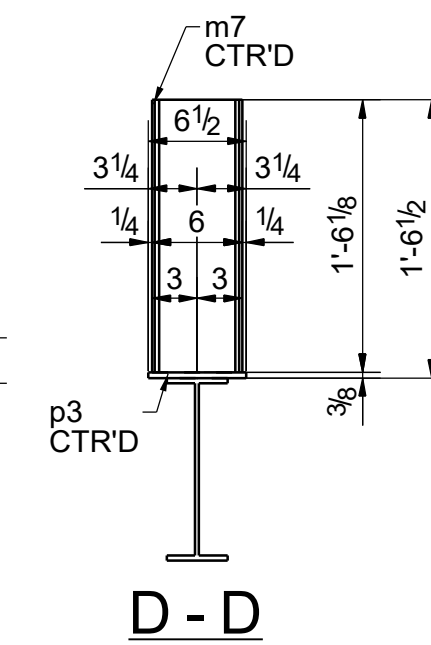
A - A



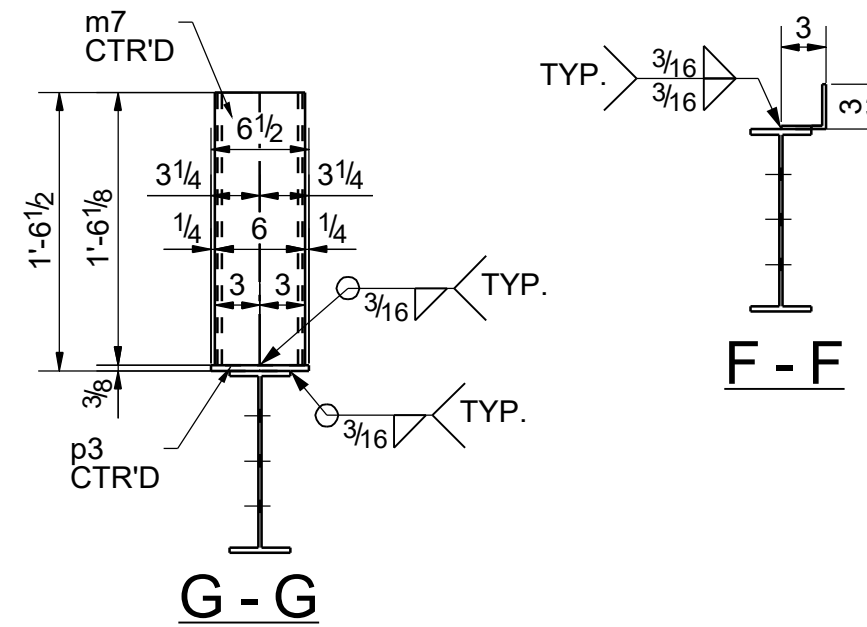
B - B



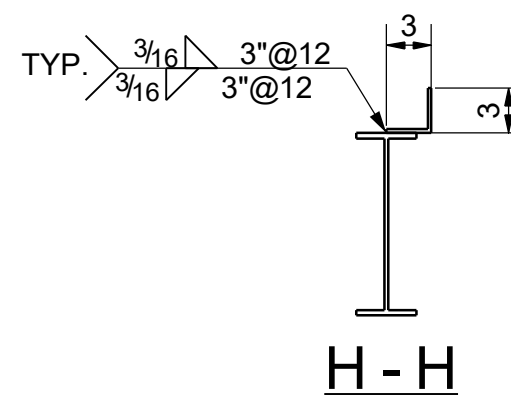
C - C



D - D



F - F



G - G



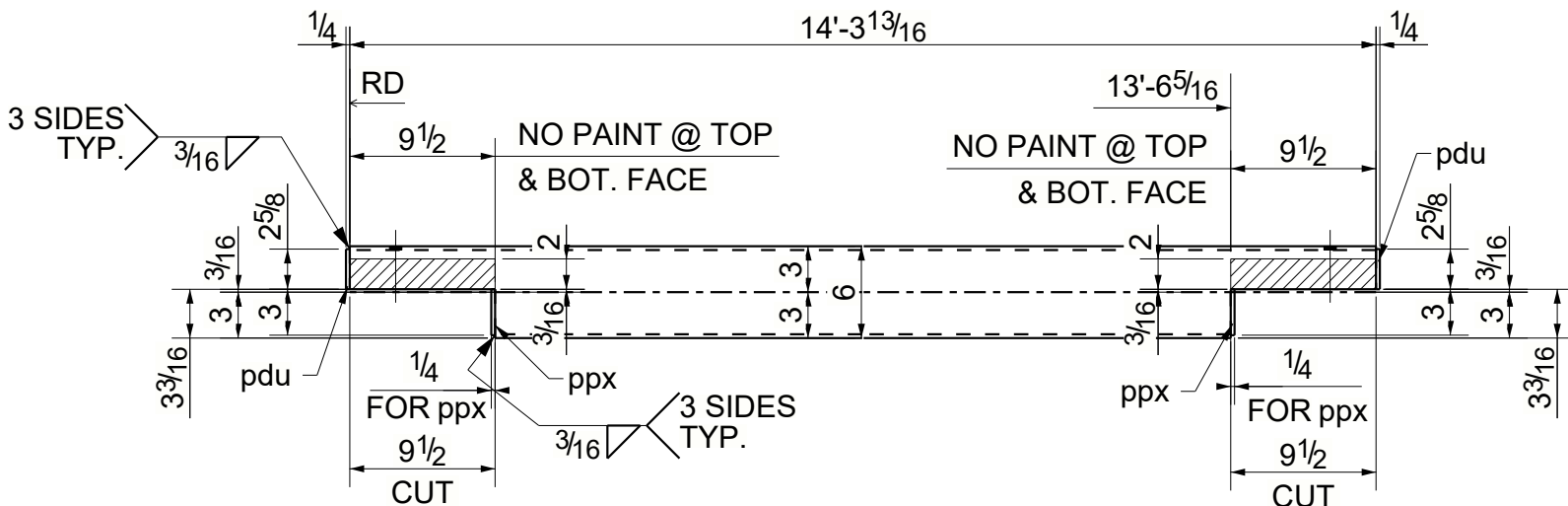
H - H

BILL OF MATERIAL

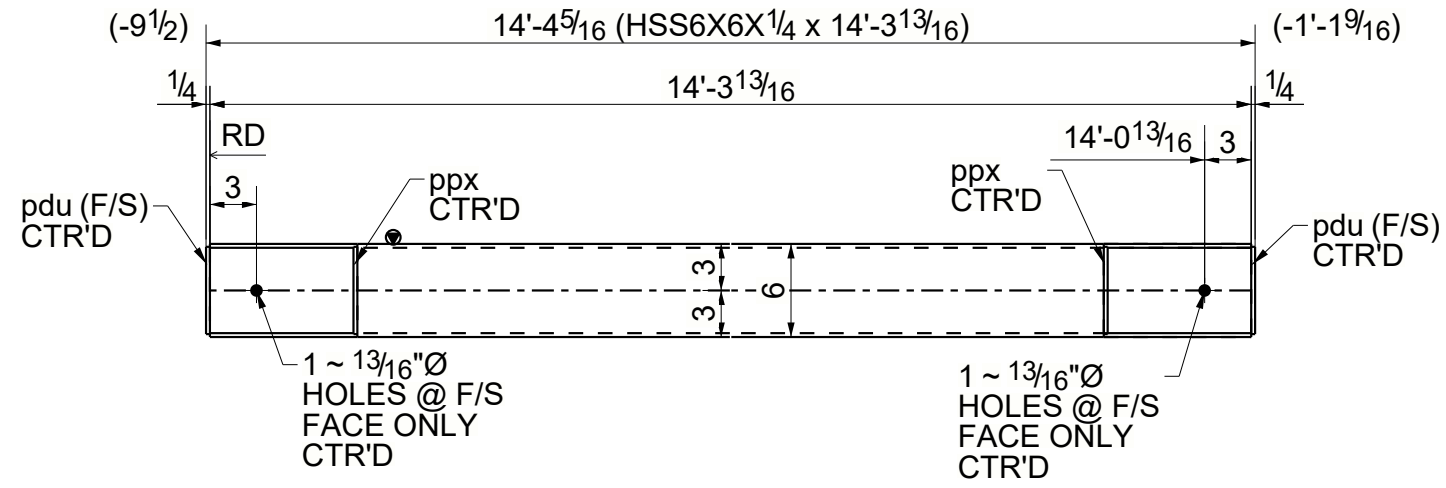
SHIP MARK	PIECE MARK	QTY	DESCRIPTION	LENGTH	REMARKS
12B1		1	W12X19	15'-4"	SO-2 A892
	a120	1	L3X3X1/4	0'-0"-3 3/8"	A36
	a126	1	L3X3X1/4	0'-0"-5 1/4"	A36
	a132	1	L3X3X1/4	3'-10 3/4"	A36
	a133	1	L3X3X1/4	3'-9 5/8"	A36
	a134	1	L3X3X1/4	2'-3 1/4"	A36
	a135	1	L3X3X1/4	3'-8 7/8"	A36
	m7	5	C6X8.2	1'-6 1/8"	A36
	p3	5	PL3/8"x2 3/8"	0'-0"-6 1/2"	A36
	p29	2	PL3/8"x8 1/16"	1'-0 3/16"	A36
12B2		1	W12X19	18'-6"	SO-2 A892
	a120	1	L3X3X1/4	0'-0"-3 3/8"	A36
	a127	1	L3X3X1/4	0'-0"-5 3/16"	A36
	a136	1	L3X3X1/4	3'-1 13/16"	A36
	a137	1	L3X3X1/4	3'-6 1/4"	A36
	a138	1	L3X3X1/4	3'-4"	A36
	a139	1	L3X3X1/4	3'-8 1/8"	A36
	a140	1	L3X3X1/4	3'-0"	A36
	m7	6	C6X8.2	1'-6 1/8"	A36
	p3	6	PL3/8"x2 3/8"	0'-0"-6 1/2"	A36

SHOP NOTES :

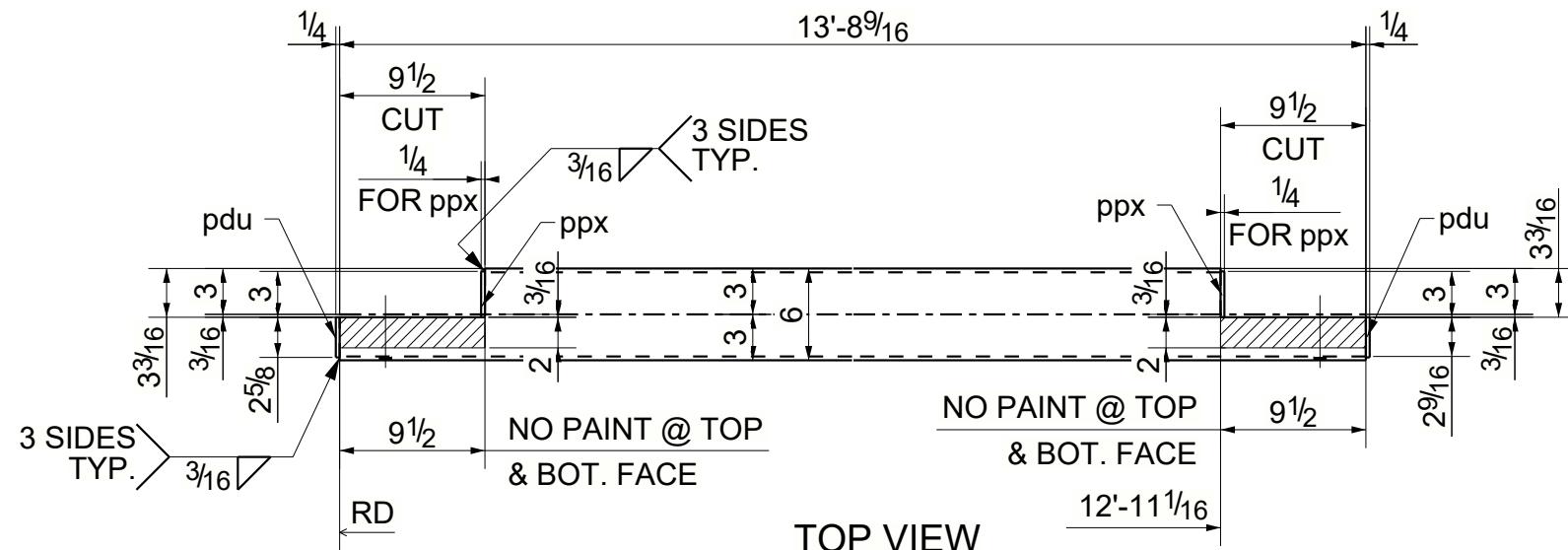
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
- TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
- ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
- ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN, ETC.
- "RD" DENOTES RUNNING DIMENSION
- ALL COPIES SHALL BE MEASURED FROM END OF BEAM.
- ALL BEAM COPIES TO BE CUT WITH 1/2" RADIUS U.N.O.
- ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
- FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC... SEE GATHER SHEETS/ SINGLE PART SHEETS.
- B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
- SHOP CLOSE ALL GAPS.



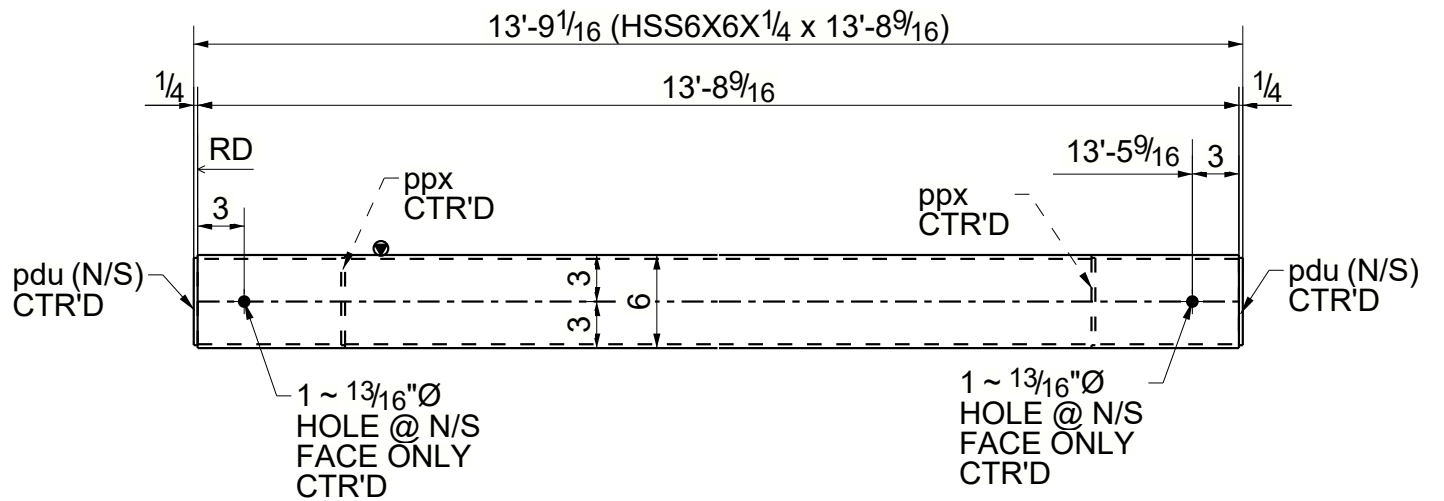
TOP VIEW



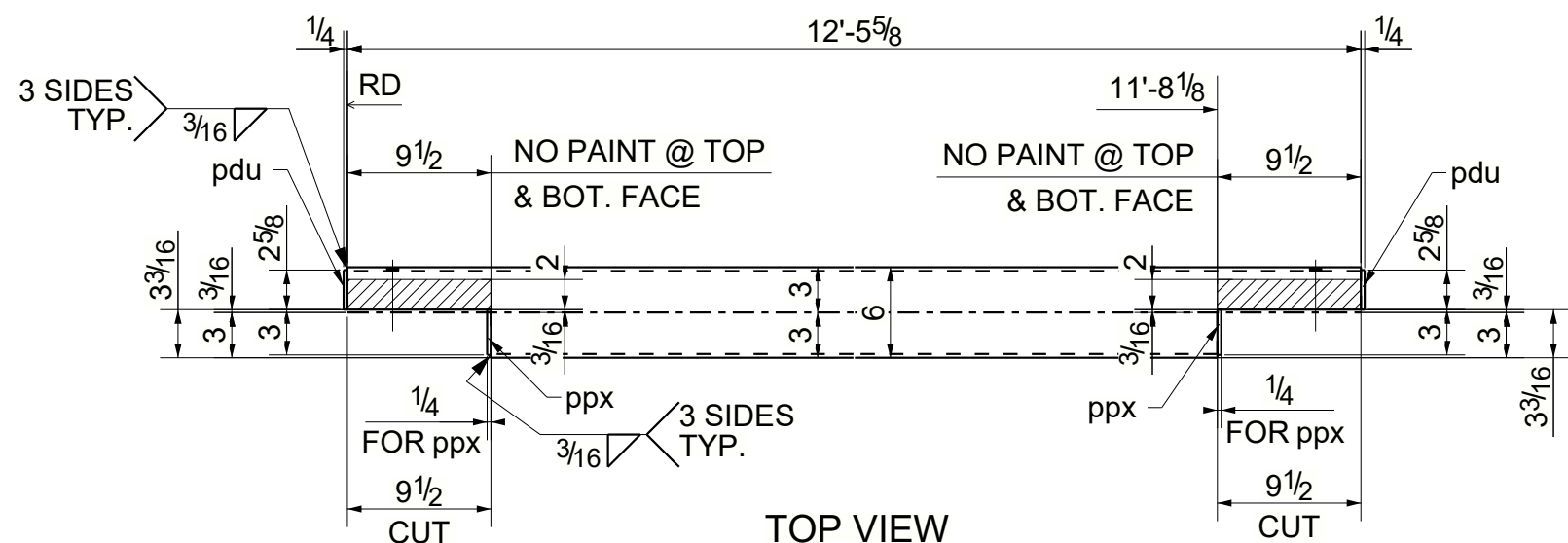
2 - VERTICAL HSS BRACE - 125D1



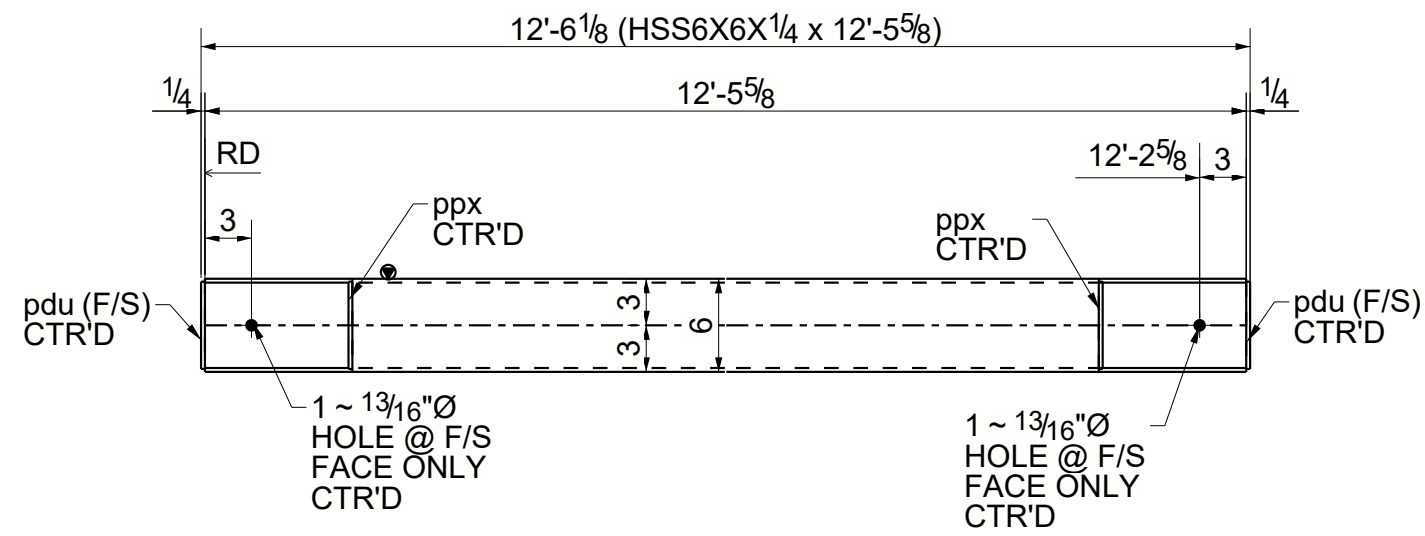
TOP VIEW



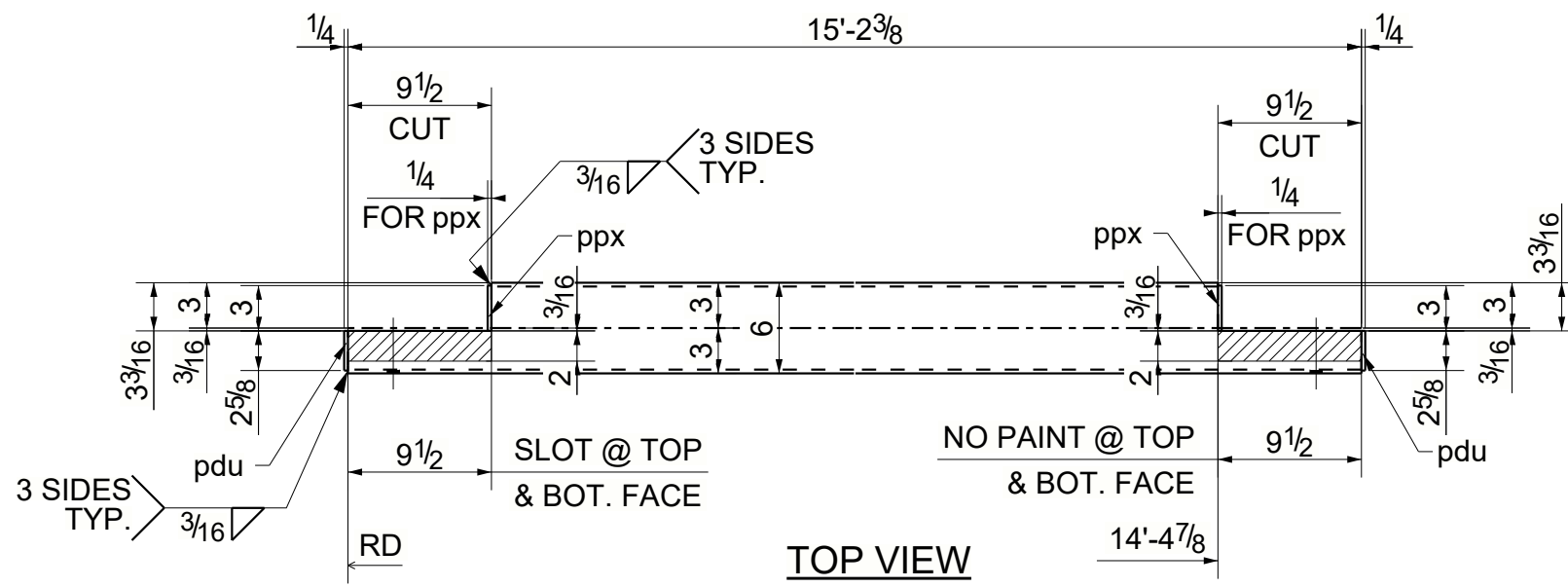
ONE - VERTICAL HSS BRACE - 125D2



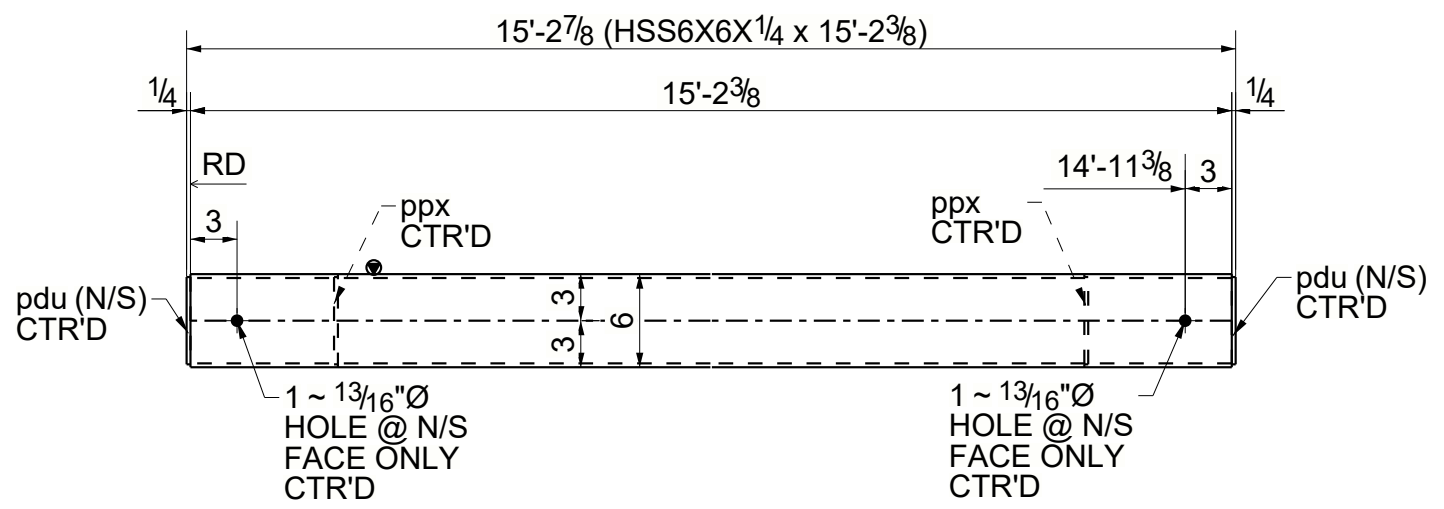
TOP VIEW



2 - VERTICAL HSS BRACE - 125D3



TOP VIEW



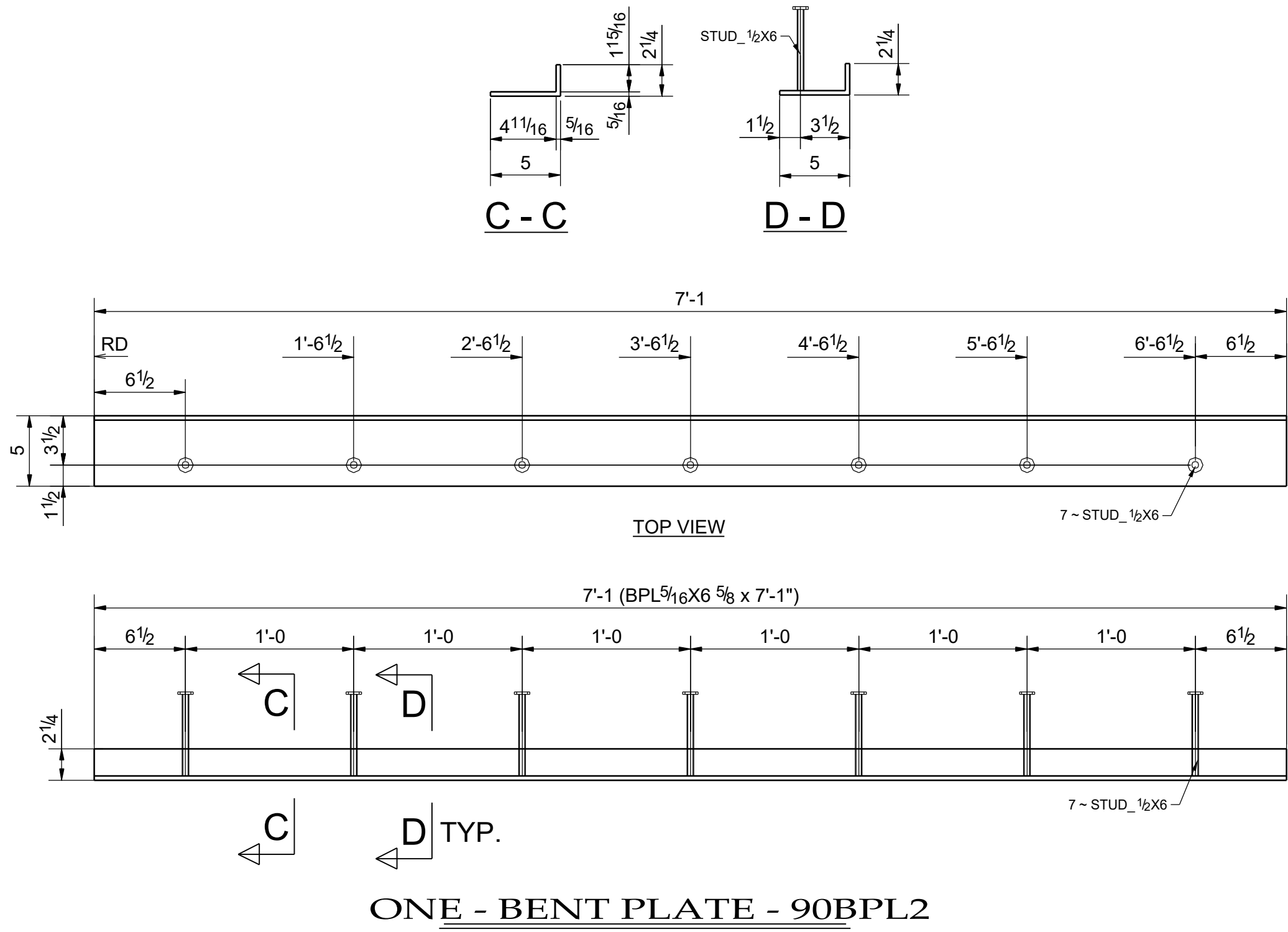
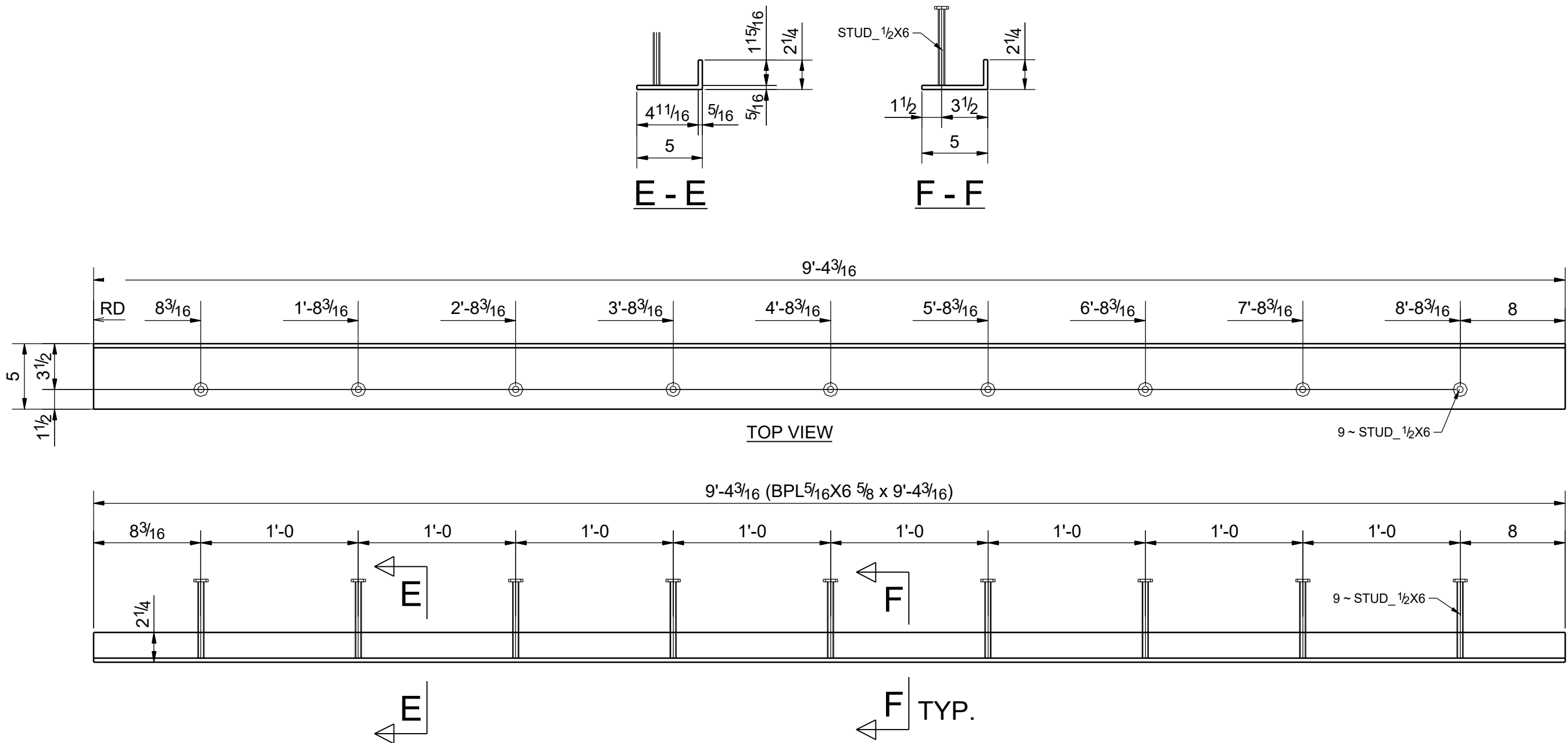
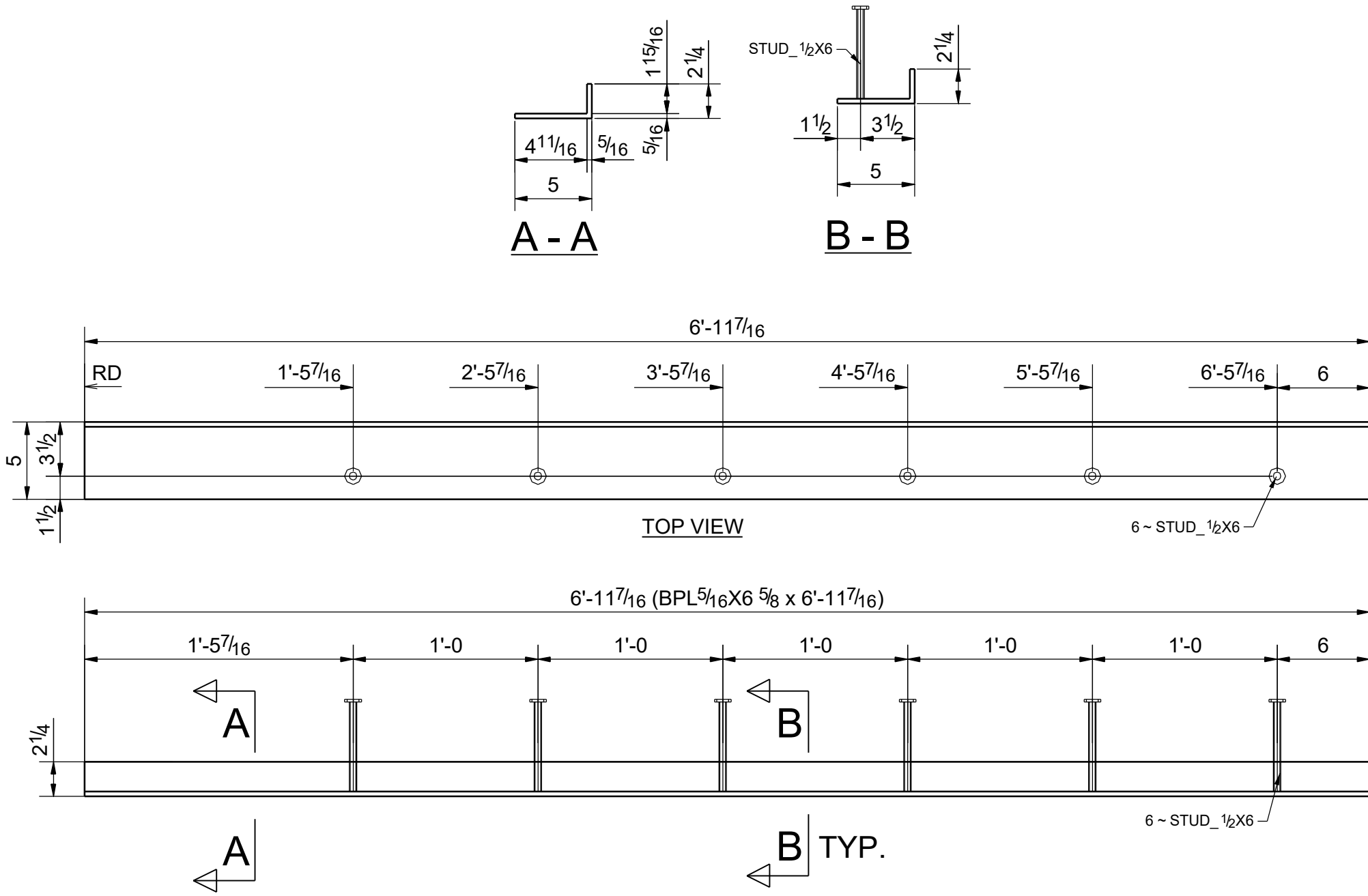
2 - VERTICAL HSS BRACE - 125D4

BILL OF MATERIAL

MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
125D1	2	VERTICAL HSS BRACES			SQ-2
125D1	2	HSS6X6X1/4	14'-3 13/16"	511	A500-GR.B
pdu	4	PL 1/4"X2 5/8"	0'-5 5/8"	4	A36
ppx	4	PL 1/4"X3"	0'-5 5/8"	5	A36
125D2	1	VERTICAL HSS BRACE			SQ-2
125D2	1	HSS6X6X1/4	13'-8 9/16"	245	A500-GR.B
pdu	2	PL 1/4"X2 5/8"	0'-5 5/8"	2	A36
ppx	2	PL 1/4"X3"	0'-5 5/8"	2	A36
125D3	2	VERTICAL HSS BRACES			SQ-2
125D3	2	HSS6X6X1/4	12'-5 5/8"	445	A500-GR.B
pdu	4	PL 1/4"X2 5/8"	0'-5 5/8"	4	A36
ppx	4	PL 1/4"X3"	0'-5 5/8"	5	A36
125D4	2	VERTICAL HSS BRACES			SQ-2
125D4	2	HSS6X6X1/4	15'-2 3/8"	542	A500-GR.B
pdu	4	PL 1/4"X2 5/8"	0'-5 5/8"	4	A36
ppx	4	PL 1/4"X3"	0'-5 5/8"	5	A36
TOTAL WEIGHT THIS DRAWING				1773	

BILL OF MATERIAL					
SHIP MARK	PIECE MARK	QTY	DESCRIPTION	LENGTH	REMARKS
90BPL1		1	BPL5/16"X6 5/8"	0'-6 5/8"	SD2 A36
		6	1/2"Ø STUD	0'-6"	
90BPL2		1	BPL5/16"X6 5/8"	0'-6 5/8"	SD2 A36
		7	1/2"Ø STUD	0'-6"	
90BPL3		1	BPL5/16"X6 5/8"	0'-6 5/8"	SD2 A36
		9	1/2"Ø STUD	0'-6"	

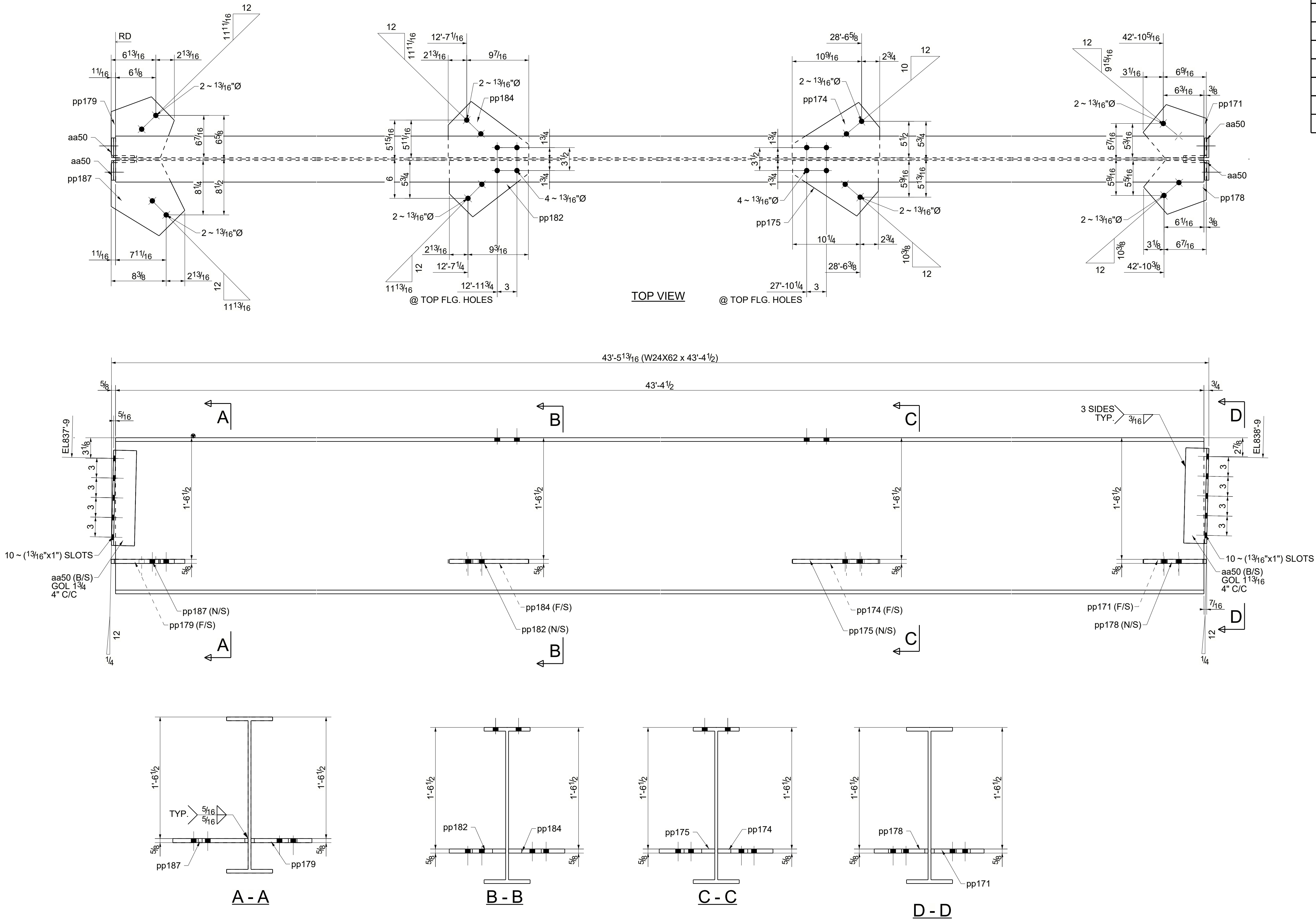
- SHOP NOTES :
- ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,..ETC.
"RD" DENOTES RUNNING DIMENSION
 - ALL COPES SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPES TO BE CUT WITH 1/2" RADIUS U.N.O.
 - ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC..., SEE GATHER SHEETS/ SINGLE PART SHEETS.
 - B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - SHOP CLOSE ALL GAPS.



QTY	NAME
1	STEEL FRAMING

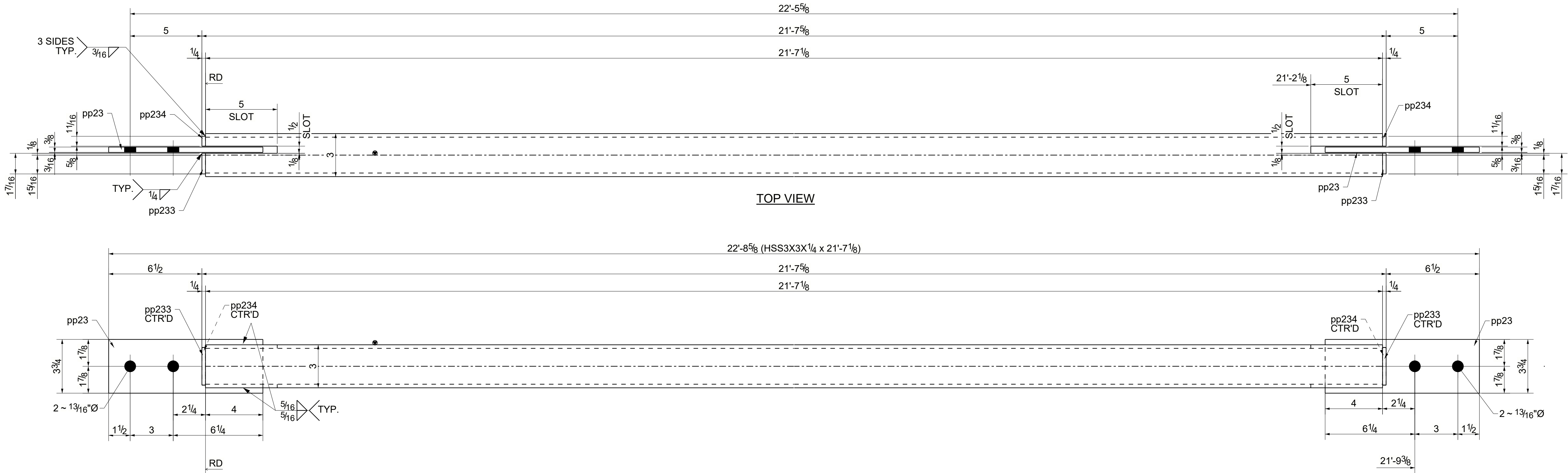
BILL OF MATERIAL					
MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
B44	1	BEAM			SQ-2
B44	1	W24X62	43'-4 1/2"	2686	A992
aa50	4	L3-1/2X3X5/16	1'-2 1/2"	32	A36
pp171	1	PL5/8"X8 11/16"	0'-9 3/4"	10	A36
pp174	1	PL5/8"X8 7/16"	1'-1 5/16"	13	A36
pp175	1	PL5/8"X8 7/16"	1'-1 1/16"	13	A36
pp178	1	PL5/8"X8 5/8"	0'-9 13/16"	10	A36
pp179	1	PL5/8"X9 7/16"	0'-10 5/8"	12	A36
pp182	1	PL5/8"X8 9/16"	1'-0"	12	A36
pp184	1	PL5/8"X8 9/16"	1'-0 1/4"	12	A36
pp187	1	PL5/8"X11 3/16"	0'-11 7/16"	16	A36
TOTAL WEIGHT THIS DRAWING				2818	

- SHOP NOTES:
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 - TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 - ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 - ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,...ETC.
 - "RD" DENOTES RUNNING DIMENSION
 - ALL COPEs SHALL BE MEASURED FROM END OF BEAM.
 - ALL BEAM COPEs TO BE CUT WITH 1/2" RADIUS U.N.O.
 - ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
 - FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
 - B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 - SHOP CLOSE ALL GAPS.
 - ALL JOINTS TO BE SEAL WELDED WITH CONT. 1/8 FILLET OR BUTT WELD AS REQ'D
 - PROVIDE VENT HOLES AS REQ'D FOR GALV.
 - PLUG VENT HOLES AFTER GALV.



ONE - BEAM - B44

QTY	NAME	BILL OF MATERIAL						
1	STEEL FRAMING	MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS	
		D1	1	VERTICAL HSS BRACE			SQ-2	
		D1	1	HSS3X3X1/4	21'-7 1/8"	179	A500-GR.C	
		pp23	2	PL3/8"X3 3/4"	0'-10 3/4"	9	A36	
		pp233	2	PL1/4"X1 7/16"	0'-2 5/8"	1	A36	
		pp234	2	PL1/4"X11/16"	0'-2 5/8"	0	A36	
				TOTAL WEIGHT THIS DRAWING			189	

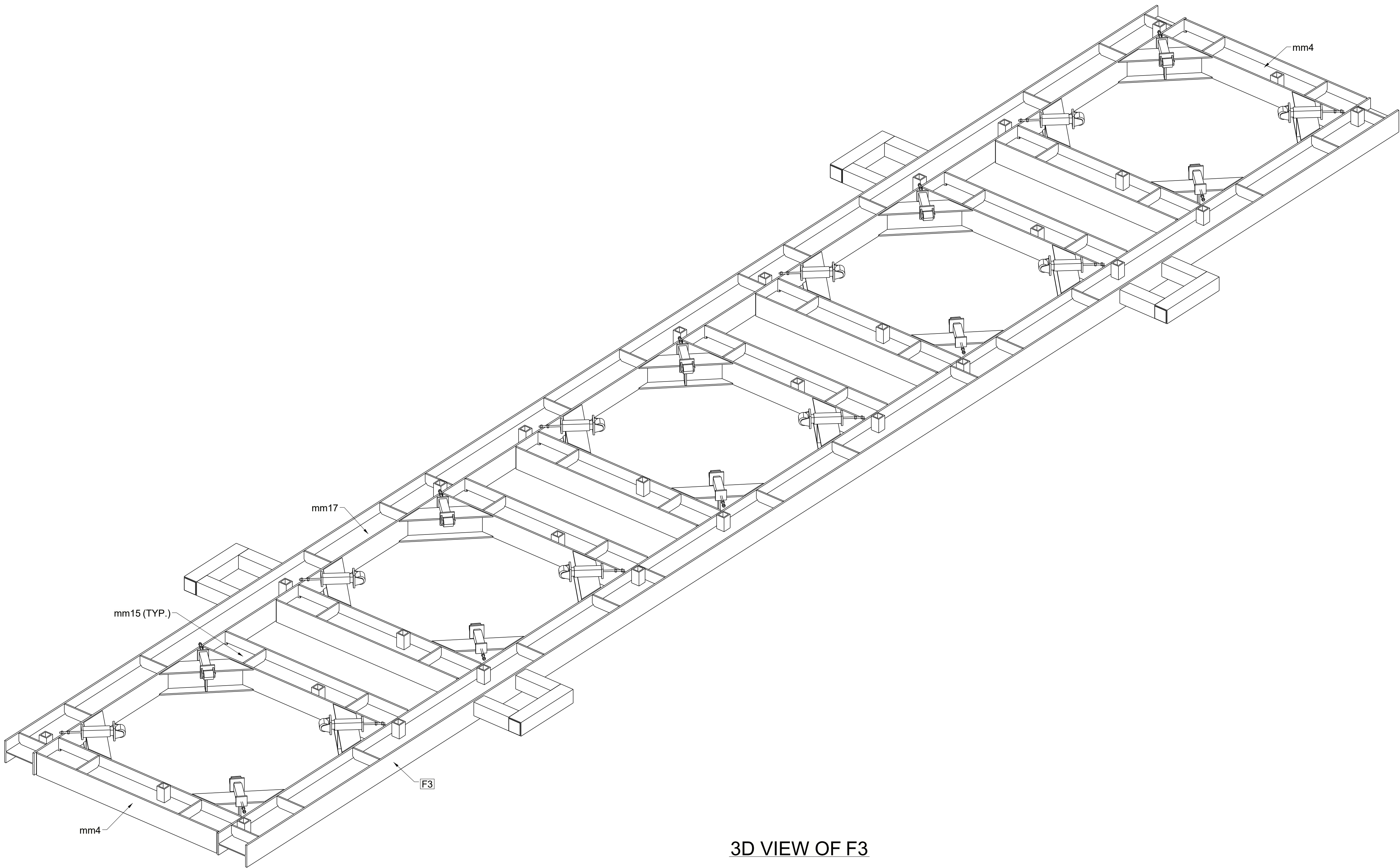


- SHOP NOTES:
1. ALL WELDS TO BE 3/16" ALL AROUND OR DOUBLE SIDED FILLET WELD UNLESS SHOWN OTHERWISE.
 2. TAG ALL MEMBERS IN A VERTICAL POSITION ON LEFT END AS DETAILED.
 3. ALL SHARP EDGES SHALL BE GROUND SMOOTH AND CORNERS ROUNDED OFF.
 4. ALL TAIL (RUNNING) DIMENSIONS SHALL BE MEASURED FROM END OF BEAM, COLUMN,..ETC. "RD" DENOTES RUNNING DIMENSION
 5. ALL COPEs SHALL BE MEASURED FROM END OF BEAM.
 6. ALL BEAM COPEs TO BE CUT WITH 1/2" RADIUS U.N.O.
 7. ALL CLIP ANGLES SHALL BE ATTACHED TO BEAM WITH 3/16" FILLET WELD ON THREE SIDES UNLESS SHOWN OTHERWISE.
 8. FOR INDIVIDUAL DETAILS OF PLATES, ANGLES, ETC., SEE SINGLE PARTS DETAIL SHEETS.
 9. B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 10. SHOP CLOSE ALL GAPS.
 11. ALL JOINTS TO BE SEAL WELDED WITH CONT. 1/8 FILLET OR BUTT WELD AS REQ'D
 12. PROVIDE VENT HOLES AS REQ'D FOR GALV.
 13. PLUG VENT HOLES AFTER GALV.

ONE - VERTICAL HSS BRACE - D1

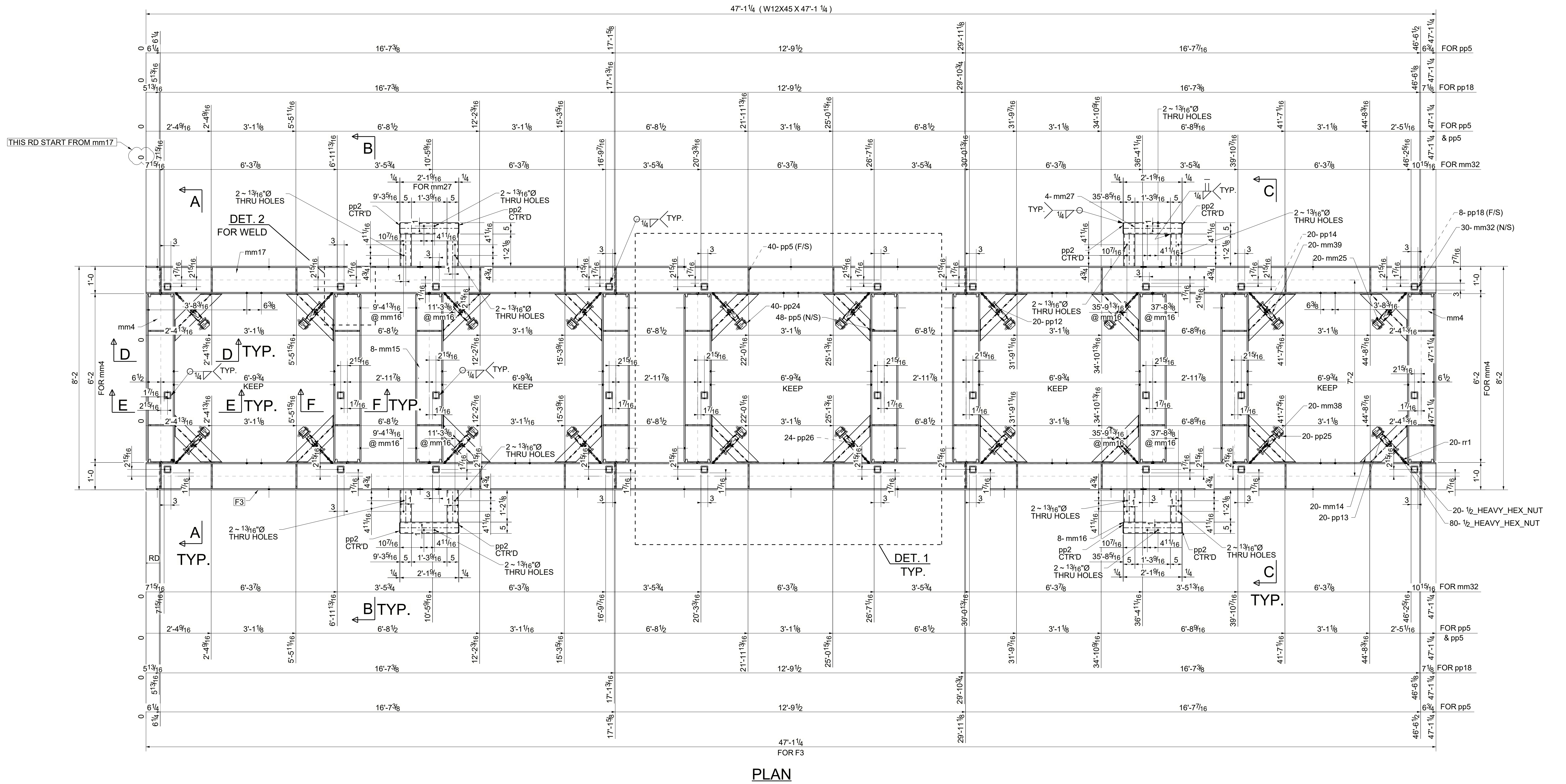
BILL OF MATERIAL					
MARK	QTY	DESCRIPTION	LENGTH	WEIGHT	REMARKS
F3	1	FRAME			SO-2
F3	1	W12X45	47'-1 1/4"	2100	A992
mm4	2	W12X45	6'-2"	550	A992
mm14	20	W8X28	2'-5 9/16"	1384	A992
mm15	8	W12X45	6'-2"	2199	A992
mm16	8	HSS5X5X3/8	1'-2 1/8"	198	A500-GR.B
mm17	1	W12X45	47'-1 1/4"	2100	A992
mm25	20	McMASTER-CARR PART: 6384K365 BCI			
mm27	4	HSS5X5X3/8	2'-1 9/16"	179	A500-GR.B
mm32	30	HSS3X3X3/8	0'-5"	144	A500-GR.B
mm38	20	HSS2X2X1/4	0'-5 3/4"	49	A500-GR.B
mm39	20	HSS2-1/2X2-1/2X3/16	0'-8 1/2"	74	A500-GR.B
pp2	8	PL1/4"X4 1/2"	0'-4 1/2"	11	A36
pp5	88	PL1/2"X3 13/16"	0'-10 3/4"	506	A36
pp12	20	PL1/2"X3 1/2"	0'-3 1/2"	35	A36
pp13	20	PL1/2"X4 1/2"	0'-4 1/2"	57	A36
pp14	20	PL1/2"X1 3/4"	0'-1 3/4"	9	A36
pp18	8	PL1/2"X3 13/16"	0'-10 3/4"	46	A36
pp24	40	PL3/8"X2 3/4"	0'-3"	25	A36
pp25	20	PL1/8"X3/8"	0'-0 3/8"	0	A36
pp26	36	PL1/2"X3 1/8"	0'-7"	107	A36
rr1	20	ROD1/2"	1'-0"	12	A36
	80	1/2_HEAVY_HEX_NUT		5	
	20	1/2_HEAVY_HEX_NUT		1	
	20	3/8" A328N TYPE-1 BOLT - W/ 4" SMOOTH SHANK	0'-5 1/4"	13	Workshop
TOTAL WEIGHT THIS DRAWING				9894	

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 9. SHOP CLOSE ALL GAPS.



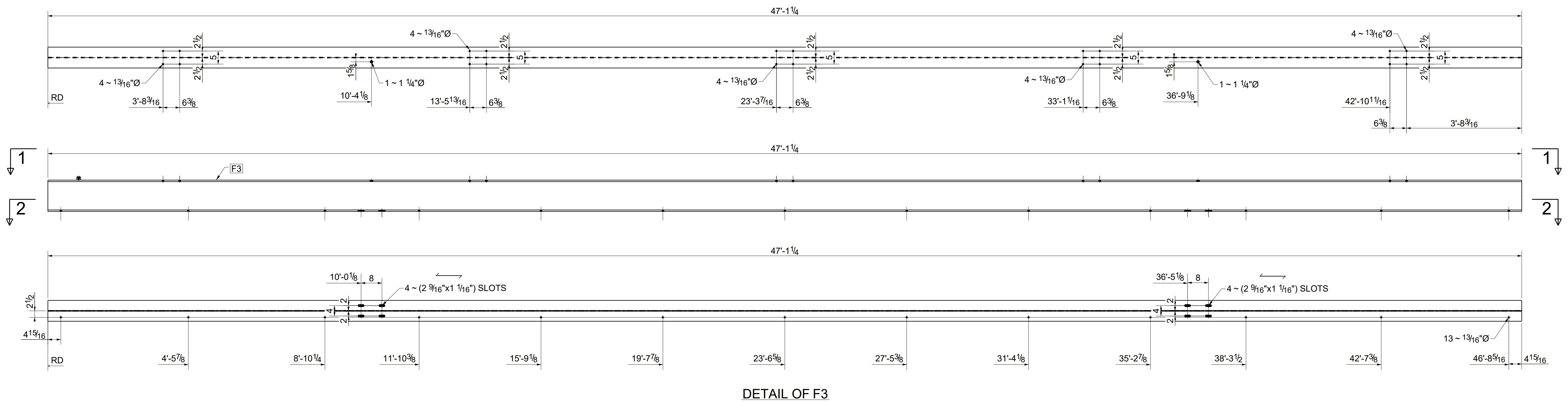
ONE - FRAME - F3

WORK THIS SHEET WITH F3-1, F3-2 & F3-3

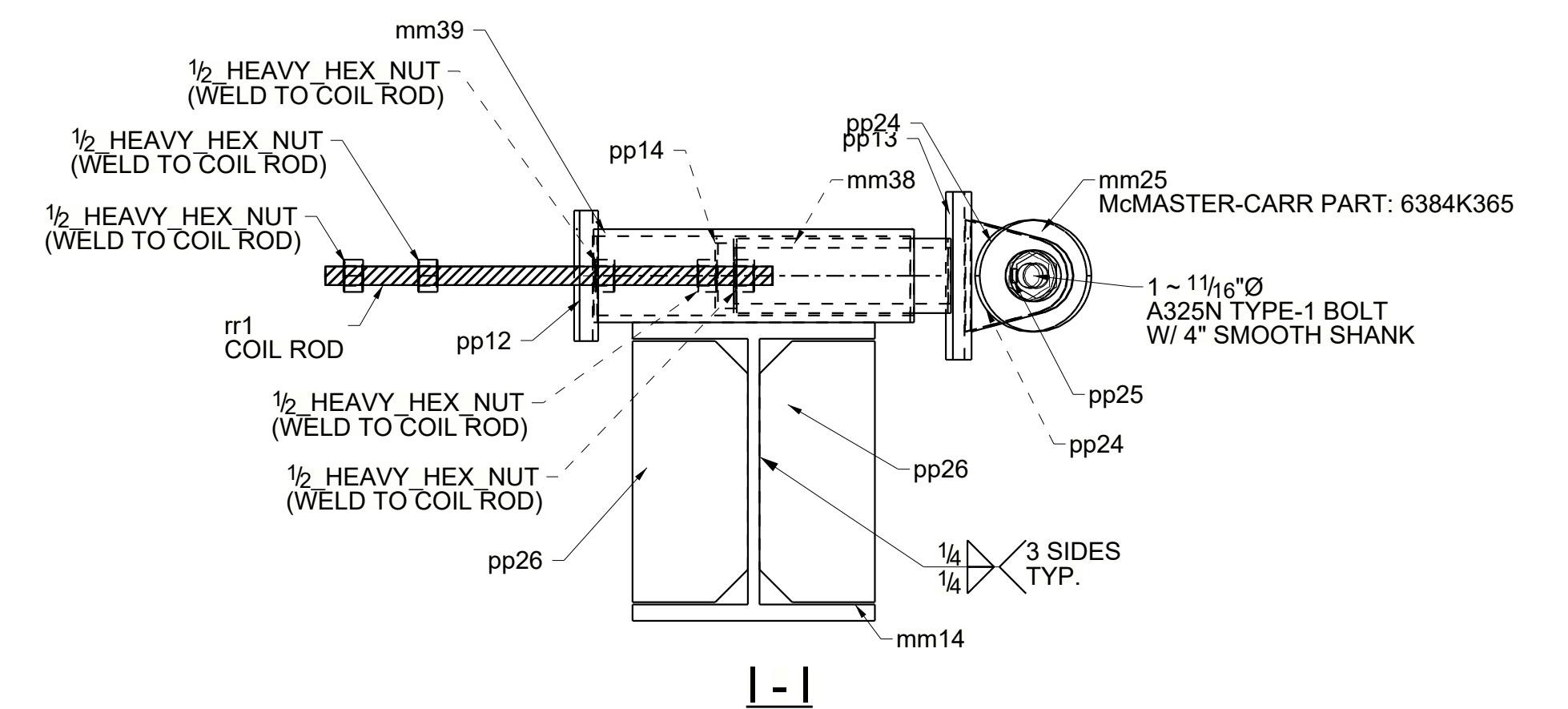
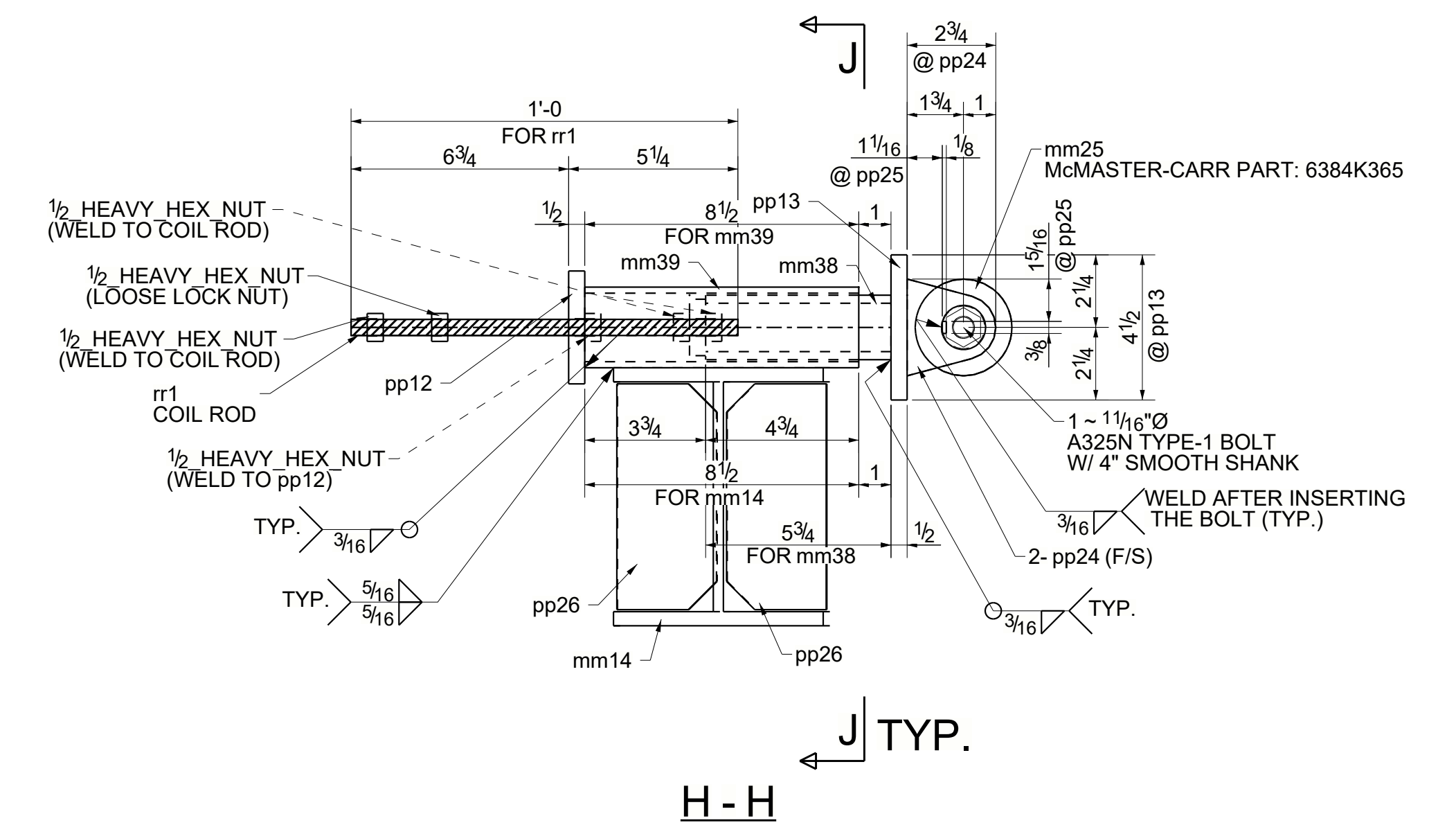
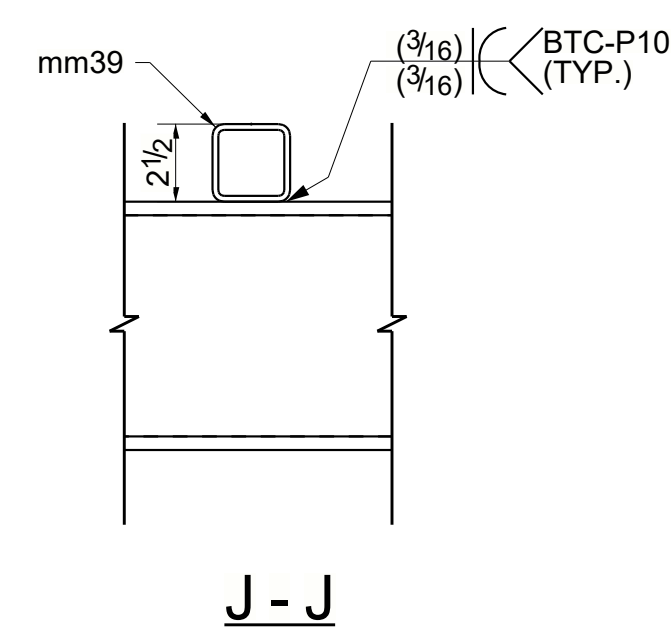
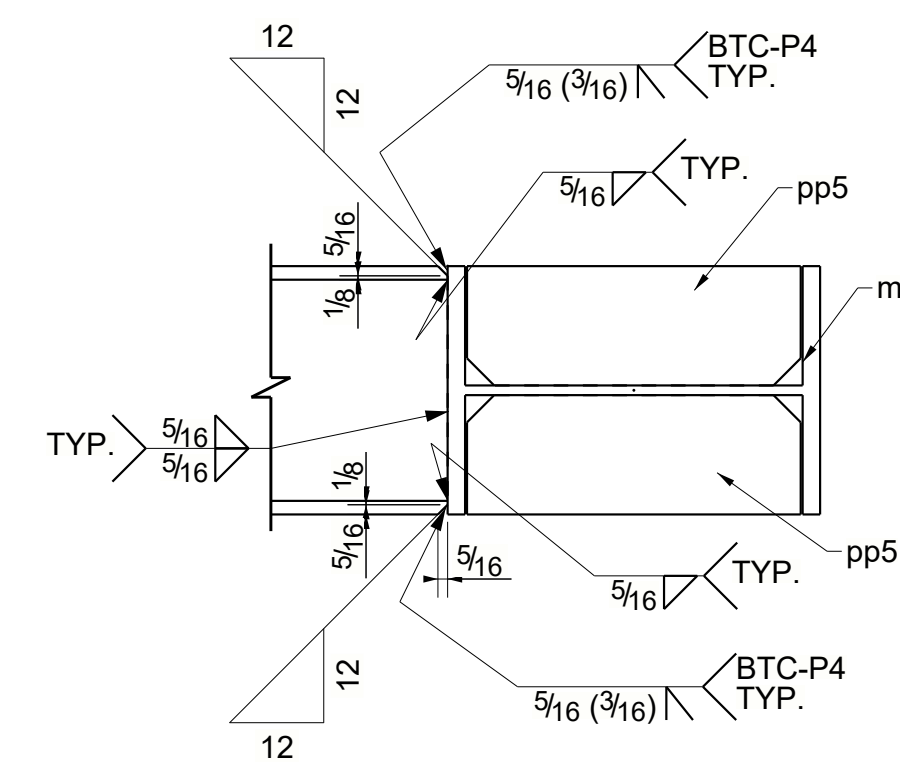
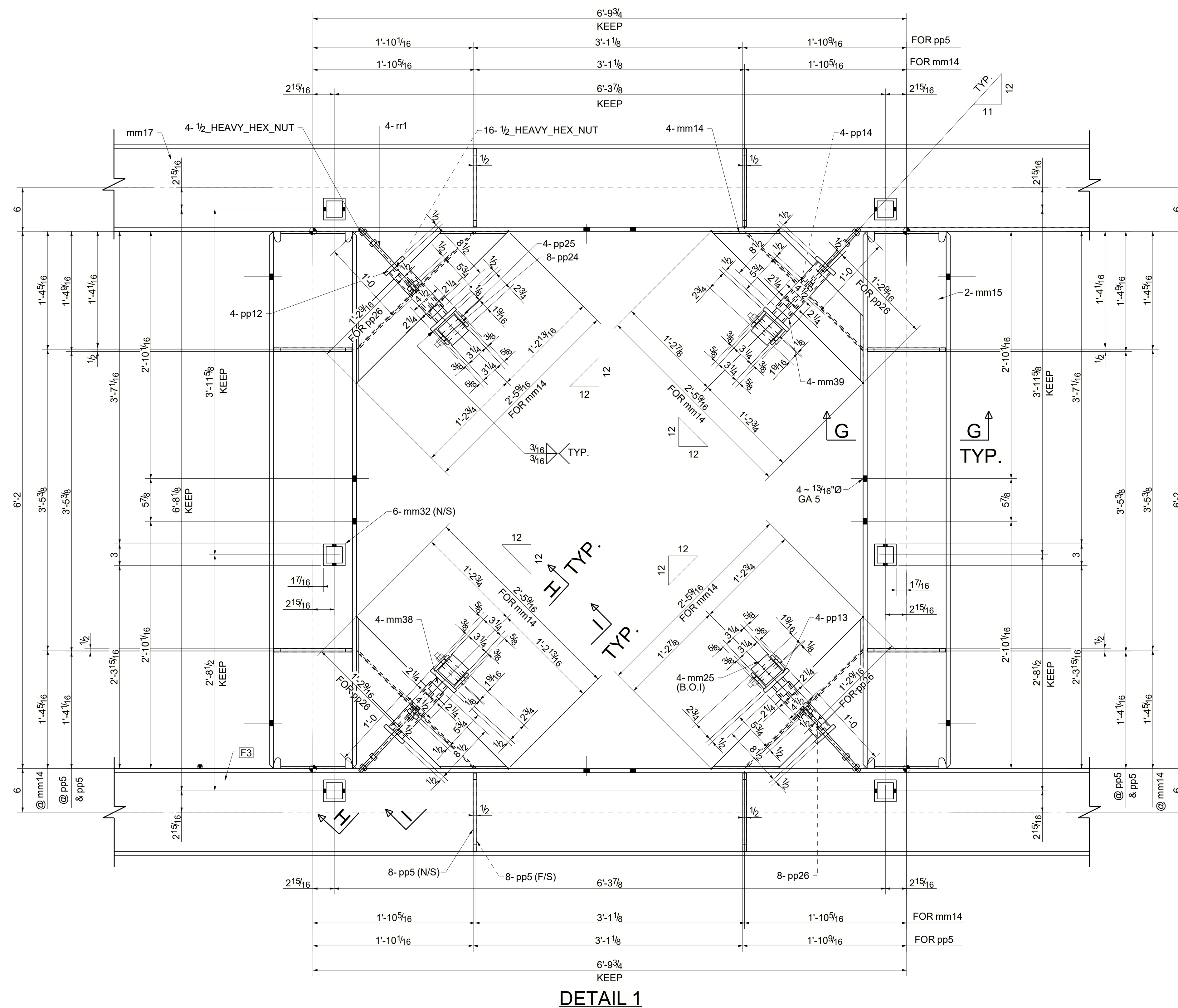


WORK THIS SHEET WITH F3, F3-2 & F3-3

ONE - FRAME - F3



WORK THIS SHEET WITH F3, F3-1 & F3-3



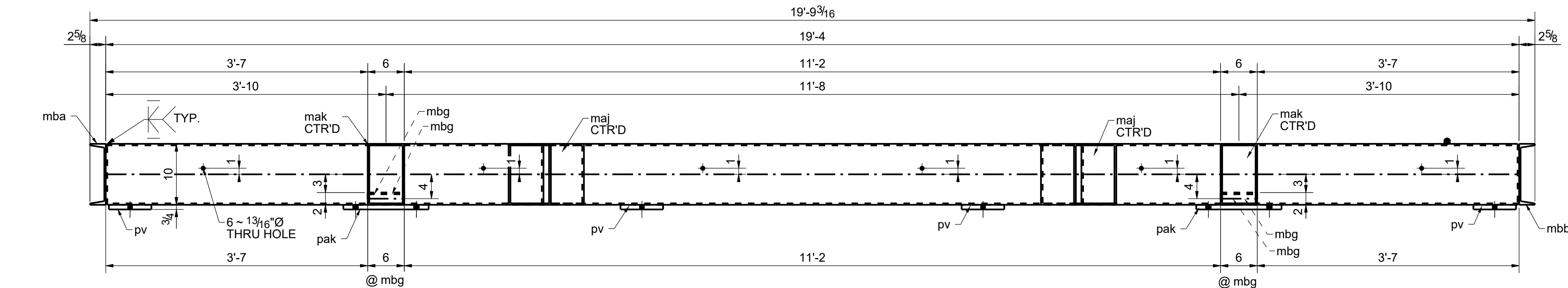
ONE - FRAME - F3

WORK THIS SHEET WITH F3, F3-1 & F3-2

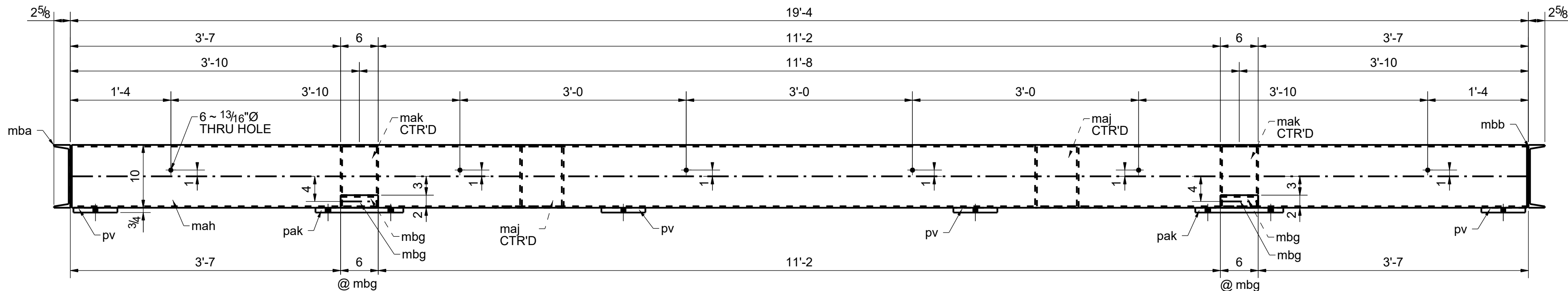
QTY	MARK	DESCRIPTION	LENGTH	GRADE	REMARKS
1	113F1	HSS10X6X1/4	19'-4"	A500-GR.B	
3	am	L4X4X3/8	0'-8"	A36	
2	at	L2X2X1/4	3'-6"	A36	
1	au	L4X4X3/8	0'-8"	A36	
1	mah	HSS10X6X1/4	19'-4"	A500-GR.B	
4	maj	HSS10X6X1/4	4'-11"	A500-GR.B	
2	mak	HSS10X6X1/4	9'-6"	A500-GR.B	
1	mba	C10X15.3	10'-6"	A36	
1	mbb	C10X15.3	10'-6"	A36	
6	mbc	C6X10.5	3'-7"	A36	
4	mbd	W10X30	1'-1 3/8"	A502	
8	mbg	HSS2X2X1/4	0'-6"	A500-GR.B	
4	pad	PL3/4"X8"	1'-0"	A36	
4	pak	PL3/4"X14"	1'-2"	A36	
8	pv	PL3/4"X7"	1'-0"	A36	

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 9. B.T.S. DENOTES BOLT TO SHIP WITH A307 BOLTS. ASSEMBLE AFTER PAINTING/GALV.
 10. SHOP CLOSE ALL GAPS.

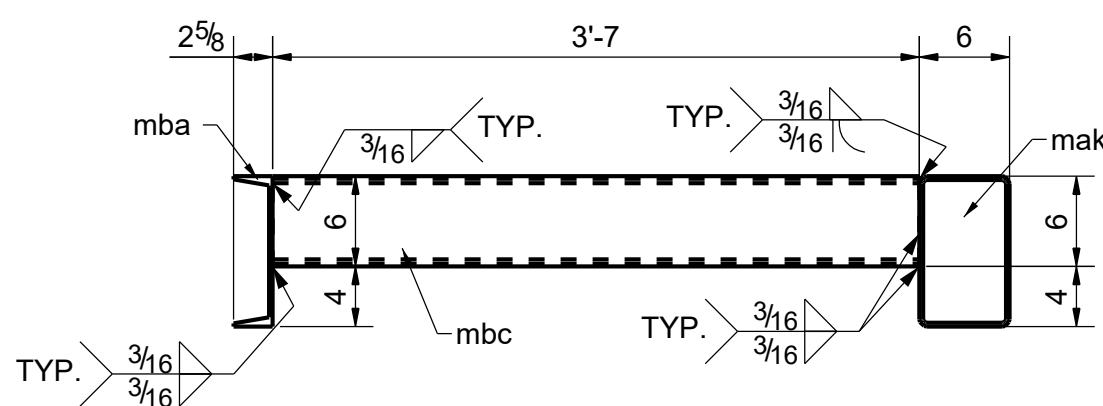
WORK THIS SHEET WITH 113,113-B



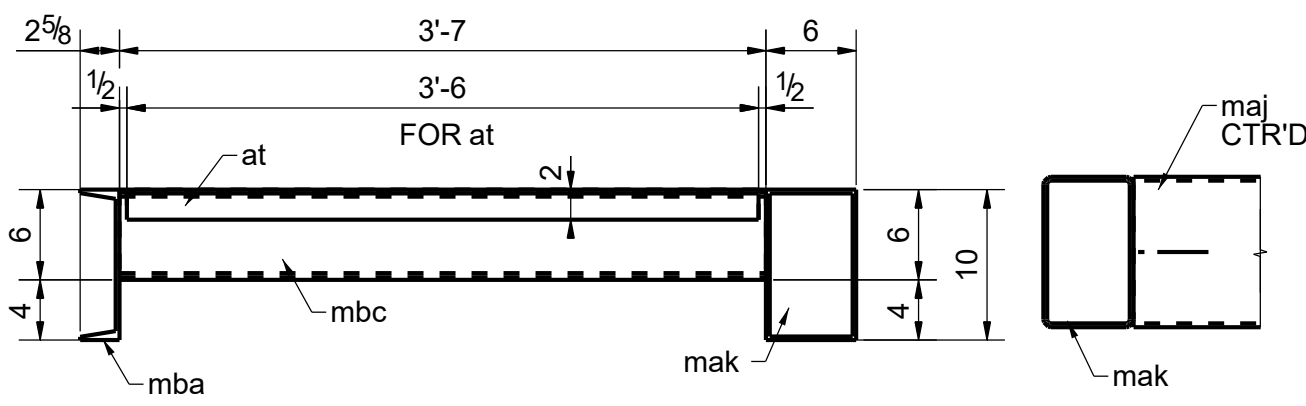
A - A



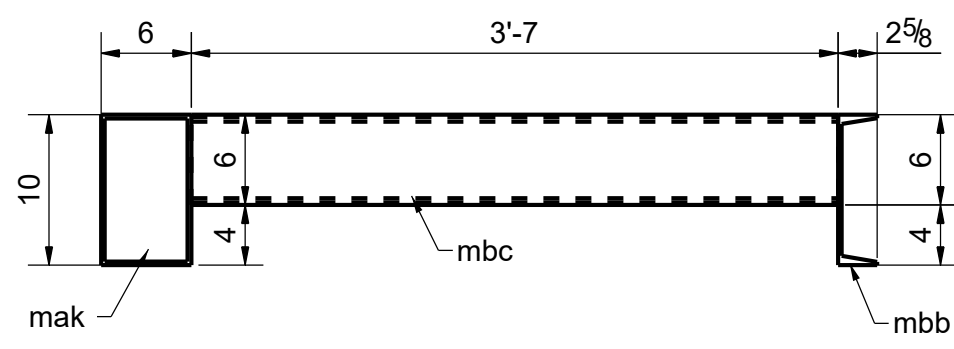
B - B



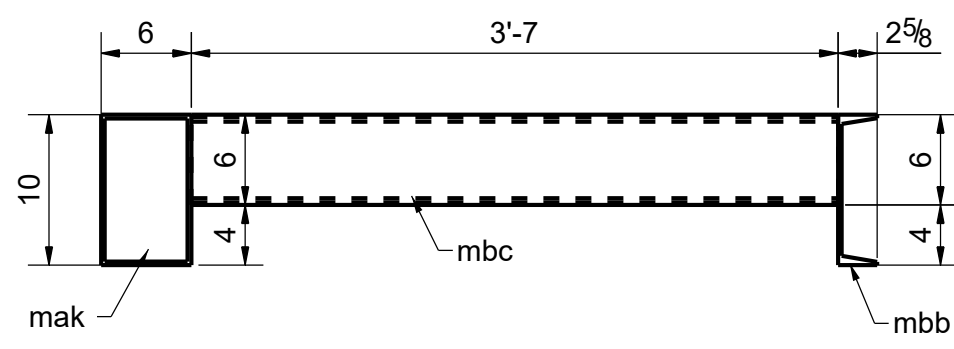
C - C



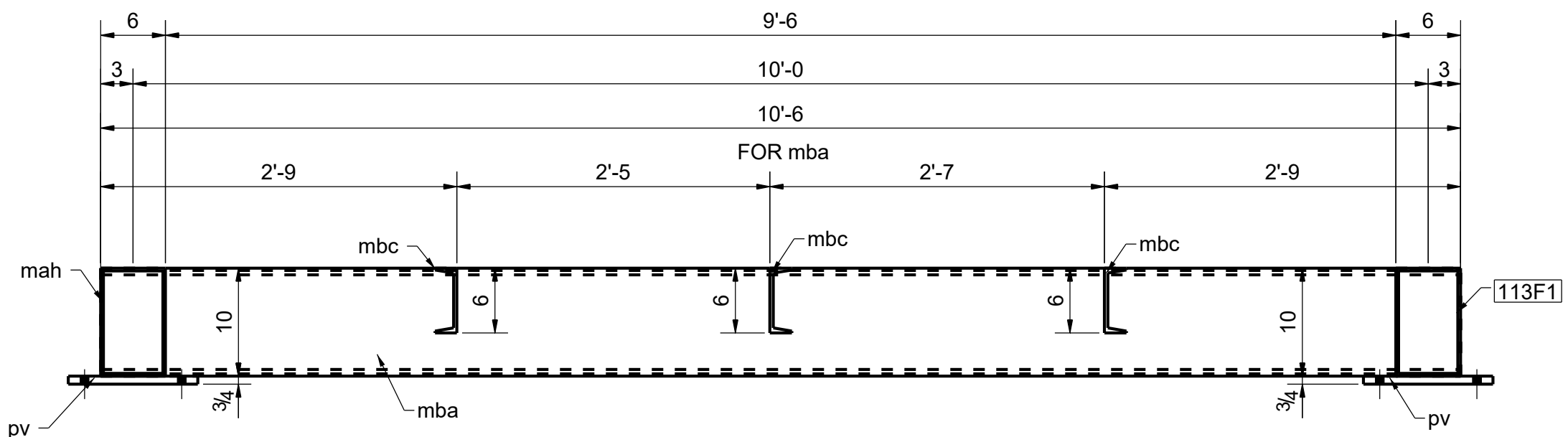
D - D



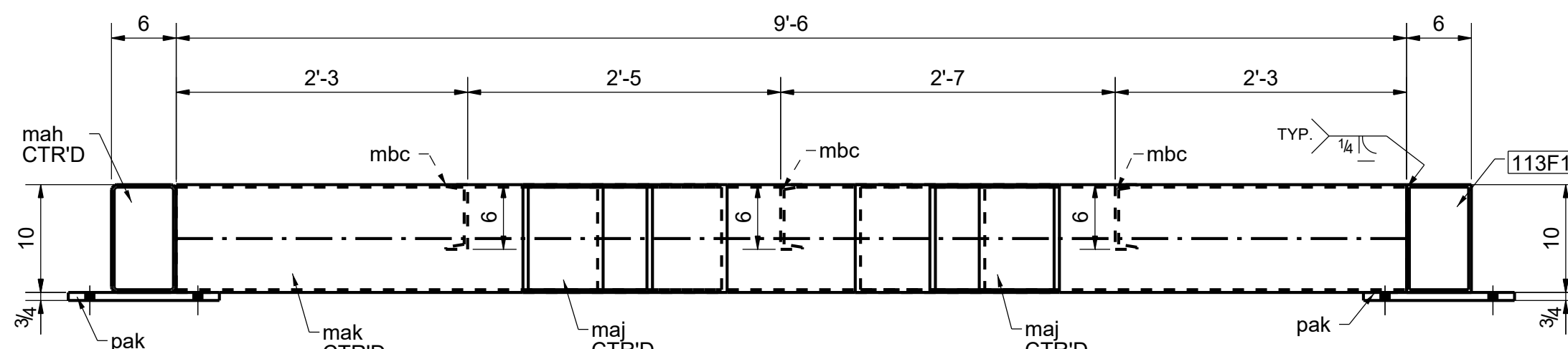
E - E



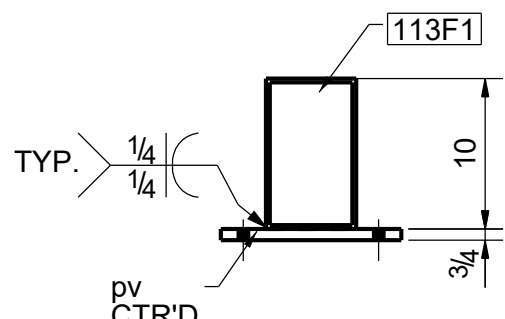
F - F



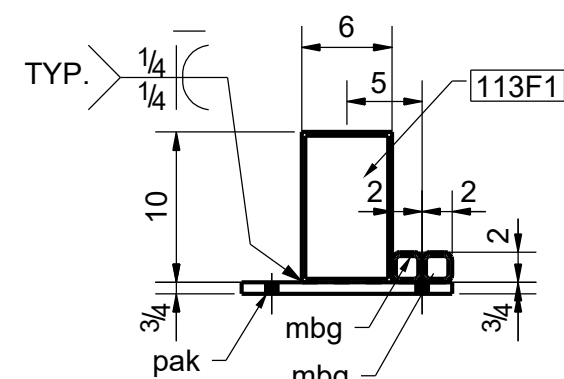
G - G



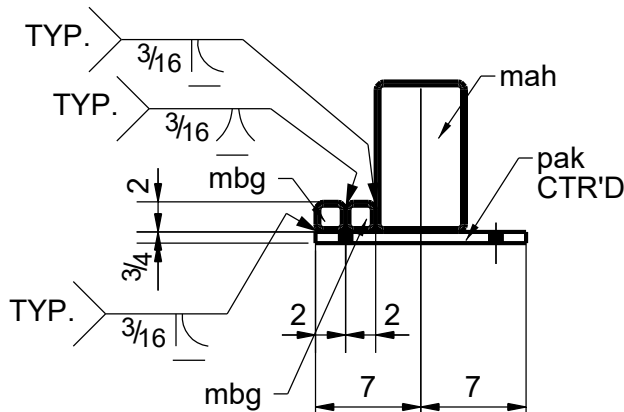
I - I



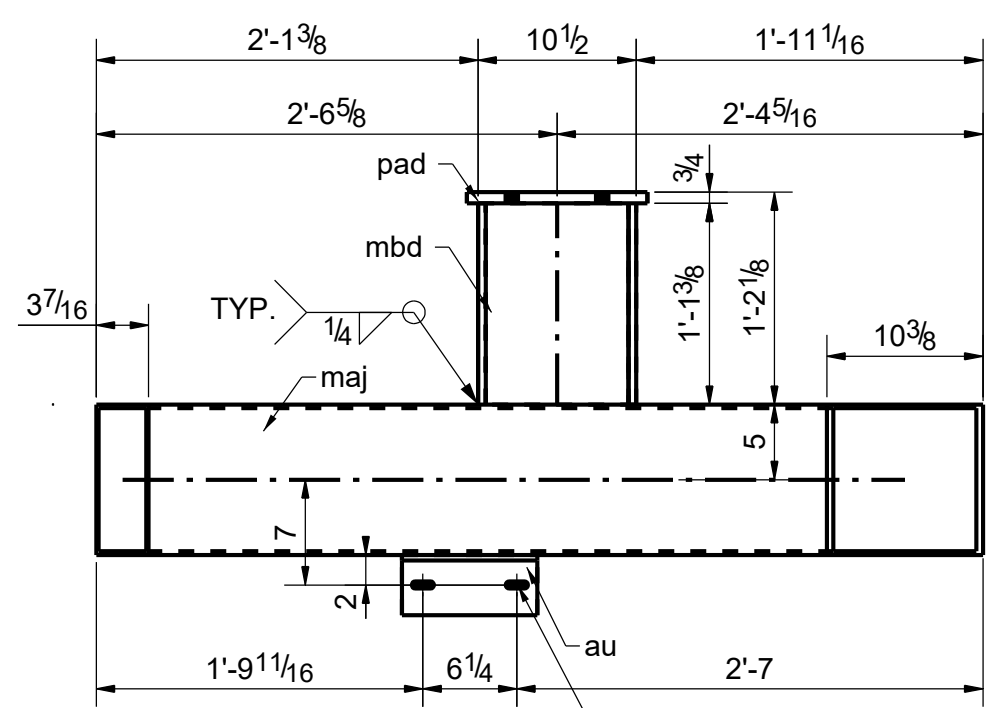
H - H



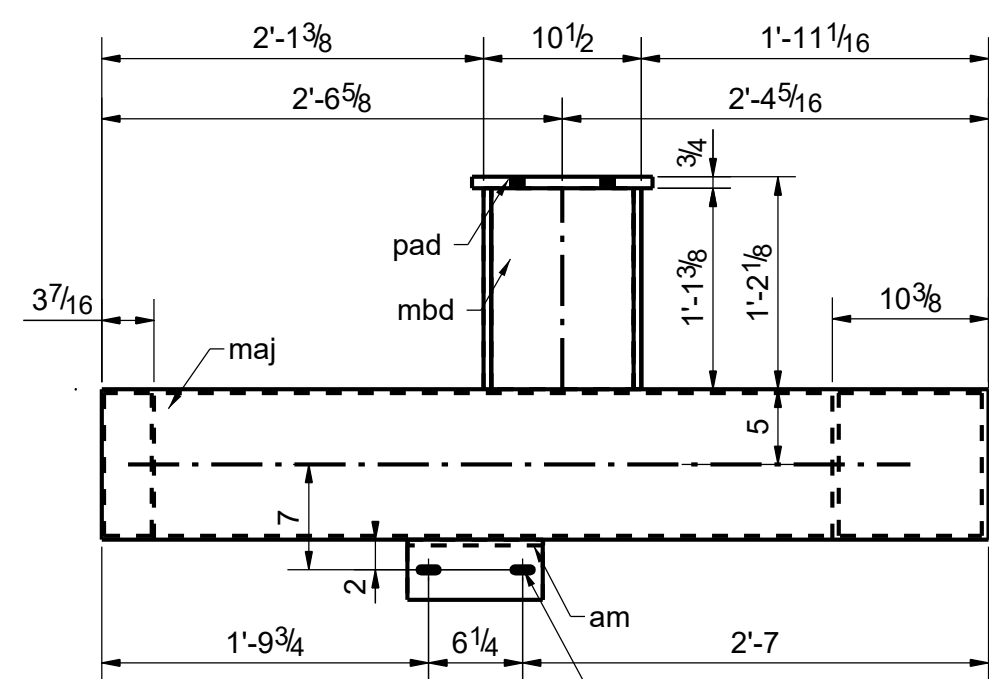
K - K



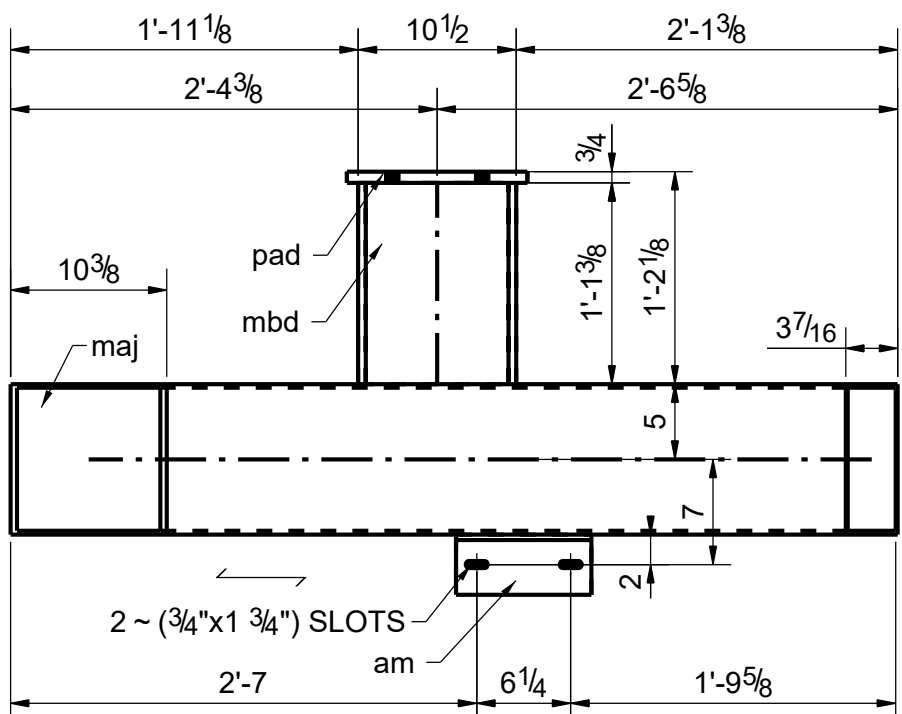
J - J



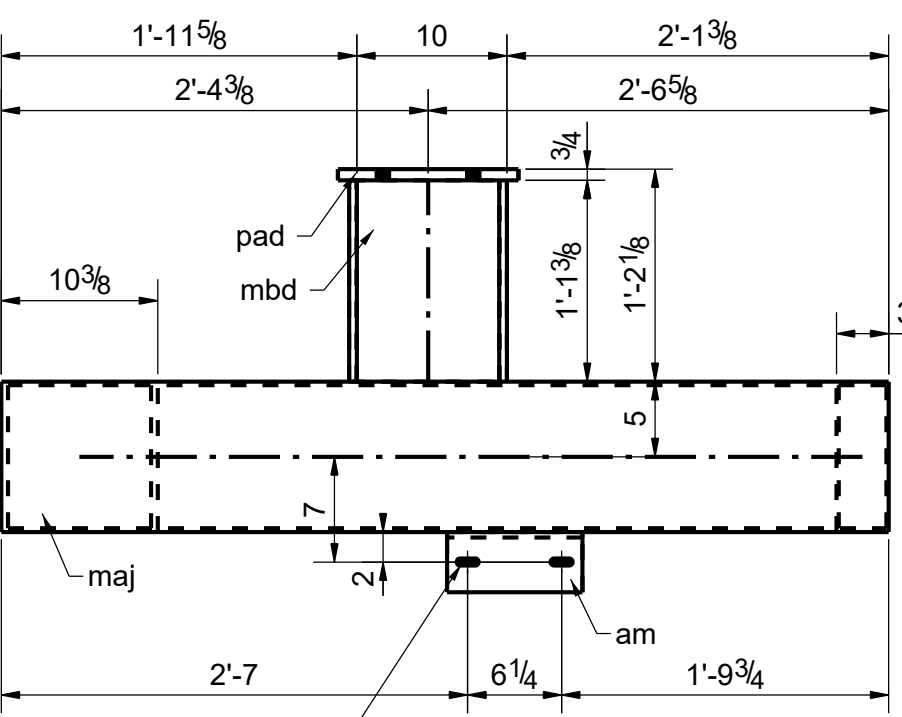
R - R



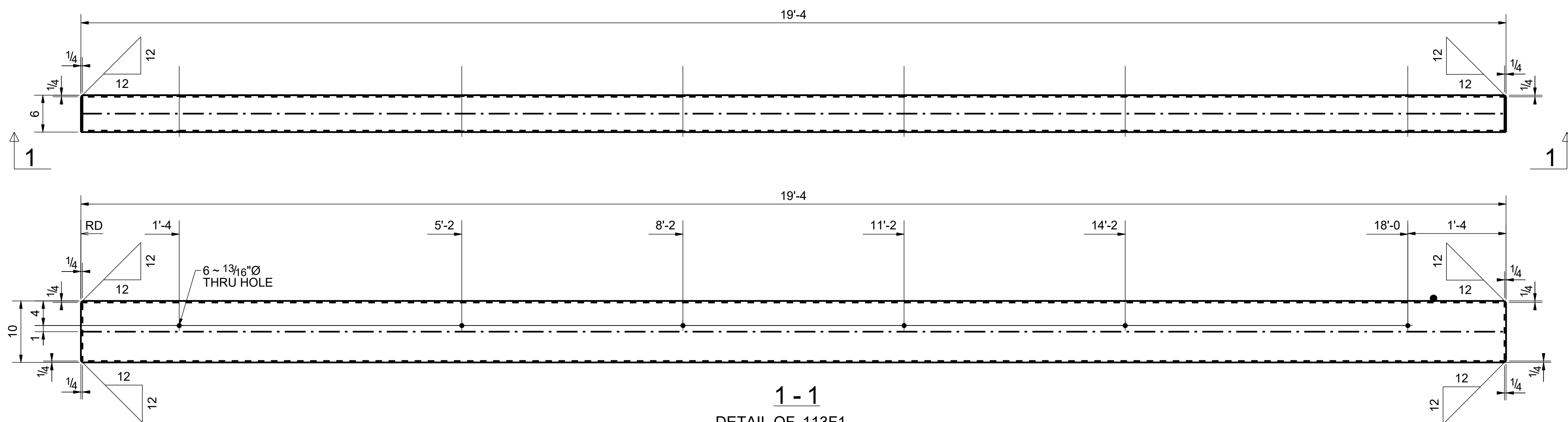
P - P



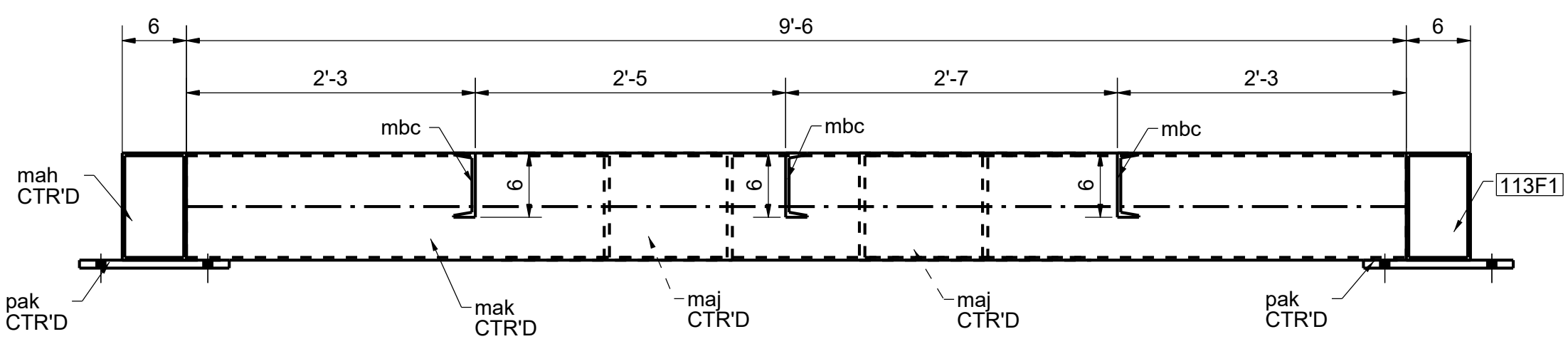
Q - Q



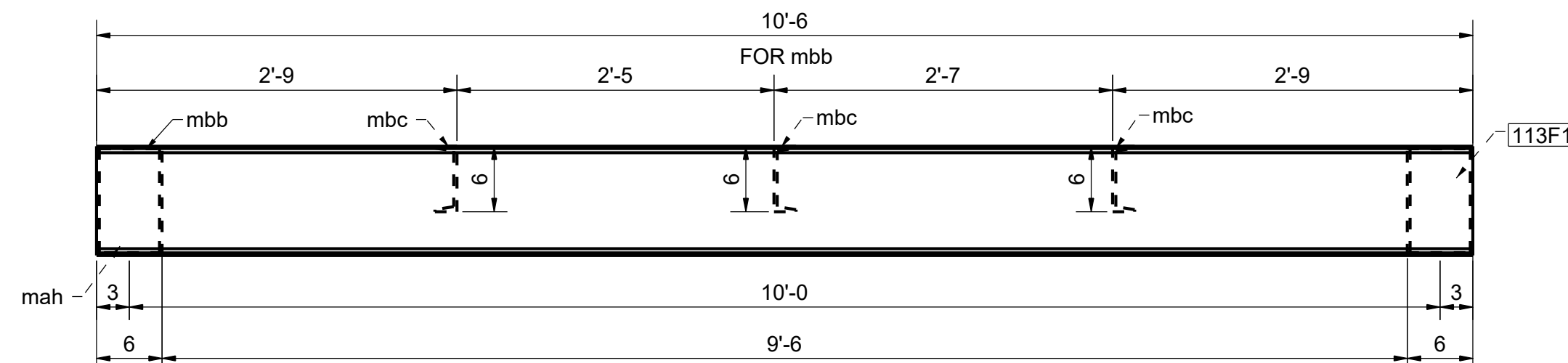
O - O



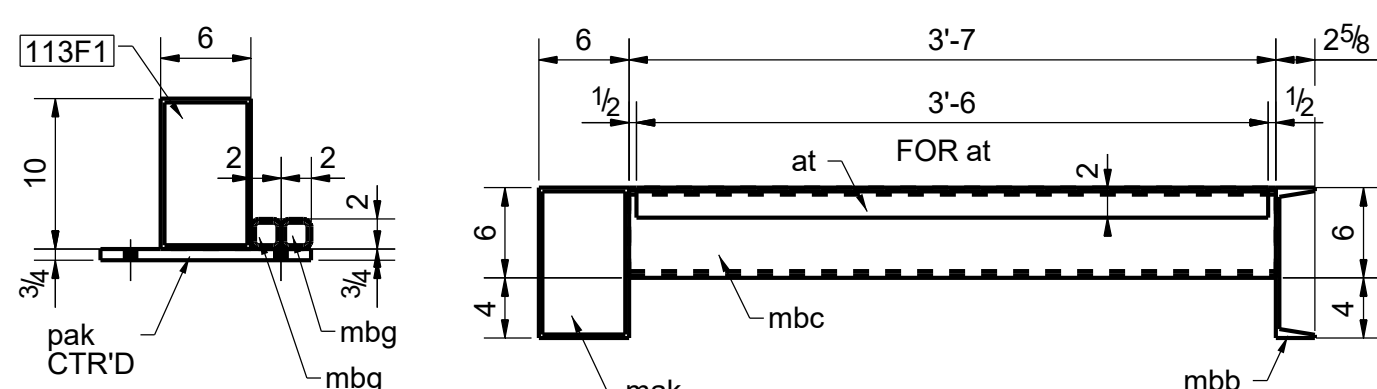
1 - 1
DETAIL OF 113F1



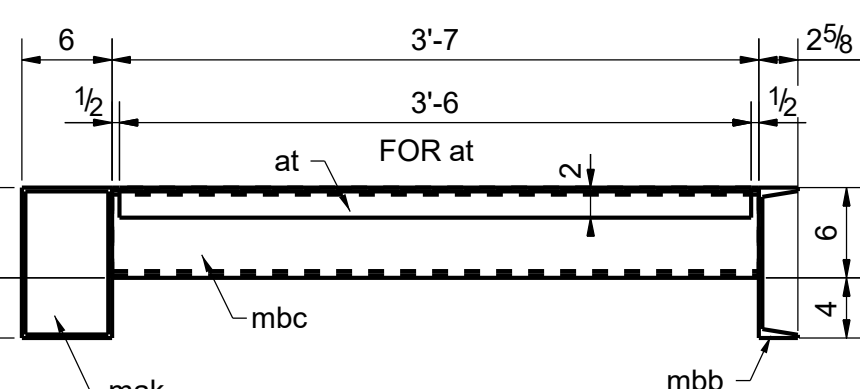
M - M



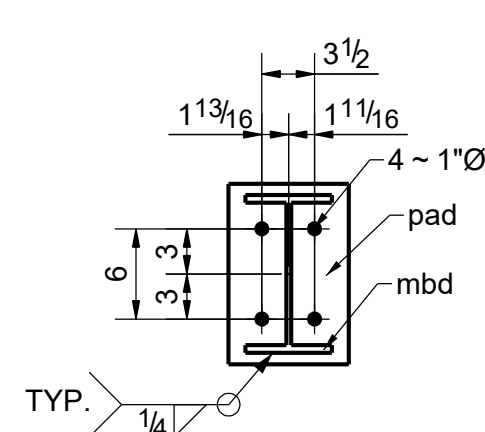
N - N



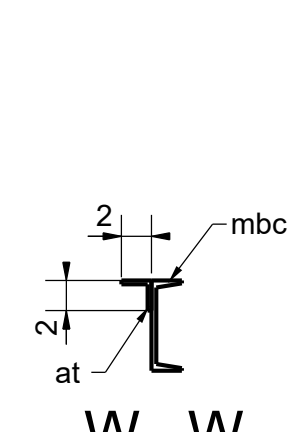
T - T



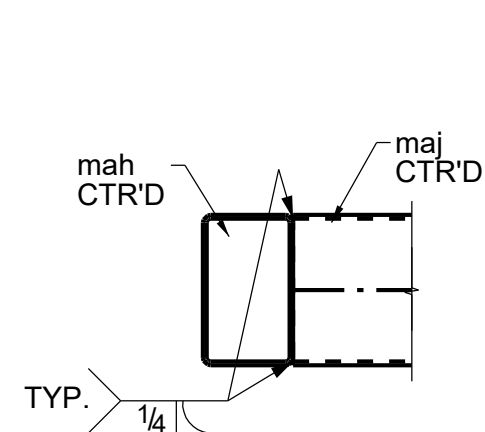
V - V



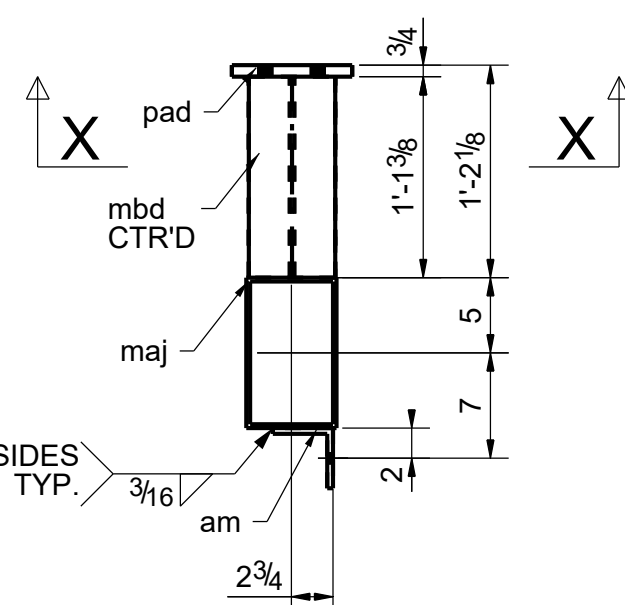
X - X



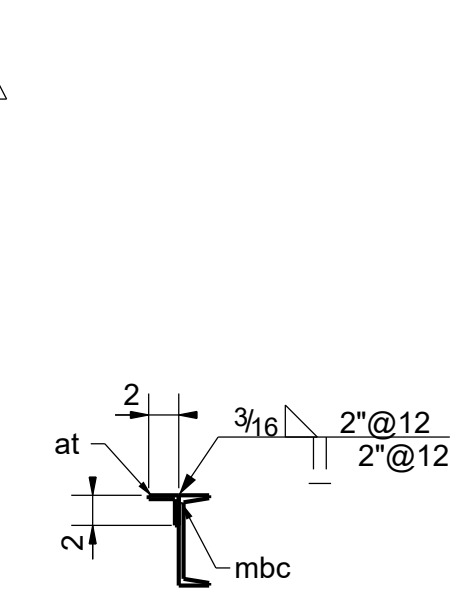
W - W



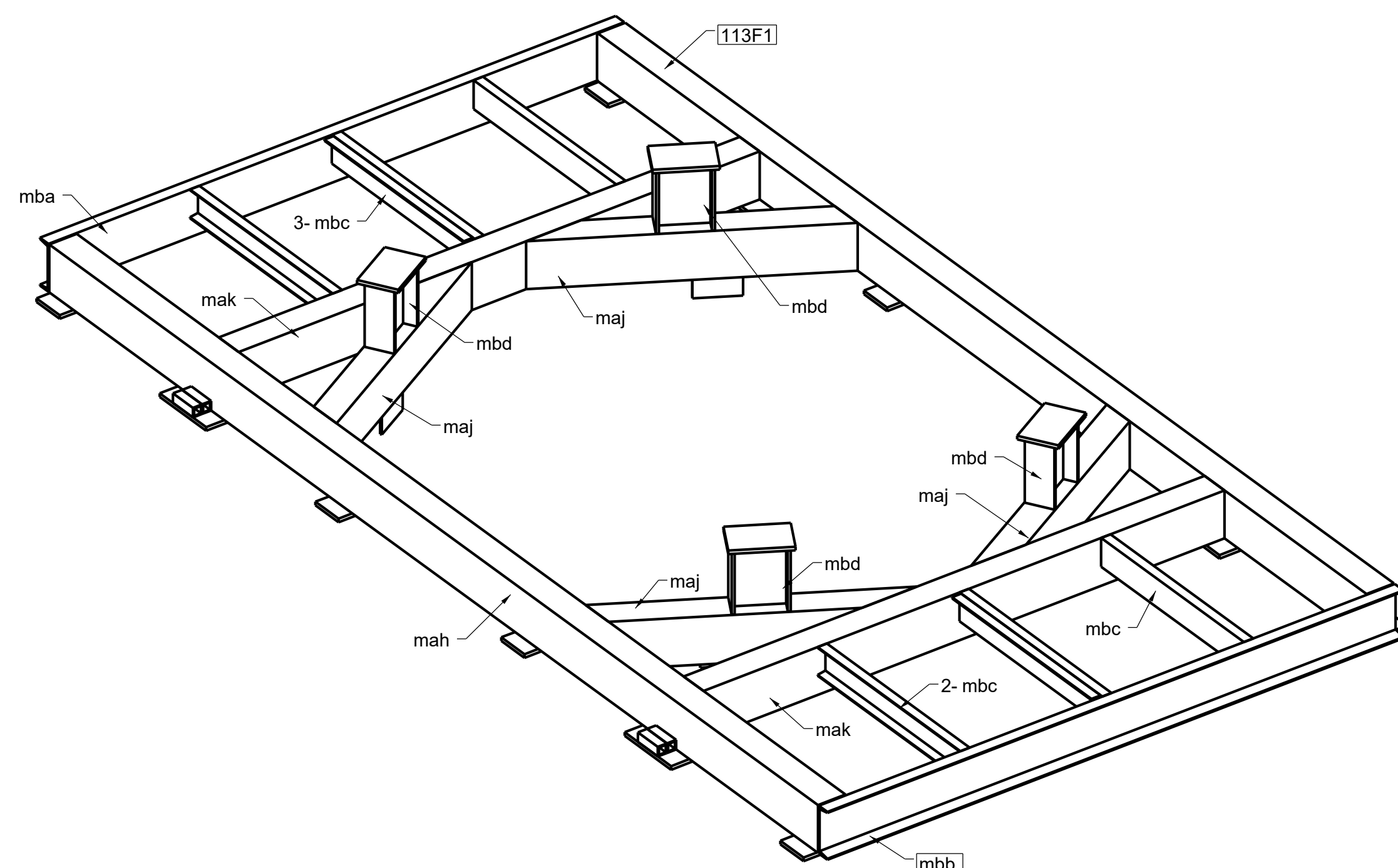
L - L



S - S



U - U



3D-VIEW OF 113F1

WORK THIS SHEET WITH 113,113-A